The working (6,7): a complete named derivation

101 graphical steps | Four explicit sphere moves | Portrait US Letter

VERY BRIEF SUMMARY. Remove the incident half twists, use a five-point disk, turn the remaining positive factors into FOUR exterior twists by PC lanterns, roses, daisies and Hurwitz transports, then cap their fourth power.

The previously missing geometric boundary closure is now supplied. The final equality is obtained from actual named substitutions and factor transports, rather than a whole-product action recognition. All four sphere moves retain their disk discrepancies, exact sphere certificates and separately checked genus-two lift.

What is still open. This detailed route has 101 retained graphical steps: five fewer than the previous 106-step named proof and shorter than the earlier 177-step archive. It does not yet meet the goal of fewer than twenty meaningful operations. Reusable named patterns and fixed-factor drags are identified, with all adjacent checks retained in the certificates.

How to read it. Read each factorization downward. Curves are paired with the braid representing that exact support; blue marked-point dots continue at braid separators. Only the changed interval is printed at a move, with the unchanged factors counted above and below. Every actual SPHERE MOVE has a yellow banner. Individual supports are not cumulative joint cut-system certificates.

Logical scope. Common frames and cyclic starts are explicit conjugations. The terminal four-exterior product is central, so reversing those conjugations recovers the source equality without assuming it was central. Marked points are not framed boundaries; signed framing balances are retained for PC substitutions. A single five-point cap has lift -I; the sixth- and fourth-power caps used here have +I.

Contents

Full diagrams, named sub-derivations and sphere discrepancies.

Complete named diagram sequence

I. The fixed-point boundary chart 4

II. Existing roots remove all half twists 17

III. The first exterior and simpler inside daisy 37

IV. Reach the second exterior twist 57

V. Return the actual borrowed root 65

VI. Align the four-point twists 71

VII. Match the actual daisy petals 77

VIII. Two daisies, four exteriors and the sphere cap 83

IX. Sphere discrepancies and certificates 87

101 graphical steps retained. Four sphere moves labeled; full discrepancies in Section IX.

What the steps will do

1. Reach the five-point chart. Split the required squares, match and absorb the incident half twists, change the exterior point explicitly, and retain then separately cap the six framing corrections.

2. Use the existing roots. Borrow Q14 from the actual four-point daisy, transport it to match the original half twists, return the same Q14 and remove the last half pair. No auxiliary half twists are introduced.

3. Extract two exterior twists. Actual cabled lanterns and borrowed-petal configurations expose two visible exterior twists. Keep both while closing the residual factors.

4. Return the borrowed root. The actual 15, 14 and 45 curves form a transported PC rose. Return the resulting transported root across the old outer and close its daisy; an untransported root with the same enclosed points would fail.

5. Align the four-point twists. Two actual pair transports align the four-point curves. Combine them and use an explicit common disk frame to make their square round.

6. Match two daisies and cap. Three explicit petal swaps and two individually matched transported daisies give four identical exterior twists. Combine their fourth power, followed by a separate labeled sphere cap with +I lift.

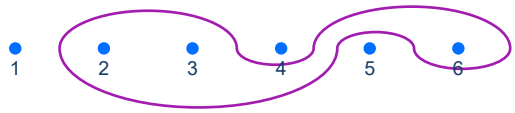
I. The fixed-point boundary chart

Actual ordered factor supports and their connected braids. All unchanged context is counted, and every retained move is shown.

Complete factorization before The fixed-point boundary chart

FULL ORDERED FACTORIZATION. Read downward. Braids remain connected, with strand colors retained; disjoint crossings may share a level.

1. F1
T3^2



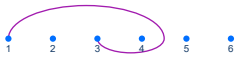
2. F2
H



3. F4
T3^2



4. F3
H



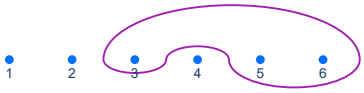
5. F5
T3^2



6. F6
H



7. F8
T3^2



8. F9
H



9. F7
H



10. F10
T3^2



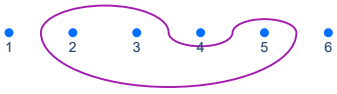
11. F11
H

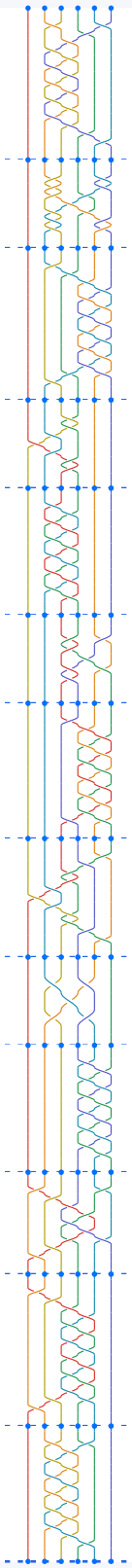


12. F12
T3^2



13. F13
T3^2





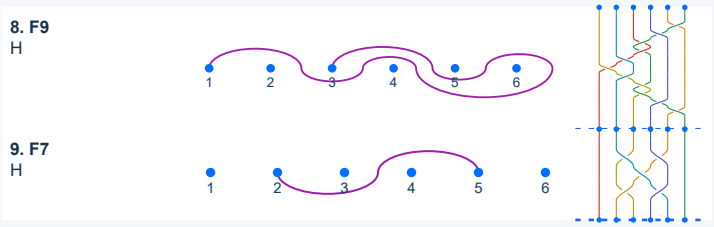
I - step 001

Change the working ordering by an inverse Hurwitz move

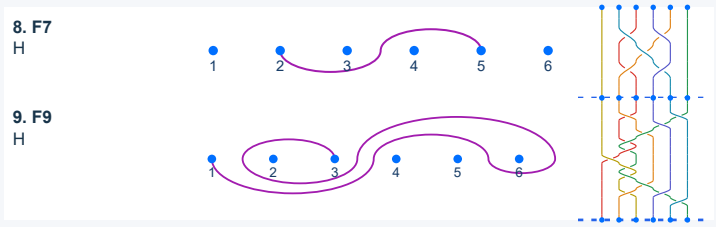
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

7 unchanged displayed factors above; 4 below. Only the changed interval is shown.

Before



After



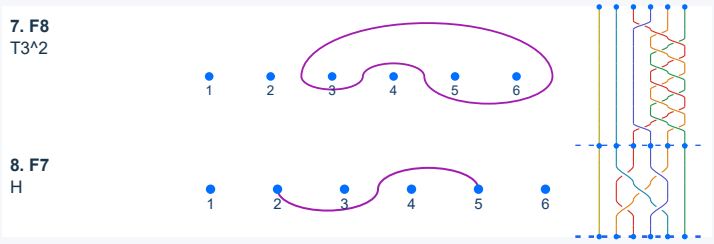
I - step 002

Change the working ordering by an inverse Hurwitz move

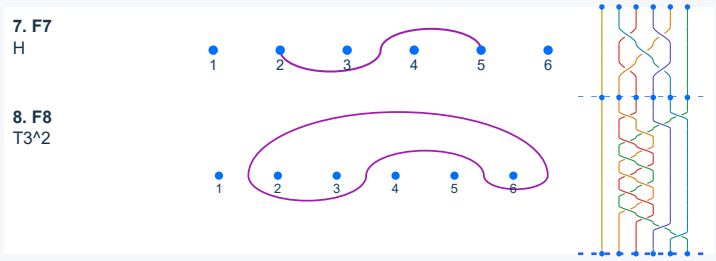
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

6 unchanged displayed factors above; 5 below. Only the changed interval is shown.

Before



After



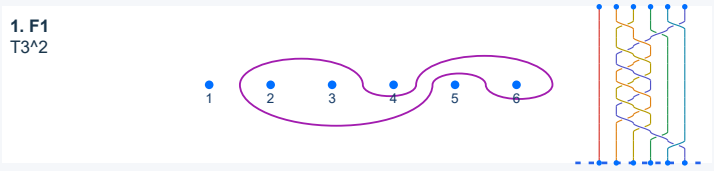
I - step 003

Split F1 into ordinary roots

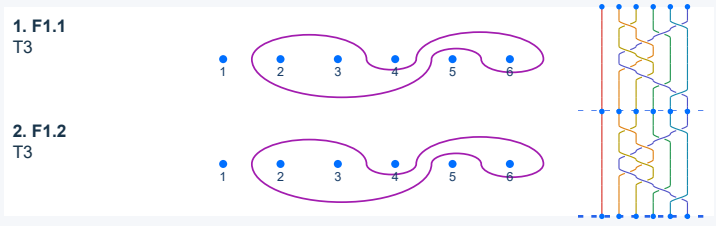
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

0 unchanged displayed factors above; 12 below. Only the changed interval is shown.

Before



After



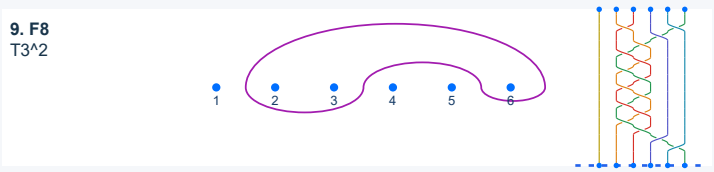
I - step 004

Split F8 into ordinary roots

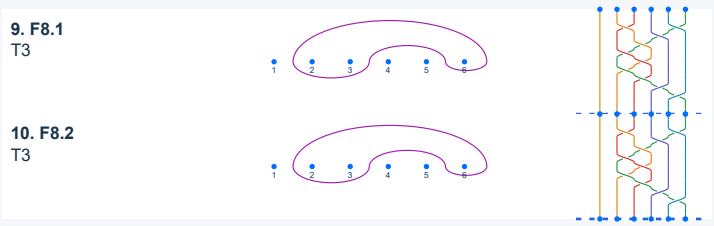
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

8 unchanged displayed factors above; 5 below. Only the changed interval is shown.

Before



After



I - step 005

Split F10 into ordinary roots

DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

11 unchanged displayed factors above; 3 below. Only the changed interval is shown.

Before



After



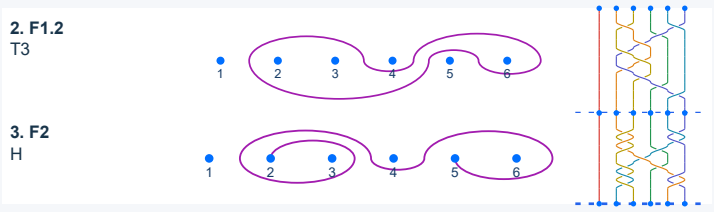
I - step 006

F1.2 down past F2

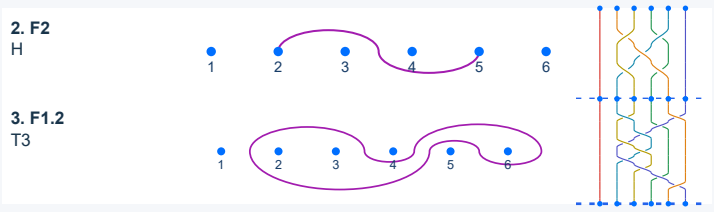
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

1 unchanged displayed factors above; 13 below. Only the changed interval is shown.

Before



After



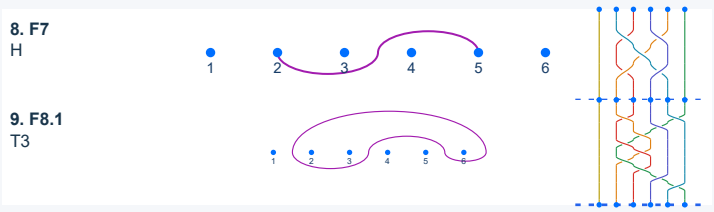
I - step 007

F8.1 up past F7

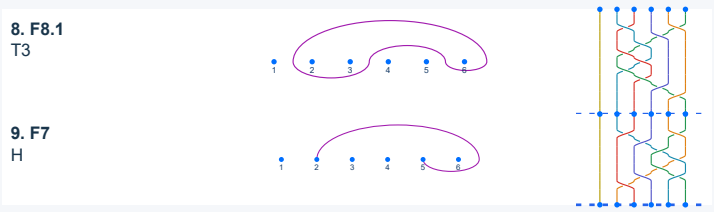
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

7 unchanged displayed factors above; 7 below. Only the changed interval is shown.

Before



After



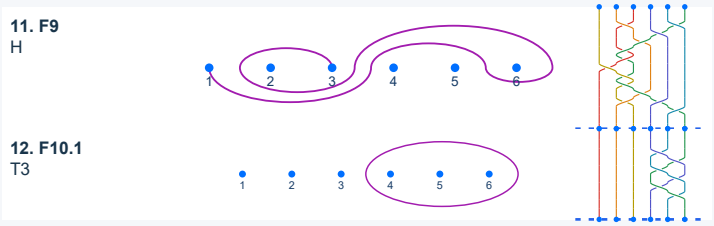
I - step 008

F10.1 up past F9

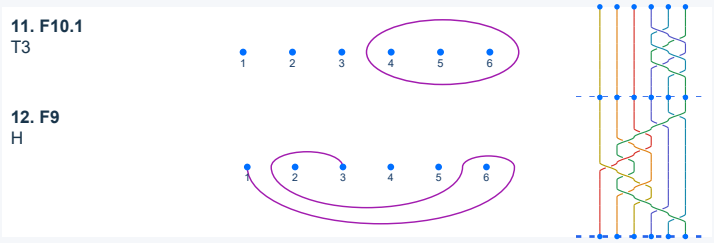
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

10 unchanged displayed factors above; 4 below. Only the changed interval is shown.

Before



After



I - step 009

F10.1 up past F8.2

DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

9 unchanged displayed factors above; 5 below. Only the changed interval is shown.

Before



After



I - step 010

F10.1 up past F7

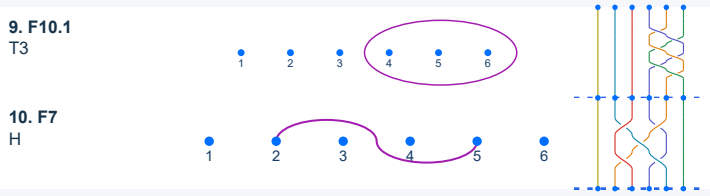
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

8 unchanged displayed factors above; 6 below. Only the changed interval is shown.

Before



After



I - step 011

Use the exact disk equality of the matched arcs

DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

9 unchanged displayed factors above; 6 below. Only the changed interval is shown.

Before



After



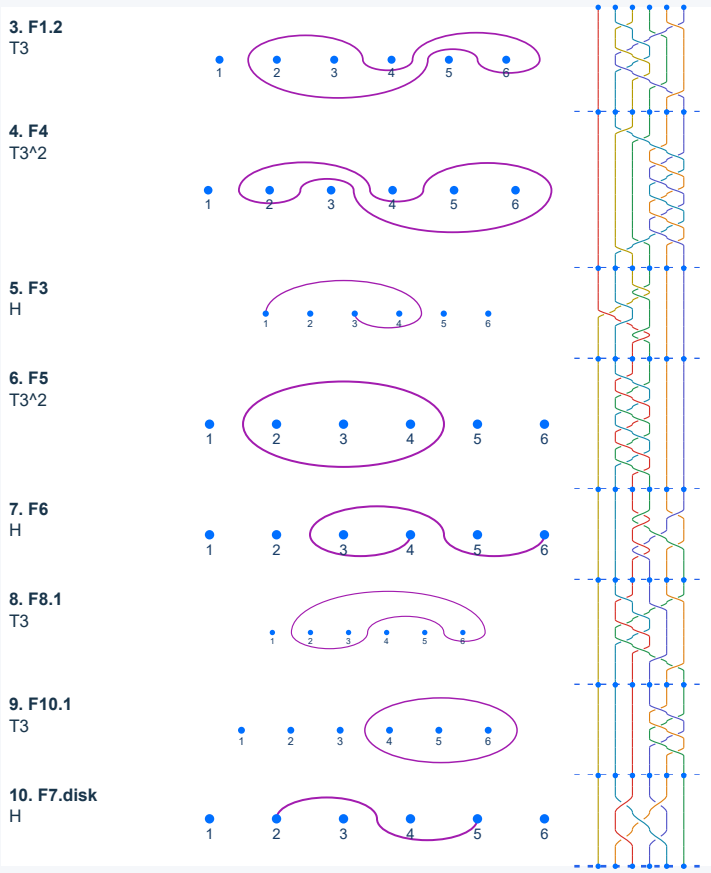
I - step 012

Drag the matched arc up across SEVEN factors (one Hurwitz drag)

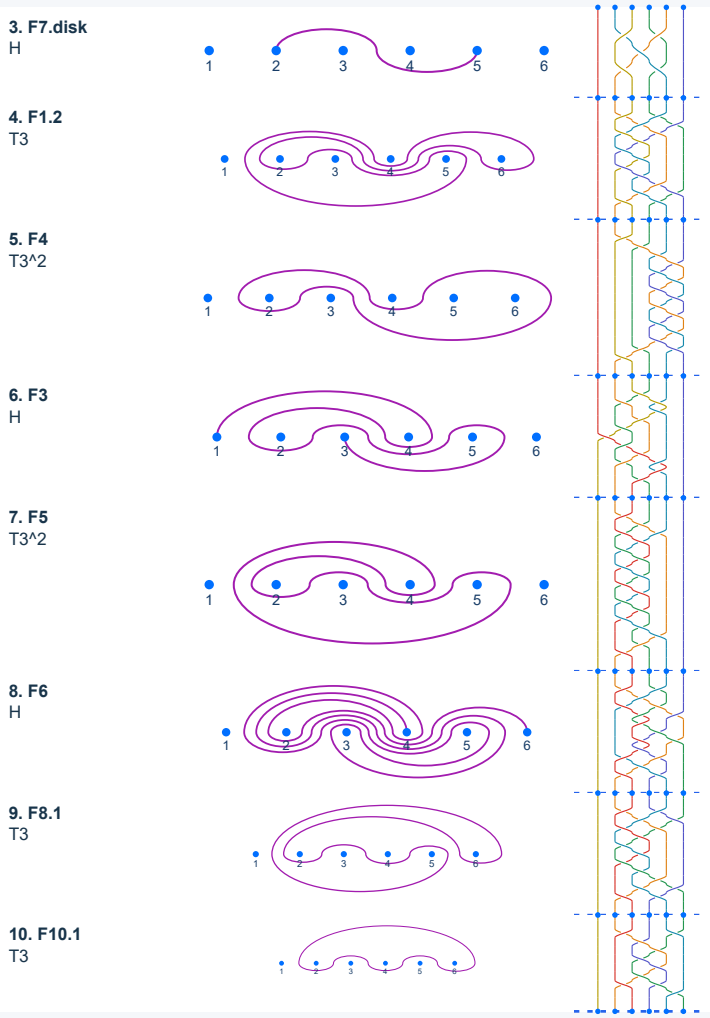
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

2 unchanged displayed factors above; 6 below. Only the changed interval is shown. Retained adjacent checks: 7.

Before



After



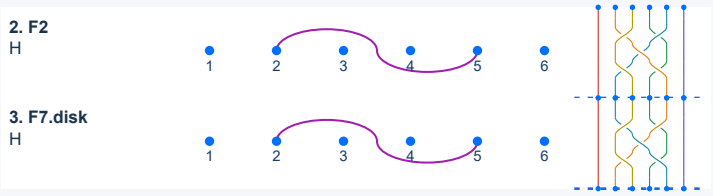
I - step 013

Combine the two matched half twists

DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

1 unchanged displayed factors above; 13 below. Only the changed interval is shown.

Before



After



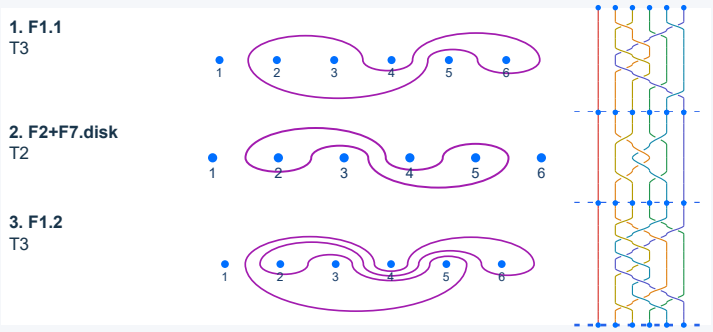
I - step 014

SPHERE LANTERN: absorb the matched pair and the two prefix roots

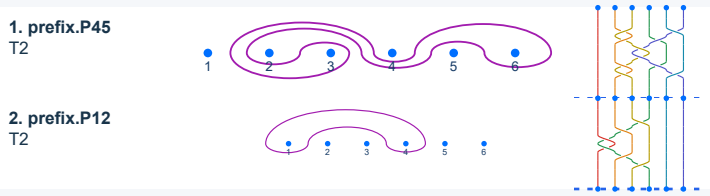
SPHERE MOVE: exact sphere action and +I lift checked; disk discrepancy retained.

0 unchanged displayed factors above; 12 below. Only the changed interval is shown. SPHERE MOVE: full disk discrepancy appears in Appendix IX; chosen lift is +I.

Before



After



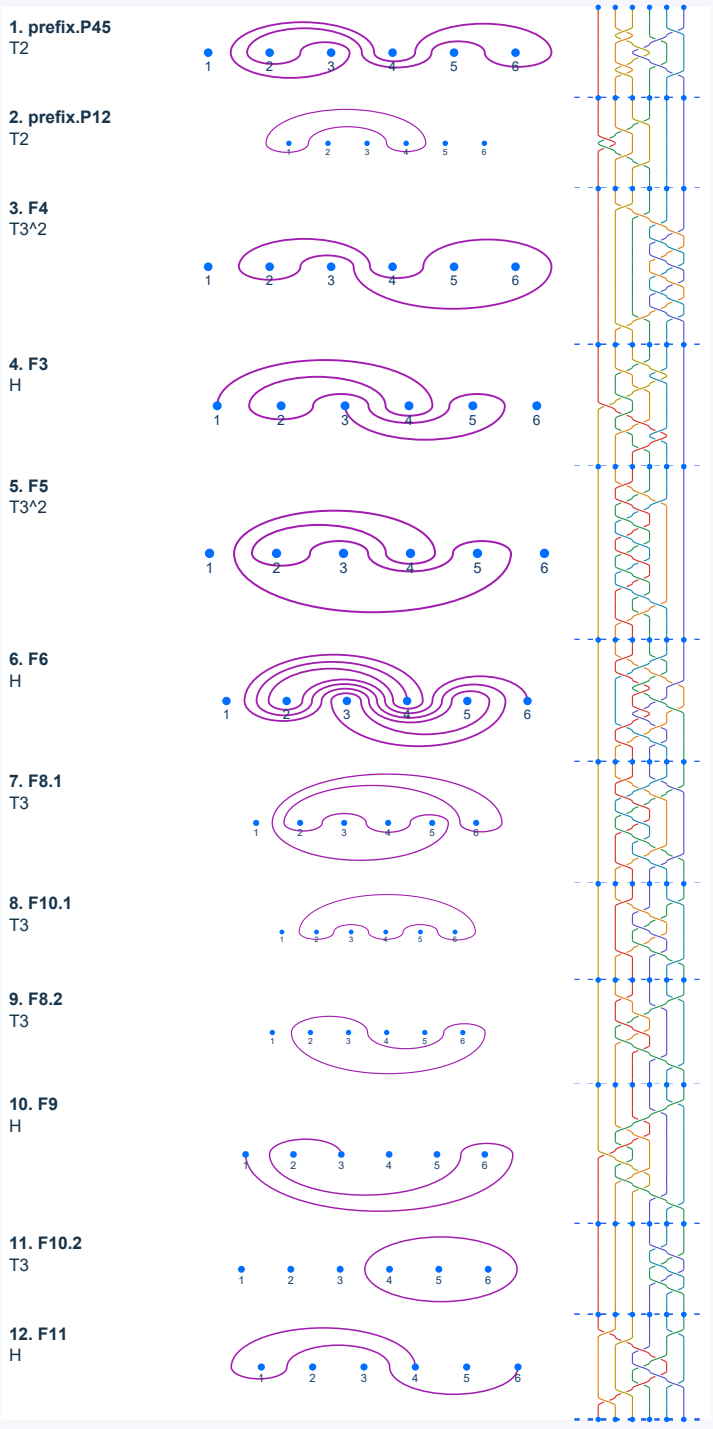
I - step 015

Global frame: fixed working point 2 becomes displayed point 6

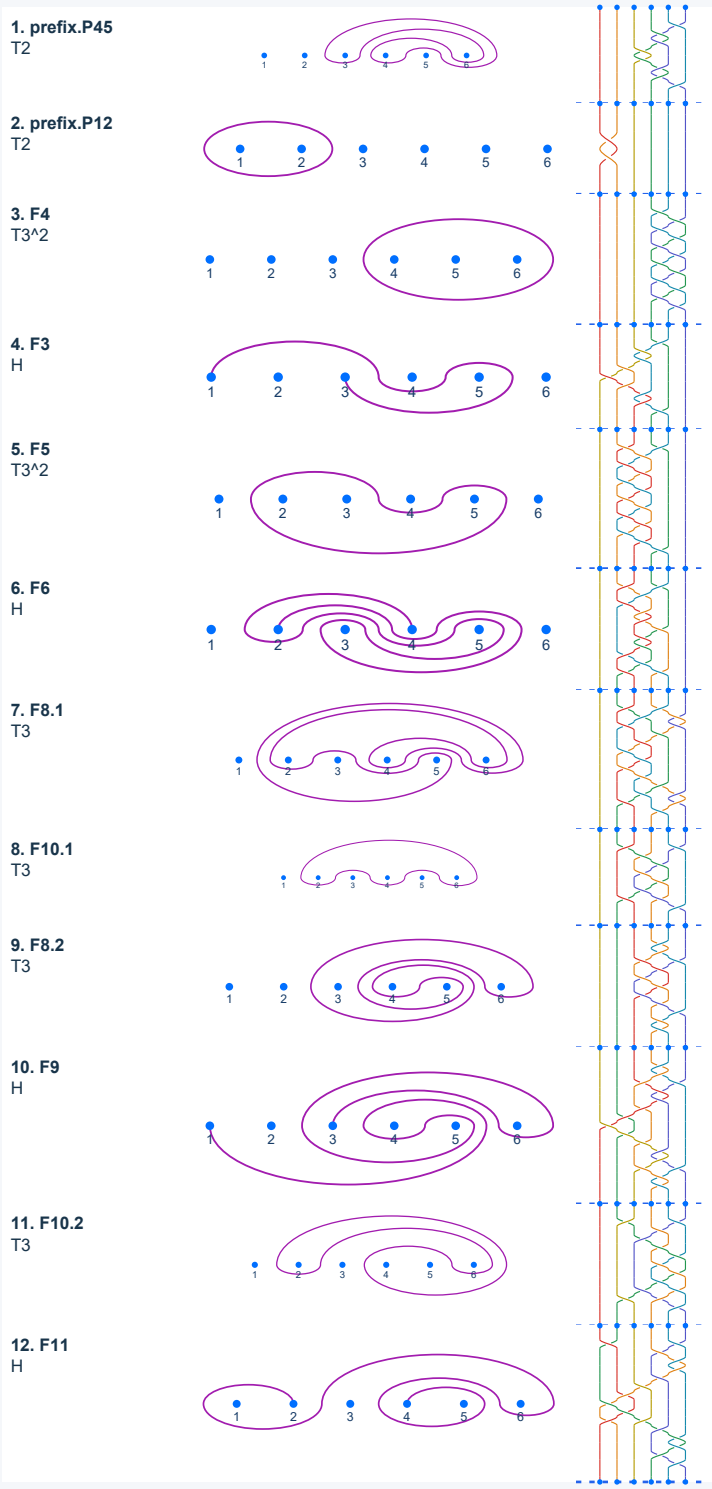
EXPLICIT GLOBAL CONJUGATION: a change of the whole product by the recorded frame.

0 unchanged displayed factors above; 0 below. Only the changed interval is shown.

Before



After



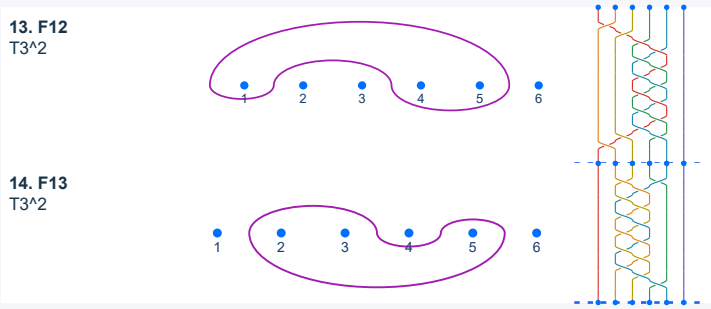
I - step 015 - continued

Global frame: fixed working point 2 becomes displayed point 6

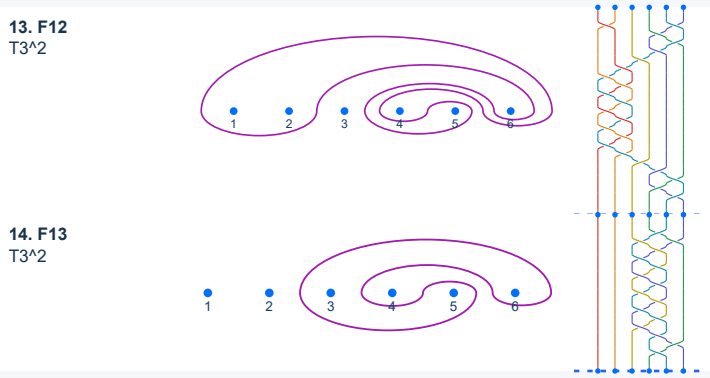
EXPLICIT GLOBAL CONJUGATION: a change of the whole product by the recorded frame.

0 unchanged displayed factors above; 0 below. Only the changed interval is shown.

Before



After



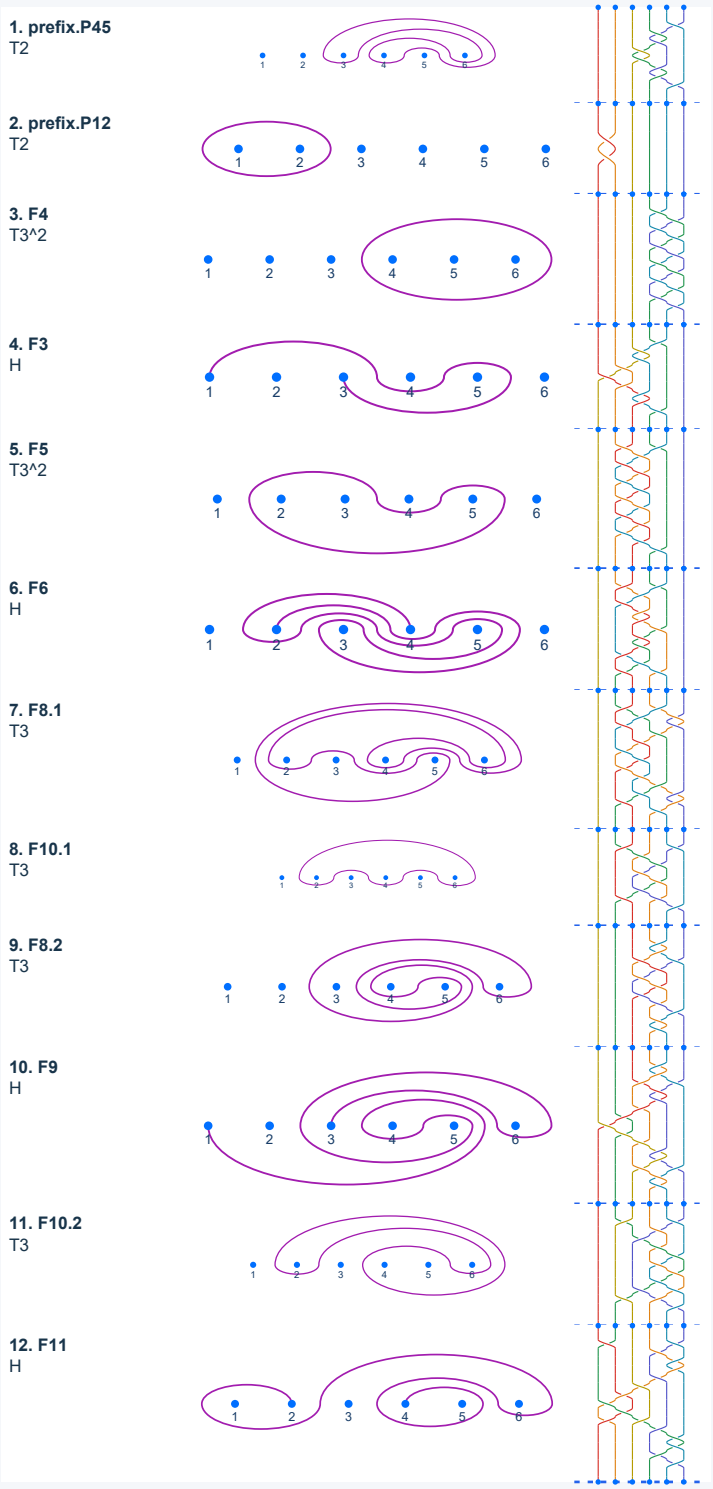
I - step 016

SPHERE RECHART: use the five-point disk; retain SIX boundary corrections

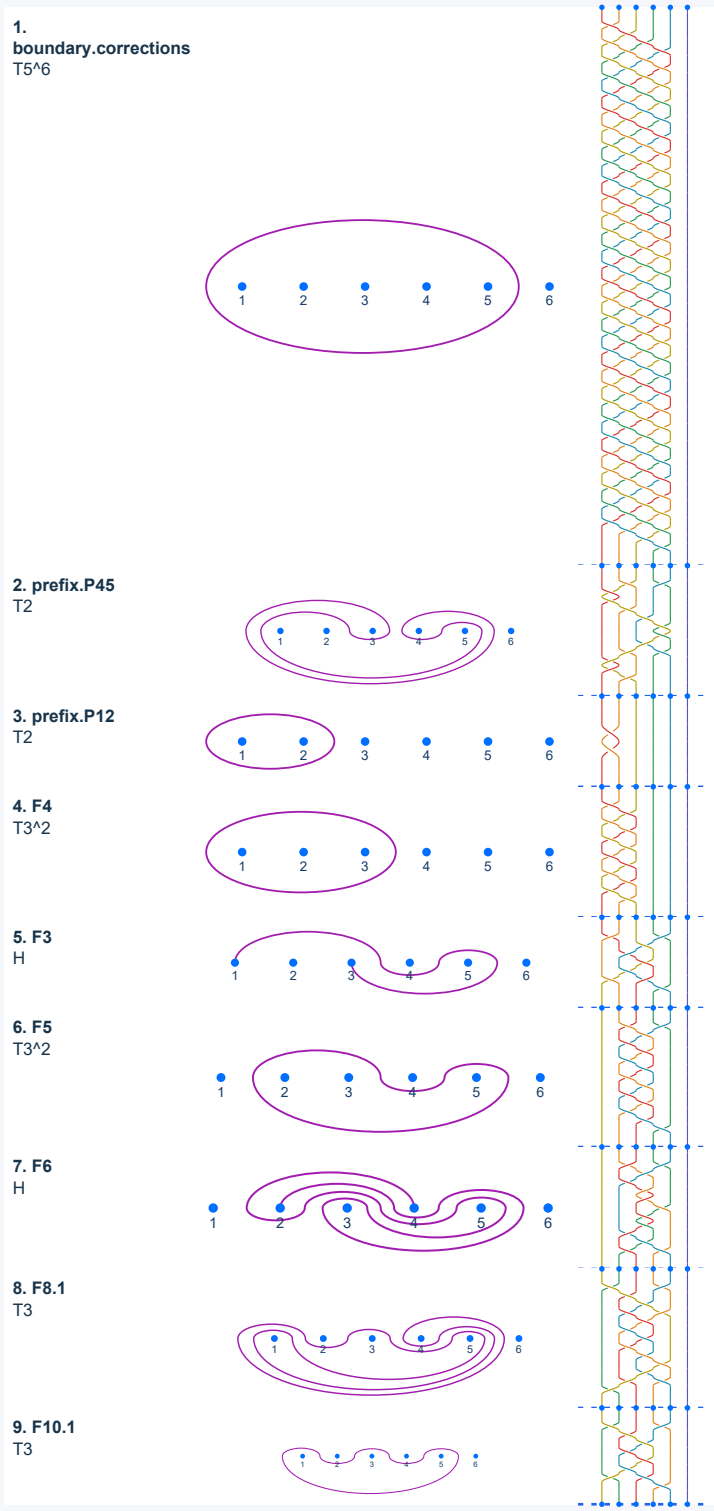
SPHERE MOVE: exact sphere action and +I lift checked; disk discrepancy retained.

0 unchanged displayed factors above; 0 below. Only the changed interval is shown. SPHERE MOVE: full disk discrepancy appears in Appendix IX; chosen lift is +I.

Before



After



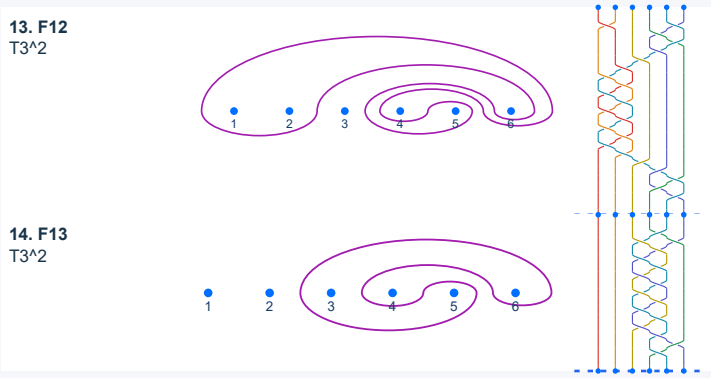
I - step 016 - continued

SPHERE RECHART: use the five-point disk; retain SIX boundary corrections

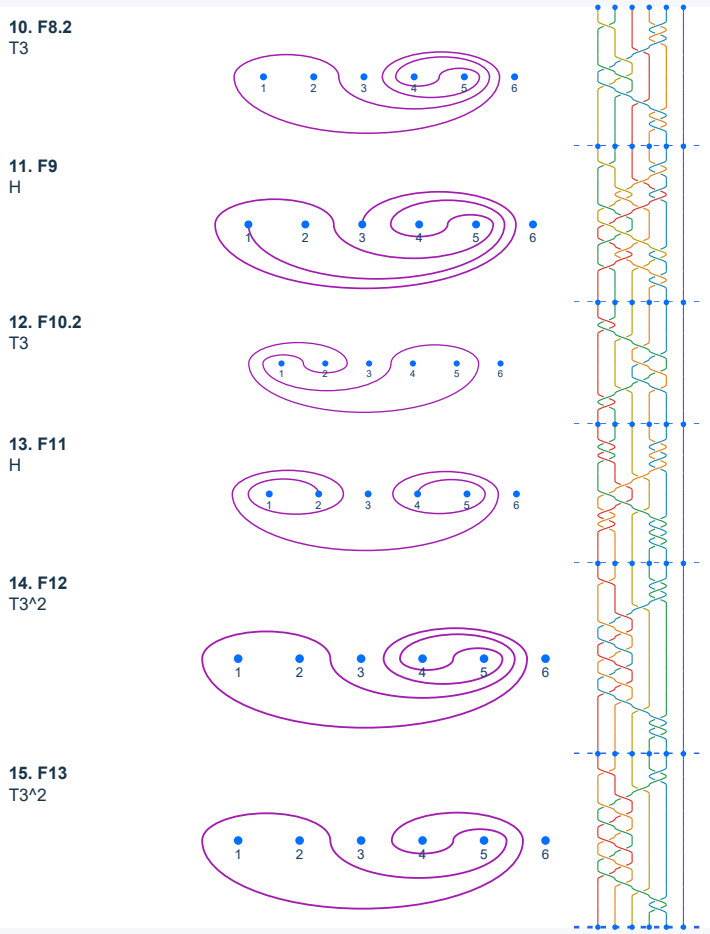
SPHERE MOVE: exact sphere action and +I lift checked; disk discrepancy retained.

0 unchanged displayed factors above; 0 below. Only the changed interval is shown. SPHERE MOVE: full disk discrepancy appears in Appendix IX; chosen lift is +I.

Before



After



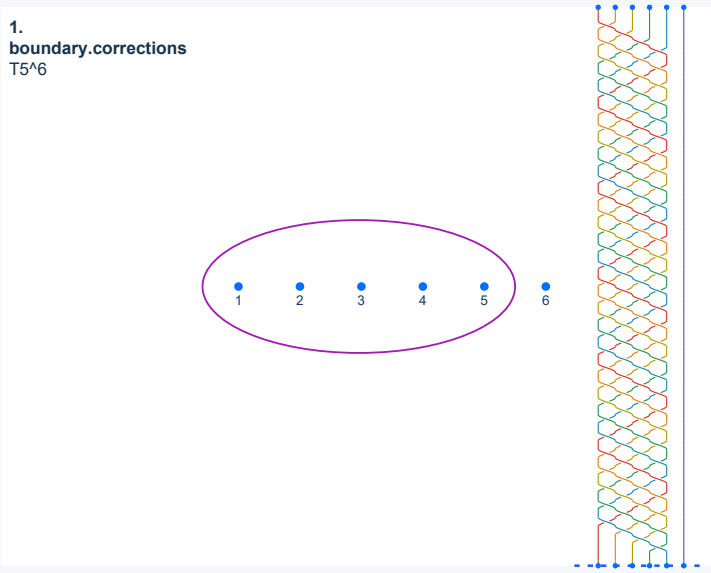
I - step 017

SPHERE MOVE: cap the SIX framing corrections

SPHERE MOVE: exact sphere action and +I lift checked; disk discrepancy retained.

0 unchanged displayed factors above; 14 below. Only the changed interval is shown. SPHERE MOVE: full disk discrepancy appears in Appendix IX; chosen lift is +I.

Before



After

Identity / no factors in this interval.

II. Existing roots remove all half twists

Actual ordered factor supports and their connected braids. All unchanged context is counted, and every retained move is shown.

II - step 018

Split F4 into its two positive units

DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

2 unchanged displayed factors above; 11 below. Only the changed interval is shown.

Before



After



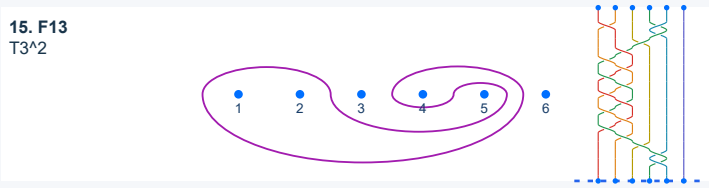
II - step 019

Split F13 into its two positive units

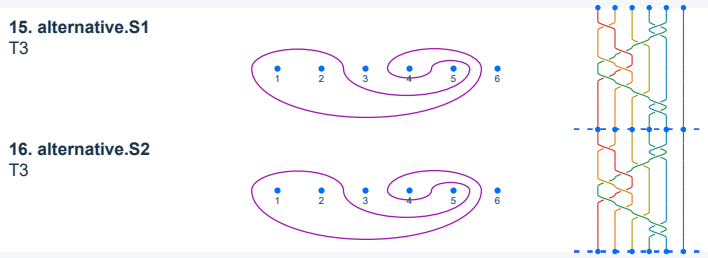
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

14 unchanged displayed factors above; 0 below. Only the changed interval is shown.

Before



After



II - step 020

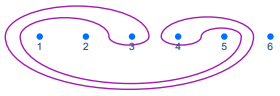
Choose the cyclic start at the existing 124 root; its whole common frame is retained

DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

0 unchanged displayed factors above; 0 below. Only the changed interval is shown. Explicit common conjugation; its full braid frame is saved.

Before

1. prefix.P45
T2



2. prefix.P12
T2



3. alternative.R1
T3



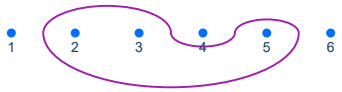
4. alternative.R2
T3



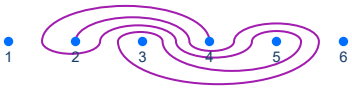
5. F3
H



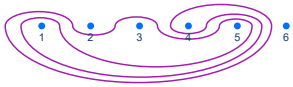
6. F5
T3^2



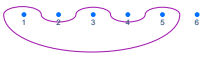
7. F6
H



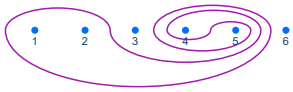
8. F8.1
T3



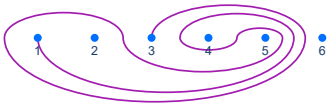
9. F10.1
T3



10. F8.2
T3

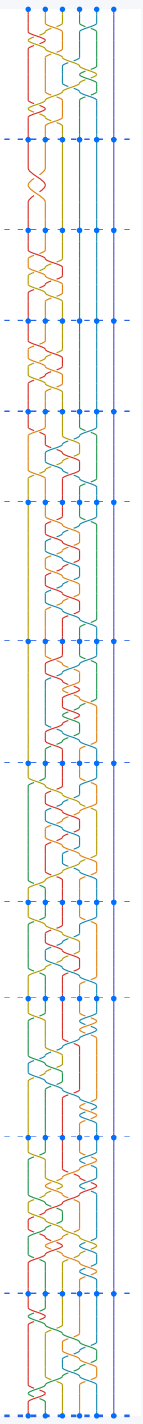


11. F9
H



12. F10.2
T3



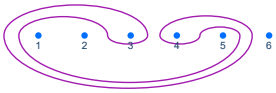


After

1. alternative.S2
T3



2. prefix.P45
T2



3. prefix.P12
T2



4. alternative.R1
T3



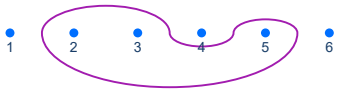
5. alternative.R2
T3



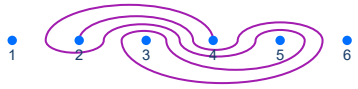
6. F3
H



7. F5
T3^2



8. F6
H



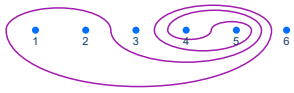
9. F8.1
T3



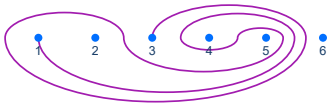
10. F10.1
T3

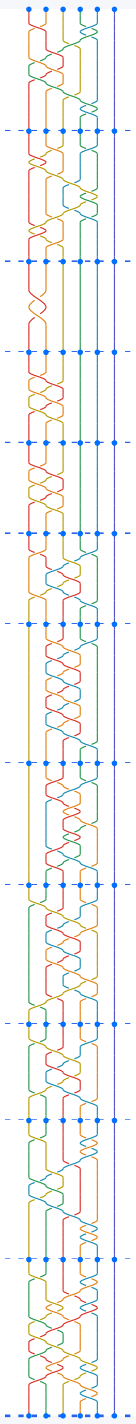


11. F8.2
T3



12. F9
H





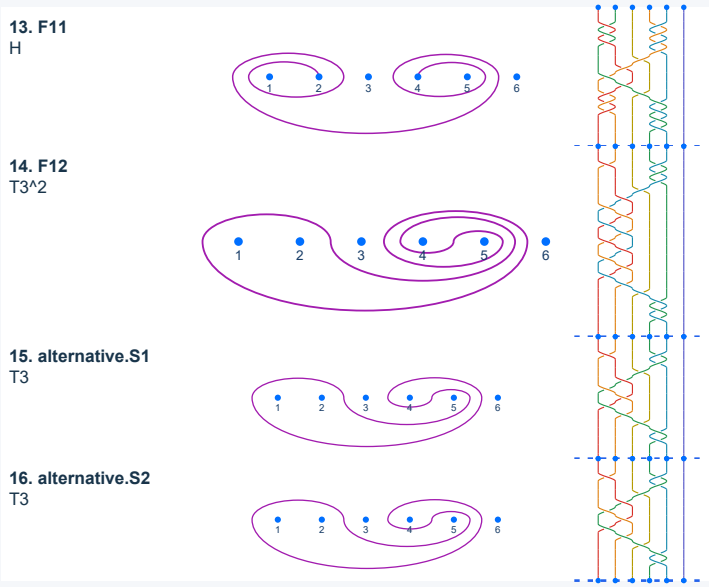
II - step 020 - continued

Choose the cyclic start at the existing 124 root; its whole common frame is retained

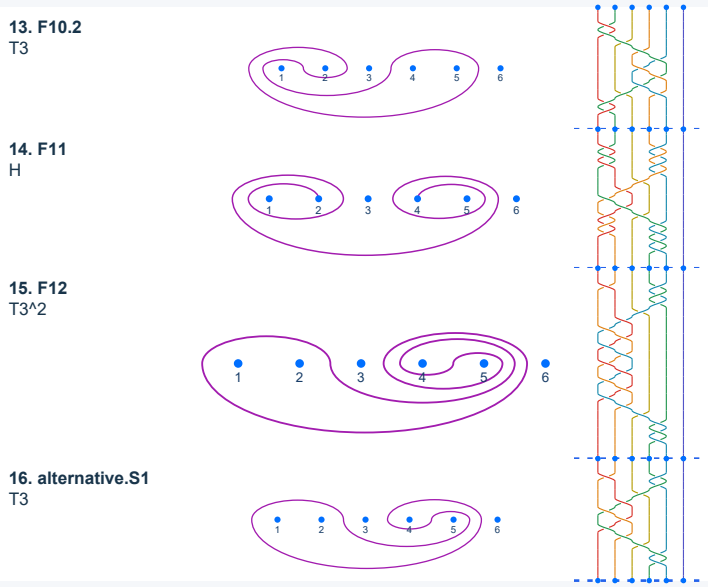
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

0 unchanged displayed factors above; 0 below. Only the changed interval is shown. Explicit common conjugation; its full braid frame is saved.

Before



After



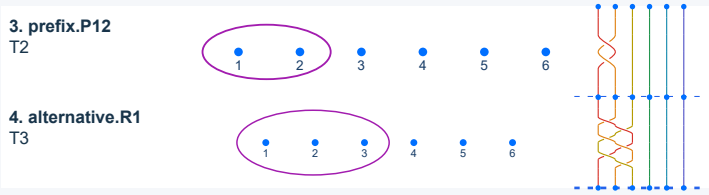
II - step 021

Commute one round T123 root above the round T12

DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

2 unchanged displayed factors above; 12 below. Only the changed interval is shown. Retained adjacent checks: 1.

Before



After



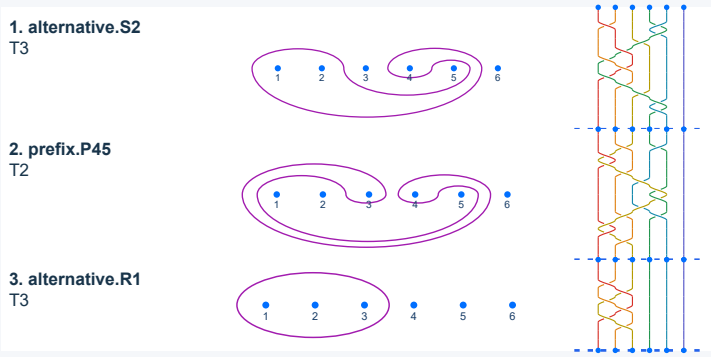
II - step 022

PC cluster lantern on the ACTUAL 124, 34 and 123 curves

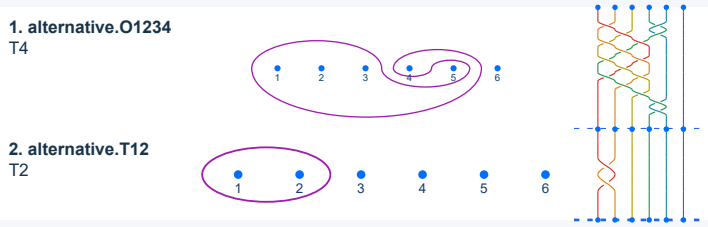
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

0 unchanged displayed factors above; 13 below. Only the changed interval is shown.

Before



After



II - step 023

Combine the matching alternative.T12 and prefix.P12

DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

1 unchanged displayed factors above; 12 below. Only the changed interval is shown.

Before



After



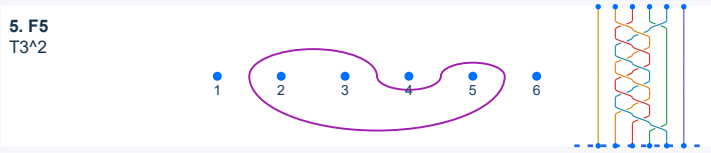
II - step 024

Split F5 into its two positive units

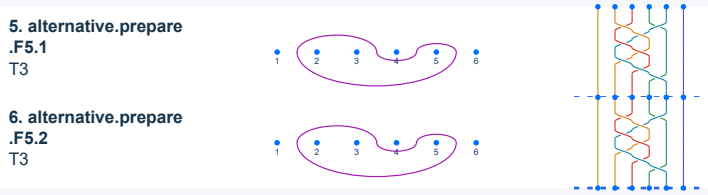
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

4 unchanged displayed factors above; 9 below. Only the changed interval is shown.

Before



After



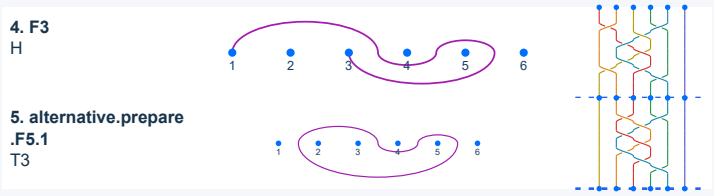
II - step 025

One 235 root up over F3

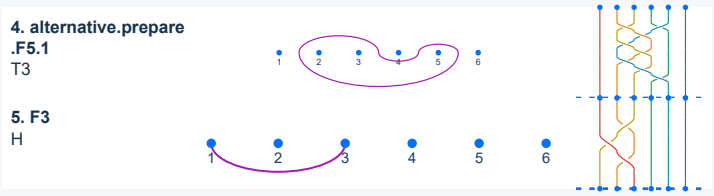
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

3 unchanged displayed factors above; 10 below. Only the changed interval is shown. Retained adjacent checks: 1.

Before



After



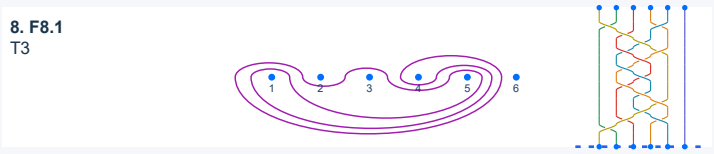
II - step 026

Original 345 daisy supplies actual P35

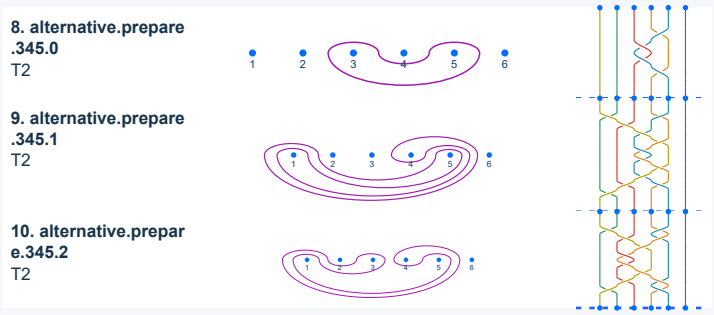
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

7 unchanged displayed factors above; 7 below. Only the changed interval is shown.

Before



After



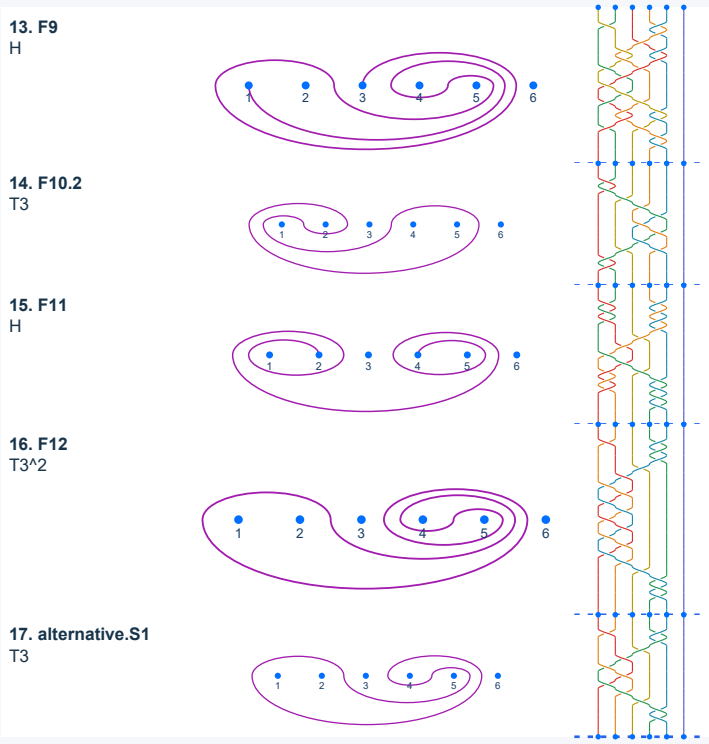
II - step 027

Remaining 124 root up over F9

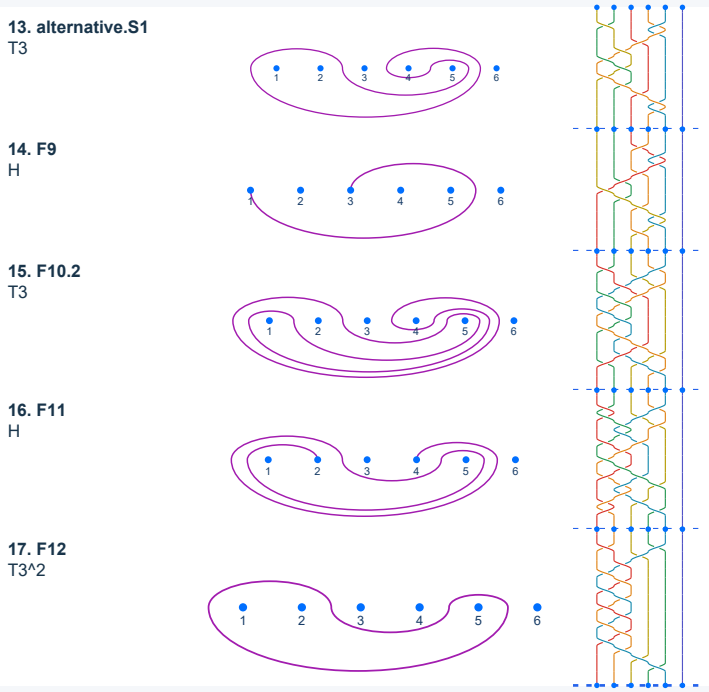
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

12 unchanged displayed factors above; 0 below. Only the changed interval is shown. Retained adjacent checks: 4.

Before



After



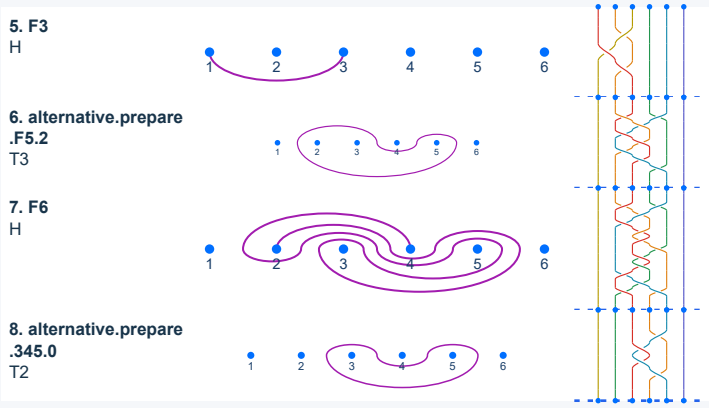
II - step 028

Carry P35 up over F3; a Q14 transport is still needed

DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

4 unchanged displayed factors above; 9 below. Only the changed interval is shown. Retained adjacent checks: 3.

Before



After



II - step 029

PC daisy lends its actual Q14 petal

DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

0 unchanged displayed factors above; 16 below. Only the changed interval is shown.

Before

1. alternative.O1234
T4



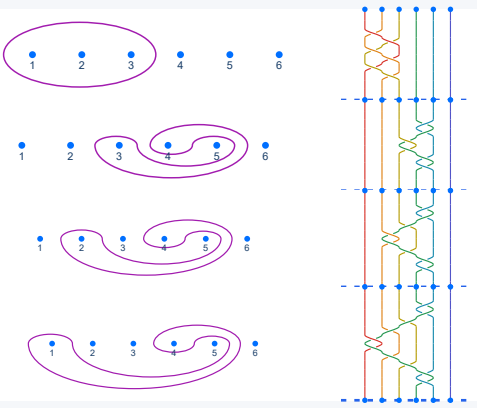
After

1.
alternative.qdaisy.0
T3

2.
alternative.qdaisy.1
T2

3.
alternative.qdaisy.2
T2

4.
alternative.qdaisy.3
T2



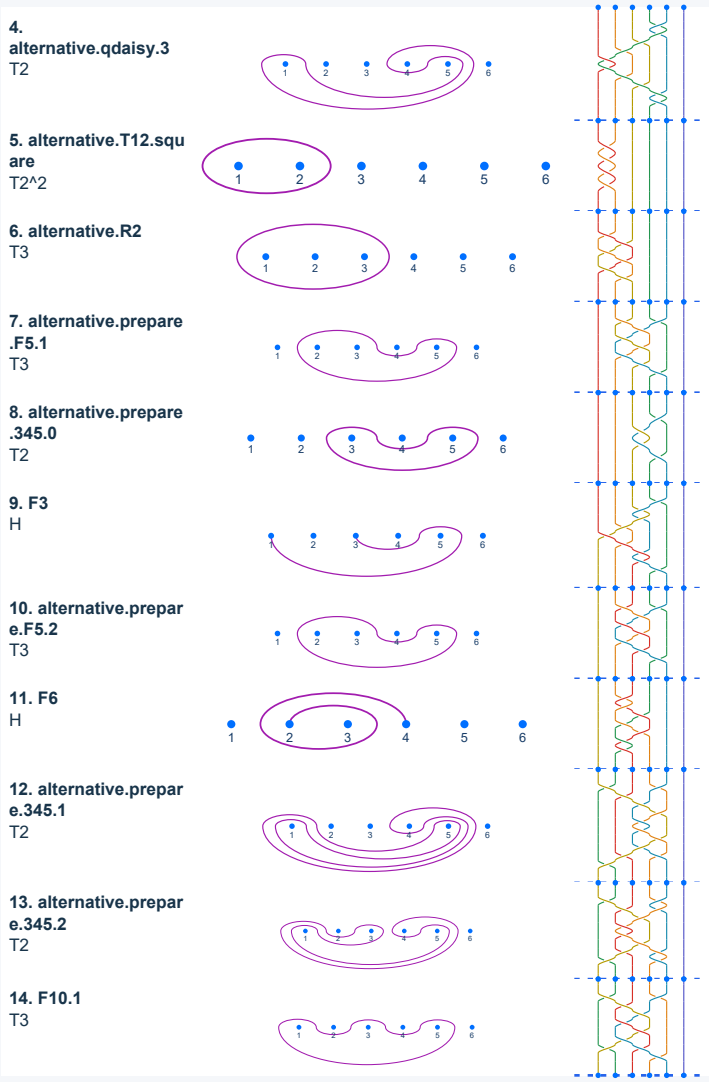
II - step 030

Carry Q14 down across F3 and the remaining 345 petals

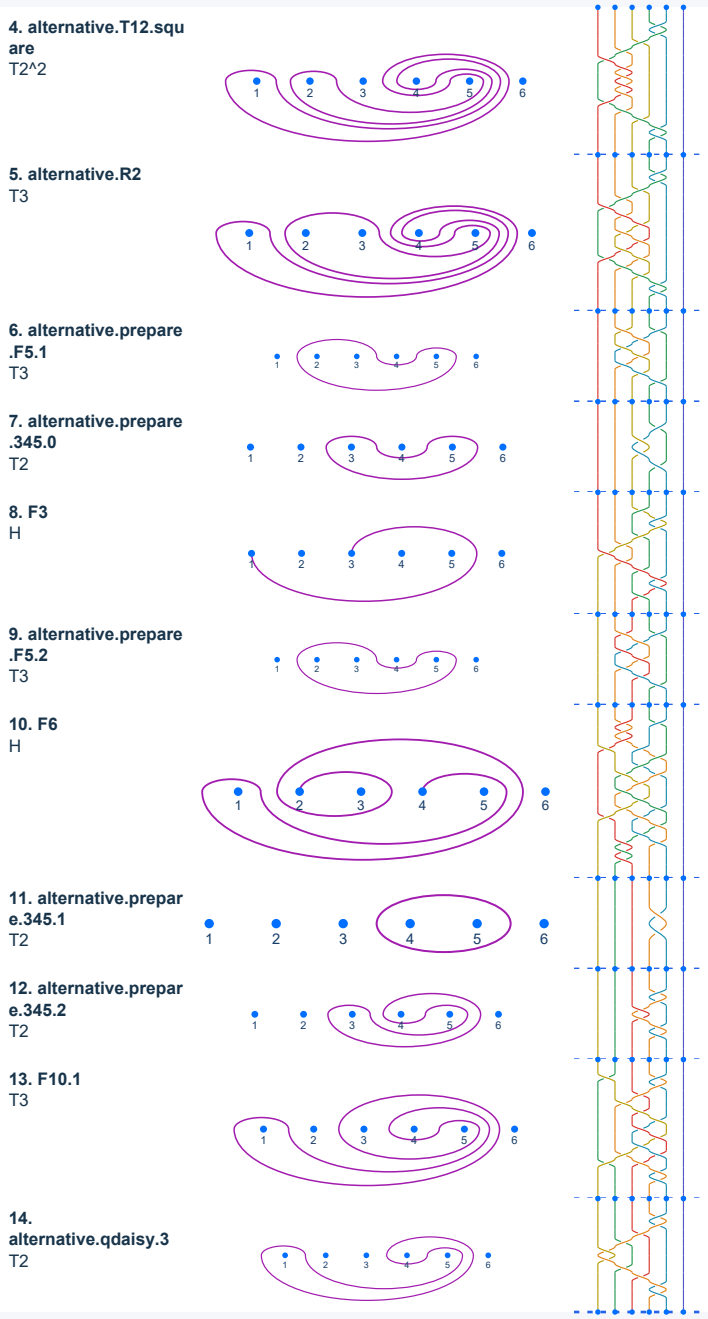
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

3 unchanged displayed factors above; 6 below. Only the changed interval is shown. Retained adjacent checks: 10.

Before



After



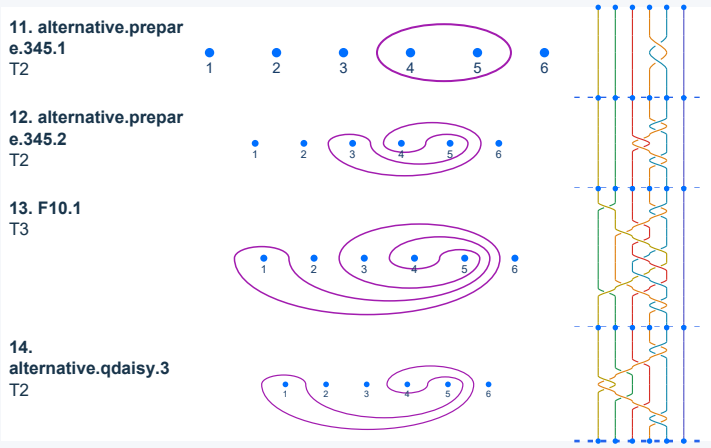
II - step 031

PC daisy contracts the ACTUAL transported petals to O1345

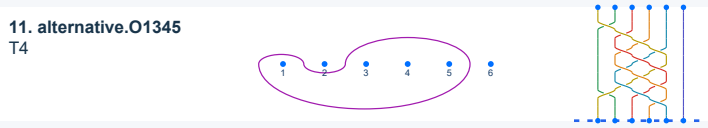
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

10 unchanged displayed factors above; 6 below. Only the changed interval is shown.

Before



After



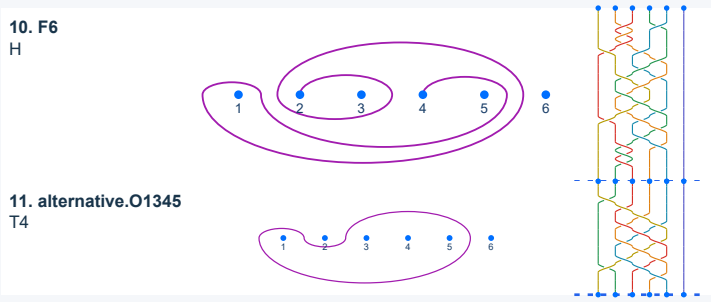
II - step 032

Carry O1345 up across F6 before the matching-half gather

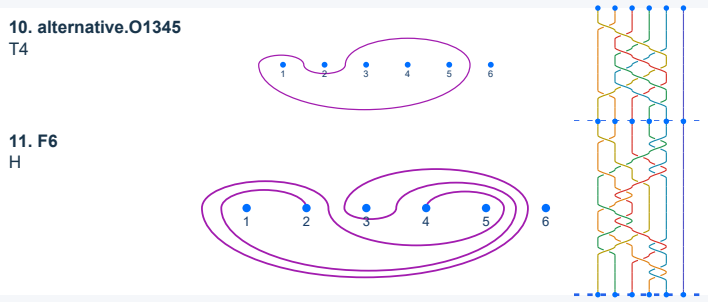
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

9 unchanged displayed factors above; 6 below. Only the changed interval is shown. Retained adjacent checks: 1.

Before



After



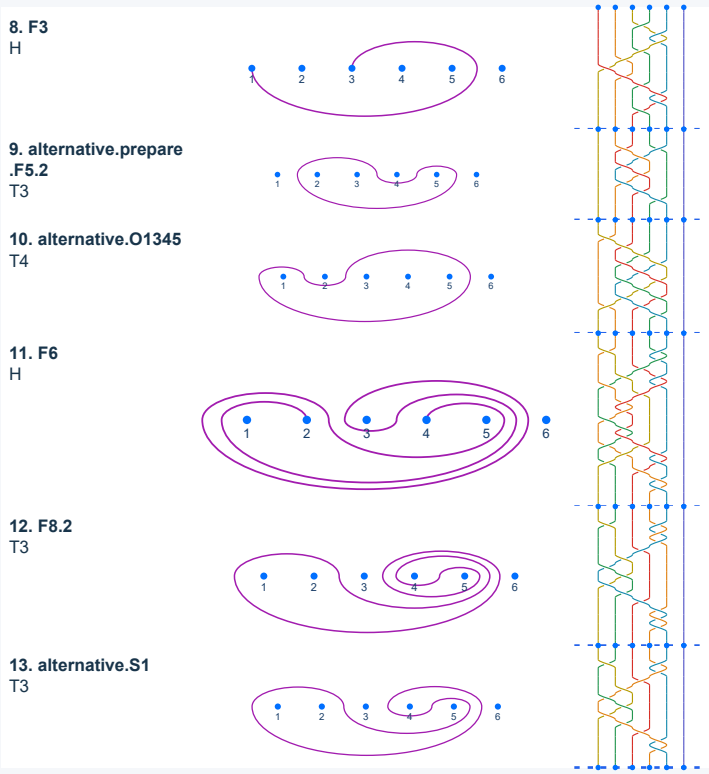
II - step 033

Carry the matched F3 down to F9; F6 unwinds

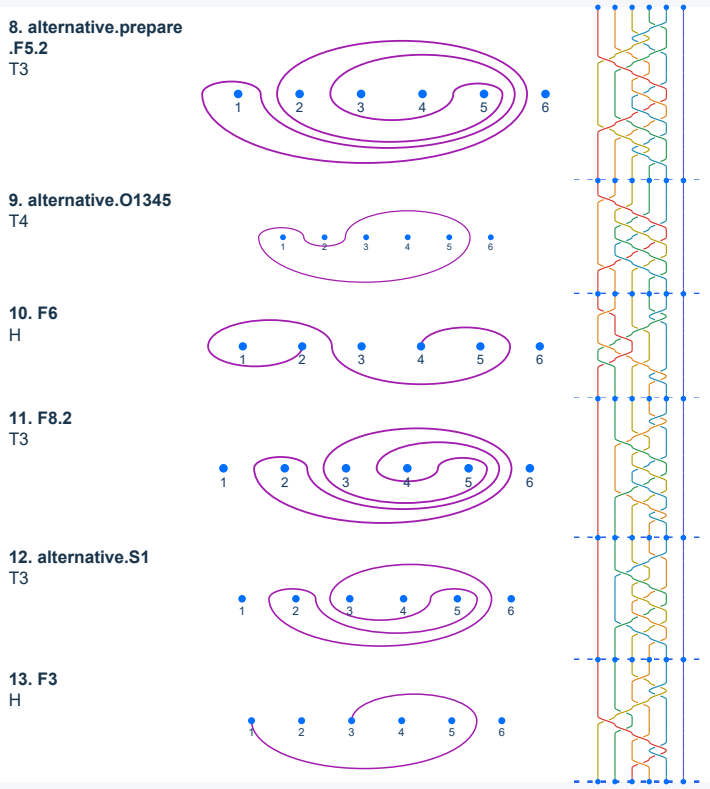
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

7 unchanged displayed factors above; 4 below. Only the changed interval is shown. Retained adjacent checks: 5.

Before



After



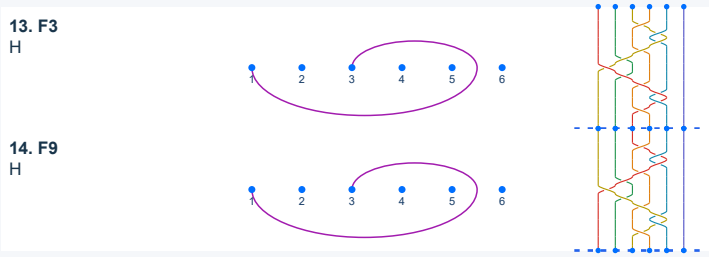
II - step 034

Combine the matching F3 and F9

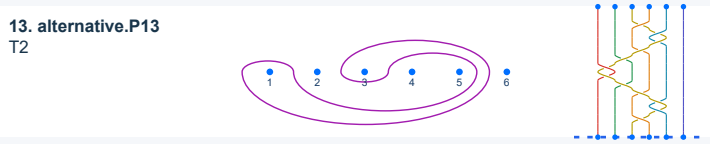
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

12 unchanged displayed factors above; 3 below. Only the changed interval is shown.

Before



After



II - step 035

The O1345 daisy returns EXACTLY the borrowed Q14

DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

8 unchanged displayed factors above; 7 below. Only the changed interval is shown.

Before

9. alternative.O1345
T4



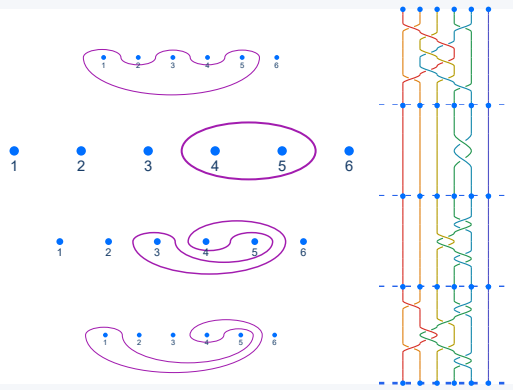
After

9. alternative.return.0
T3

10. alternative.return.1
T2

11. alternative.return.2
T2

12. alternative.return.3
T2



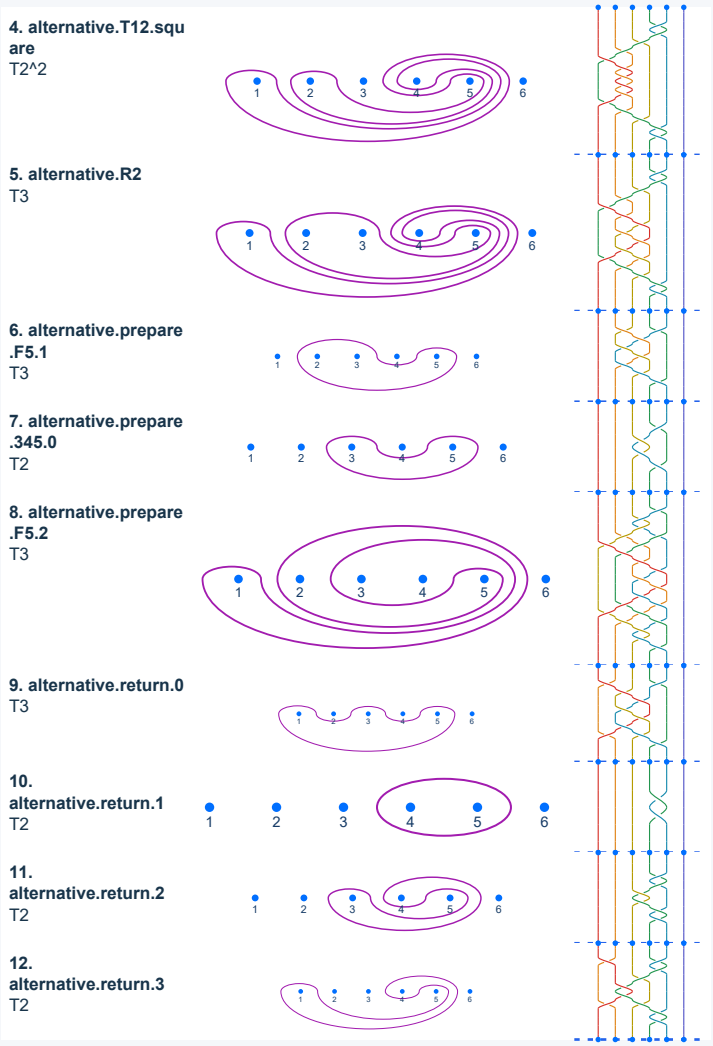
II - step 036

Return Q14 up to restore the round T12 square and T123

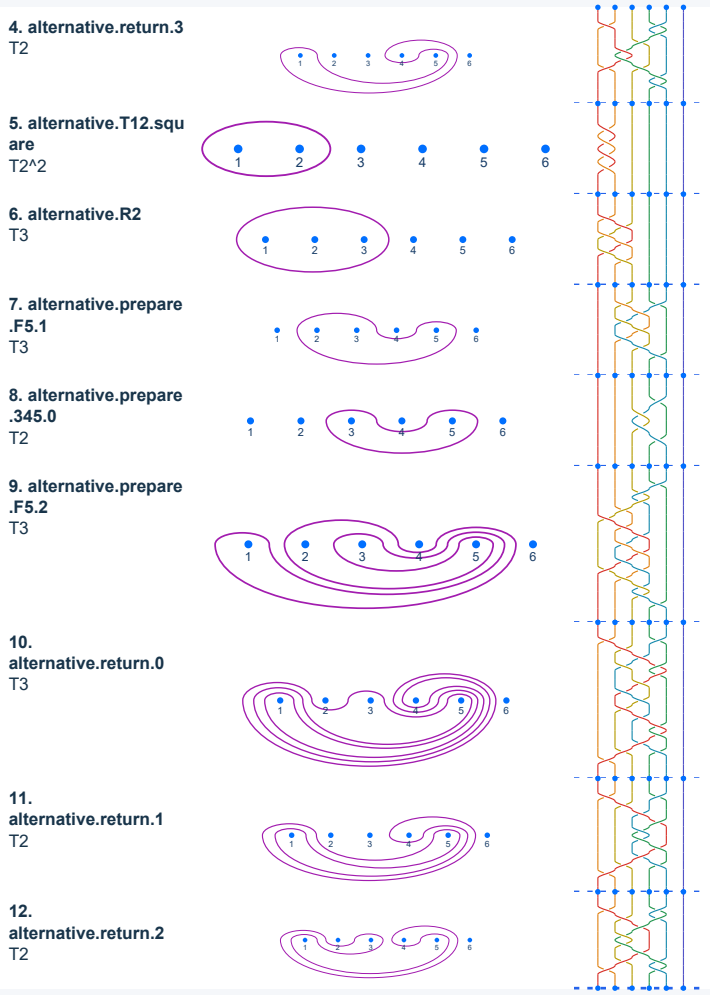
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

3 unchanged displayed factors above; 7 below. Only the changed interval is shown. Retained adjacent checks: 8.

Before



After



II - step 037

Close the original daisy after its actual Q14 returns

DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

0 unchanged displayed factors above; 15 below. Only the changed interval is shown.

Before



After



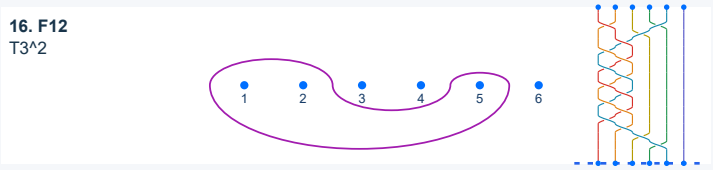
II - step 038

Split F12 into its two positive units

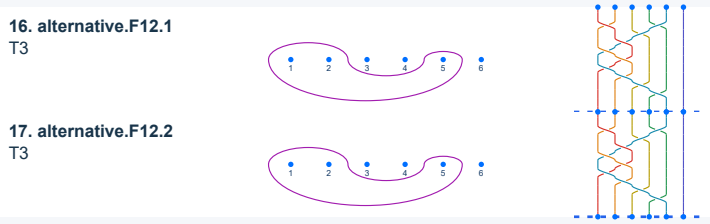
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

15 unchanged displayed factors above; 0 below. Only the changed interval is shown.

Before



After



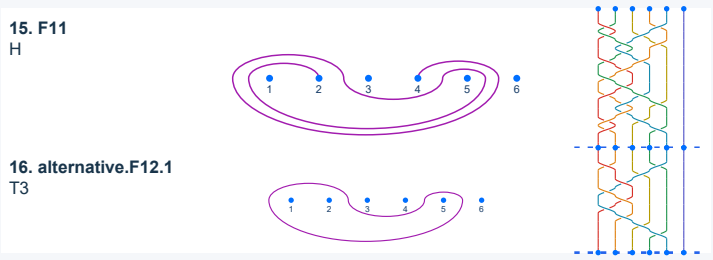
II - step 039

One C12 root shortens F11

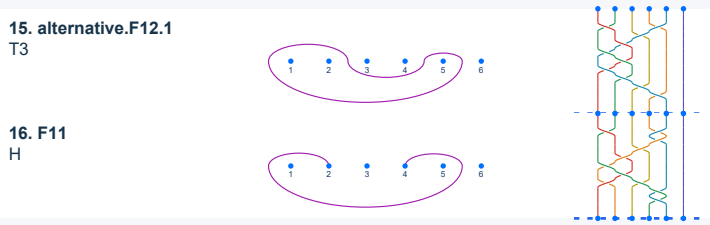
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

14 unchanged displayed factors above; 1 below. Only the changed interval is shown. Retained adjacent checks: 1.

Before



After



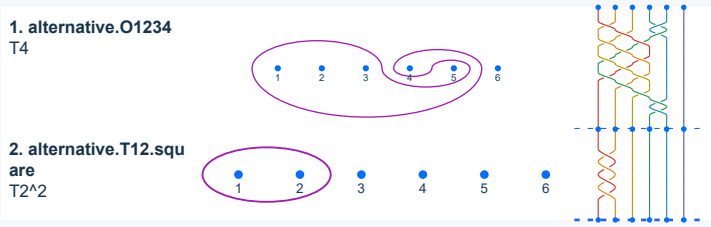
II - step 040

Commute the recovered round square to the top

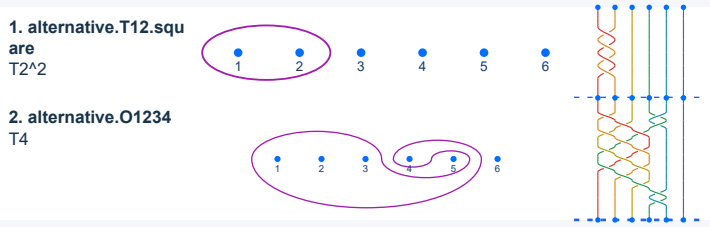
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

0 unchanged displayed factors above; 15 below. Only the changed interval is shown. Retained adjacent checks: 1.

Before



After



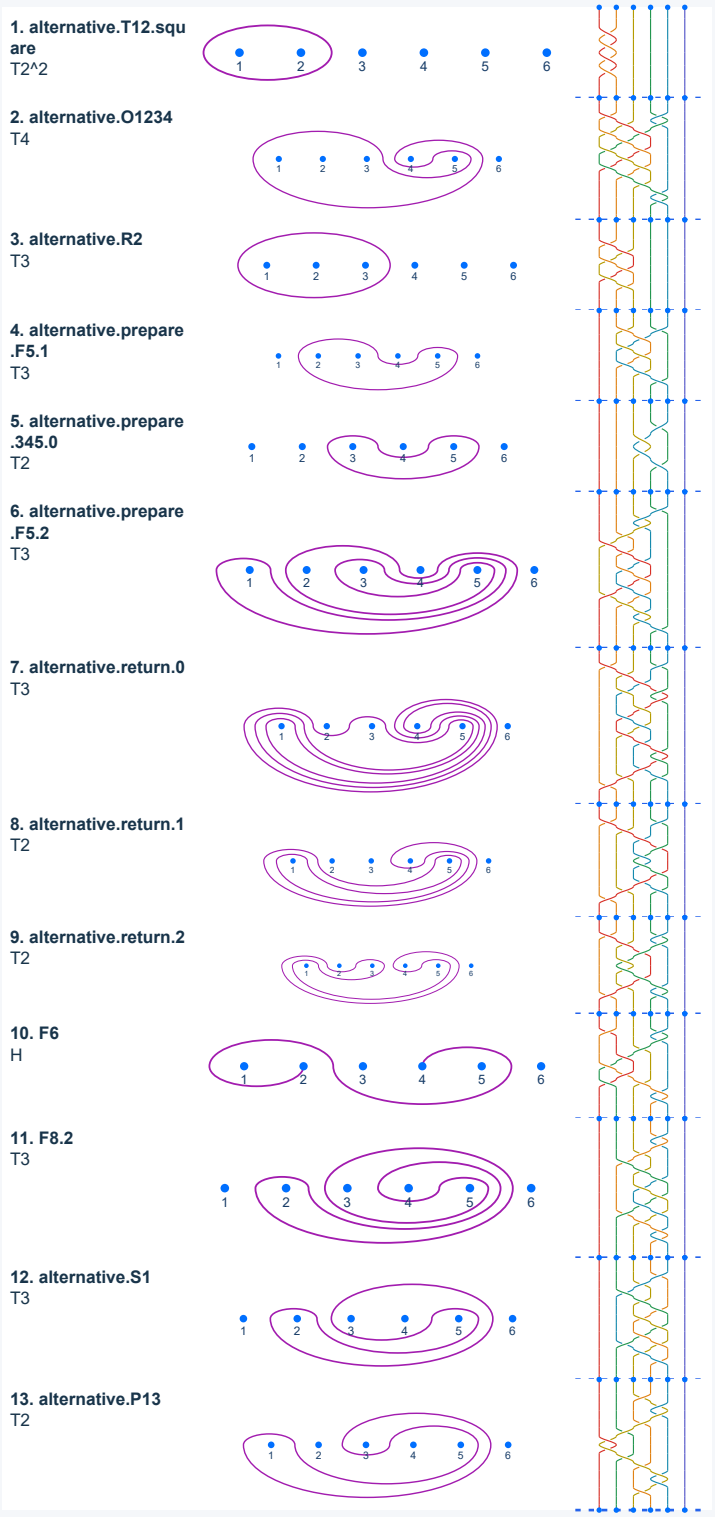
II - step 041

Choose the cyclic start just after the round square

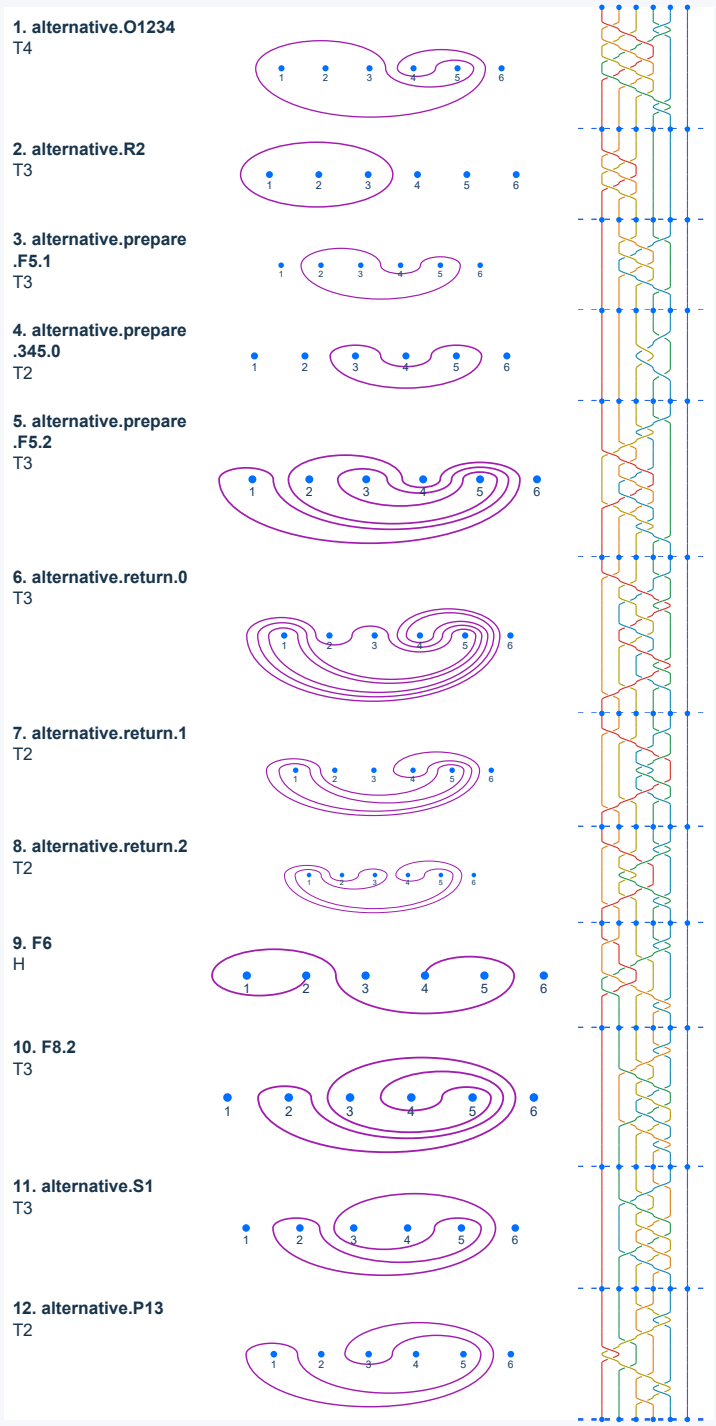
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

0 unchanged displayed factors above; 0 below. Only the changed interval is shown. Explicit common conjugation; its full braid frame is saved.

Before



After



II - step 041 - continued

Choose the cyclic start just after the round square

DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

0 unchanged displayed factors above; 0 below. Only the changed interval is shown. Explicit common conjugation; its full braid frame is saved.

Before



After



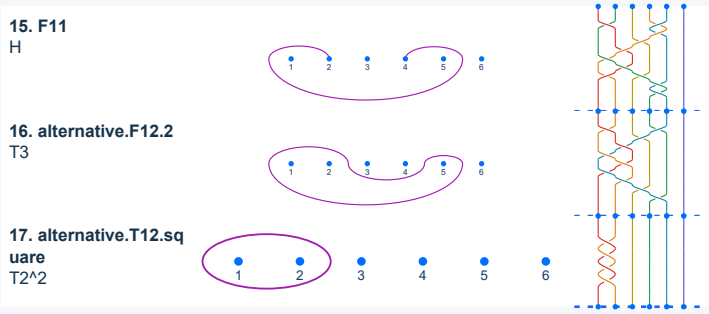
II - step 042

Carry the square up the short side; F11 matches F6

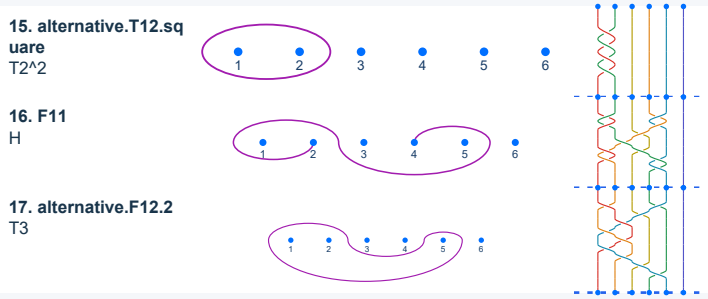
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

14 unchanged displayed factors above; 0 below. Only the changed interval is shown. Retained adjacent checks: 2.

Before



After



II - step 043

Common disk frame makes the first four-point curve round

EXPLICIT GLOBAL CONJUGATION: a change of the whole product by the recorded frame.

0 unchanged displayed factors above; 0 below. Only the changed interval is shown.

Before

1. alternative.O1234
T4



2. alternative.R2
T3



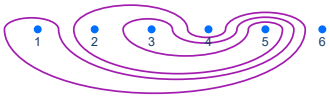
3. alternative.prepare .F5.1
T3



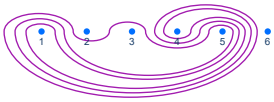
4. alternative.prepare .345.0
T2



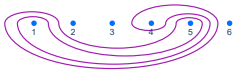
5. alternative.prepare .F5.2
T3



6. alternative.return.0
T3



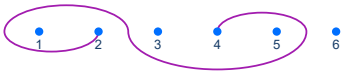
7. alternative.return.1
T2



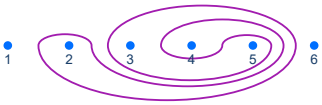
8. alternative.return.2
T2



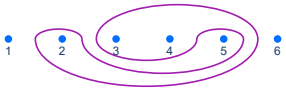
9. F6
H



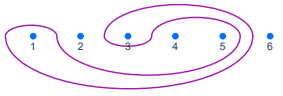
10. F8.2
T3

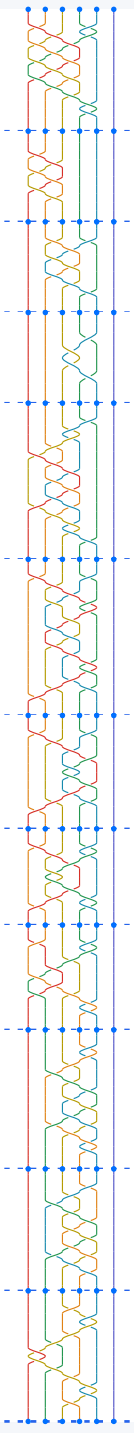


11. alternative.S1
T3



12. alternative.P13
T2





After

1. alternative.O1234
T4



2. alternative.R2
T3



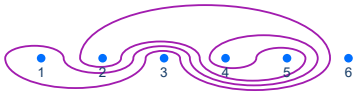
3. alternative.prepare .F5.1
T3



4. alternative.prepare .345.0
T2



5. alternative.prepare .F5.2
T3



6. alternative.return.0
T3



7. alternative.return.1
T2



8. alternative.return.2
T2



9. F6
H



10. F8.2
T3



11. alternative.S1
T3



12. alternative.P13
T2

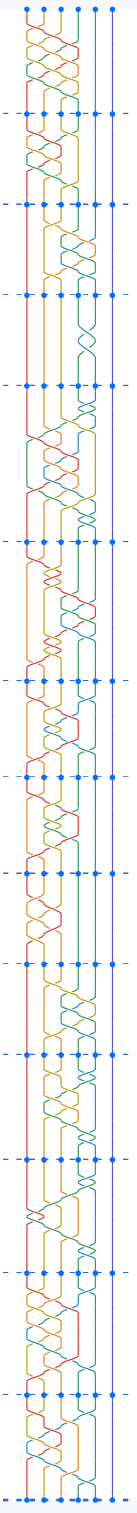


13. F10.2
T3



14. alternative.F12.1
T3





II - step 043 - continued

Common disk frame makes the first four-point curve round

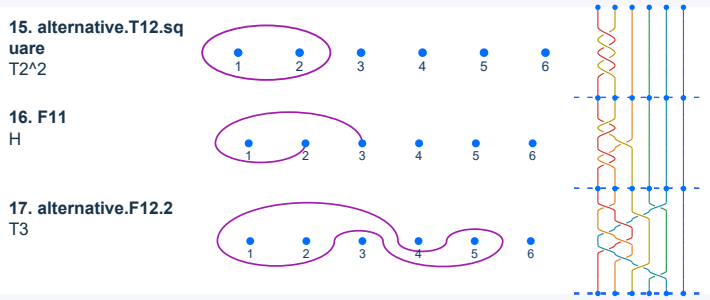
EXPLICIT GLOBAL CONJUGATION: a change of the whole product by the recorded frame.

0 unchanged displayed factors above; 0 below. Only the changed interval is shown.

Before



After



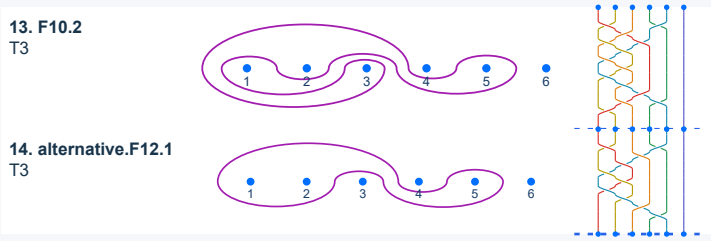
II - step 044

Reorder C12 before the final matching-half gather

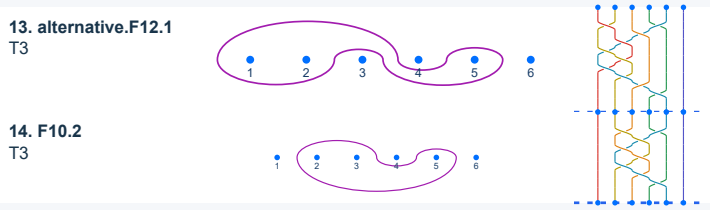
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

12 unchanged displayed factors above; 3 below. Only the changed interval is shown. Retained adjacent checks: 1.

Before



After



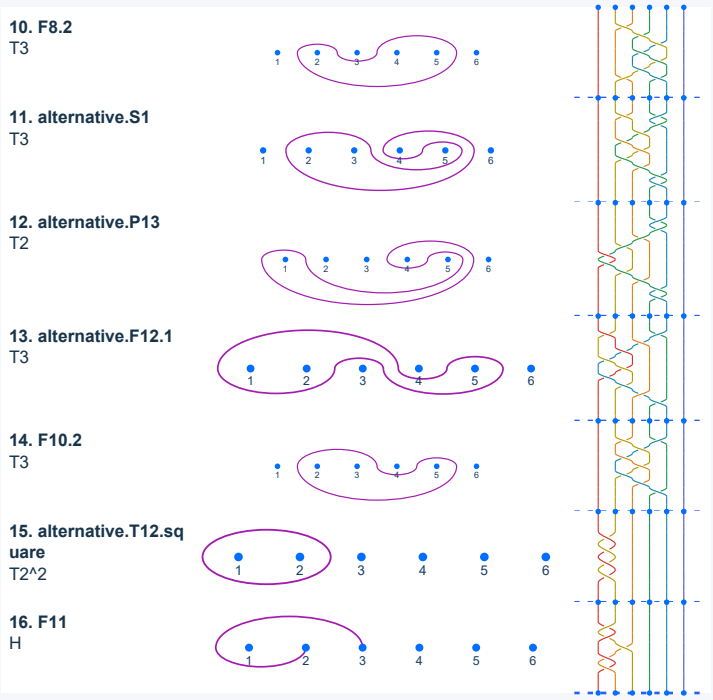
II - step 045

Gather the final matching original halves

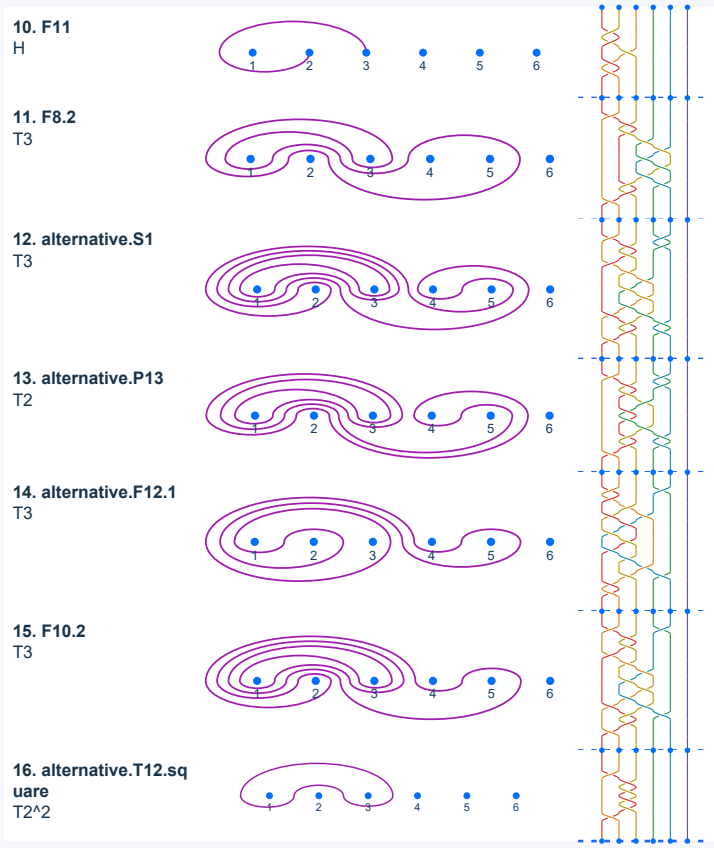
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

9 unchanged displayed factors above; 1 below. Only the changed interval is shown. Retained adjacent checks: 6.

Before



After



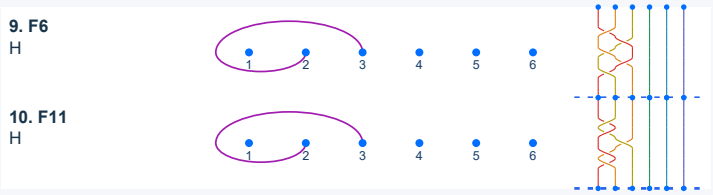
II - step 046

Combine the matching F6 and F11

DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

8 unchanged displayed factors above; 7 below. Only the changed interval is shown.

Before



After



III. The first exterior and simpler inside daisy

Actual ordered factor supports and their connected braids. All unchanged context is counted, and every retained move is shown.

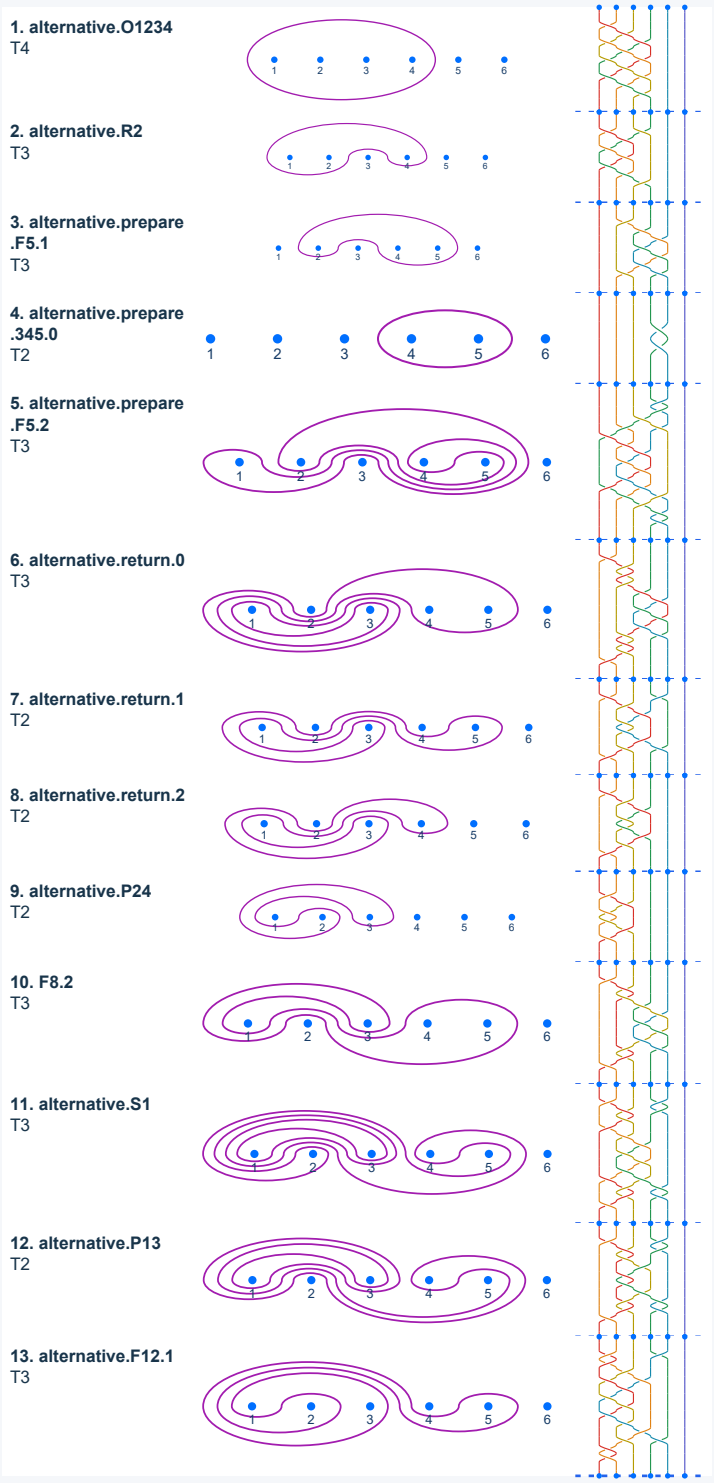
III - step 047

Common disk frame for the next actual Q15 supplier

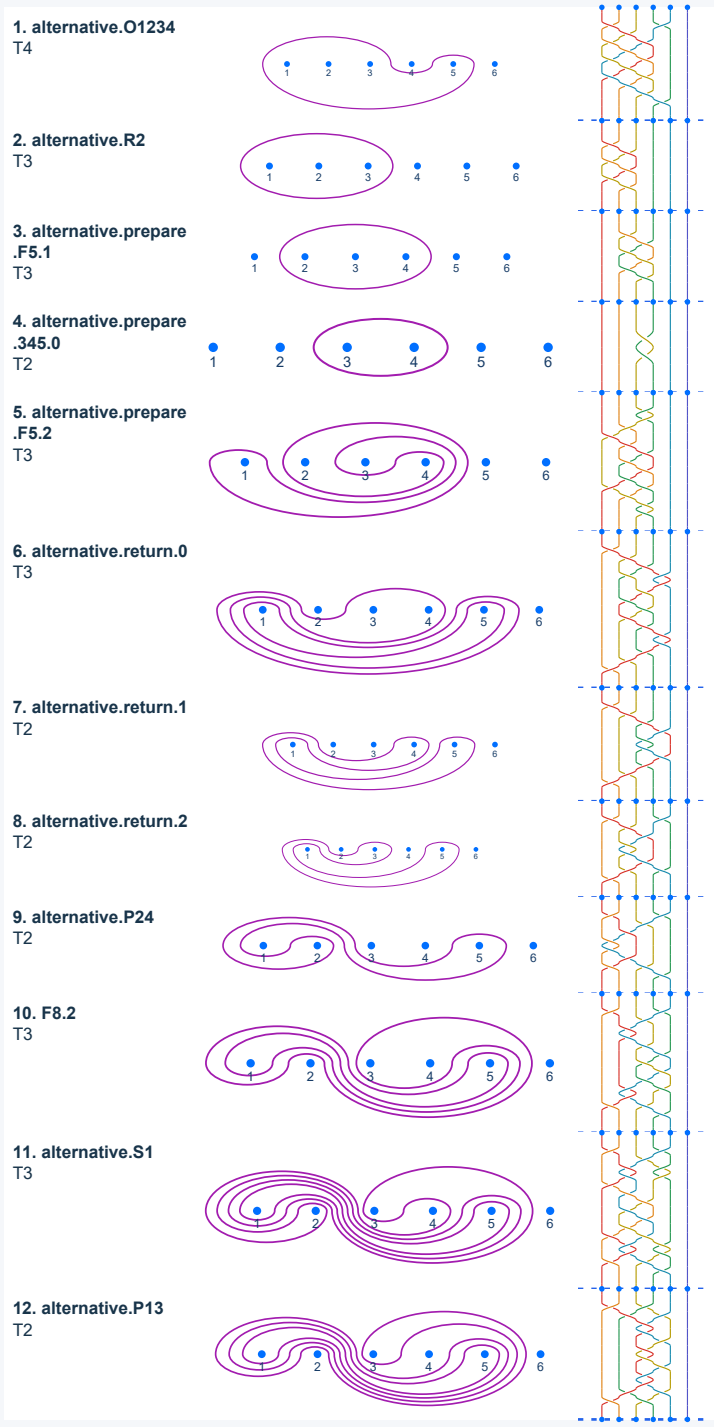
EXPLICIT GLOBAL CONJUGATION: a change of the whole product by the recorded frame.

0 unchanged displayed factors above; 0 below. Only the changed interval is shown.

Before



After



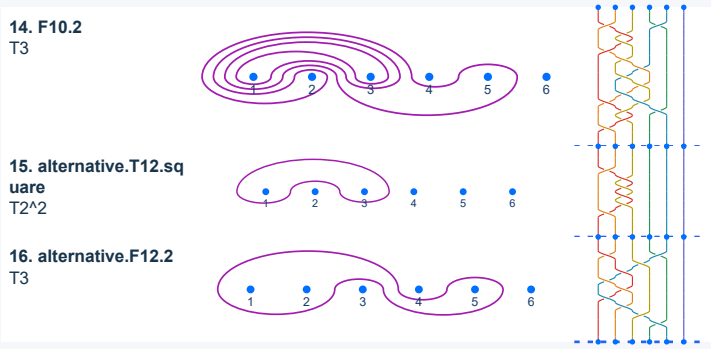
III - step 047 - continued

Common disk frame for the next actual Q15 supplier

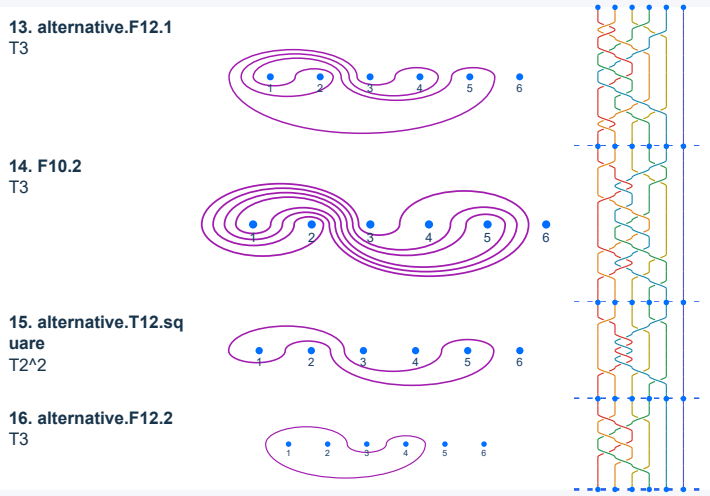
EXPLICIT GLOBAL CONJUGATION: a change of the whole product by the recorded frame.

0 unchanged displayed factors above; 0 below. Only the changed interval is shown.

Before



After



III - step 048

C12 daisy supplies the actual Q15

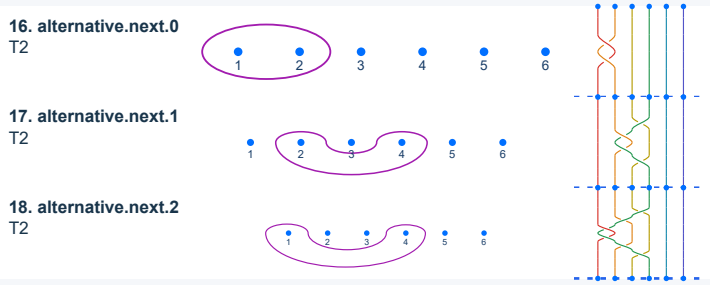
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

15 unchanged displayed factors above; 0 below. Only the changed interval is shown.

Before



After



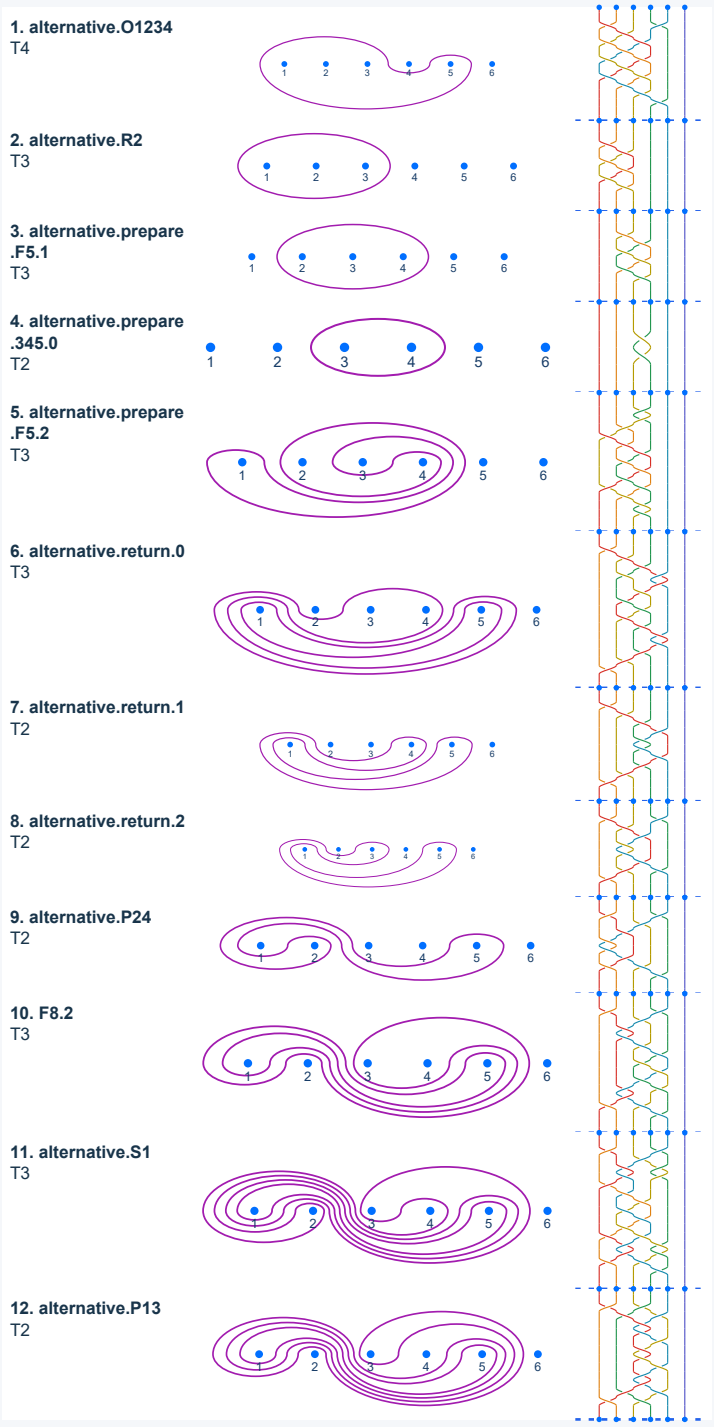
III - step 049

Choose the cyclic start at the actual Q15

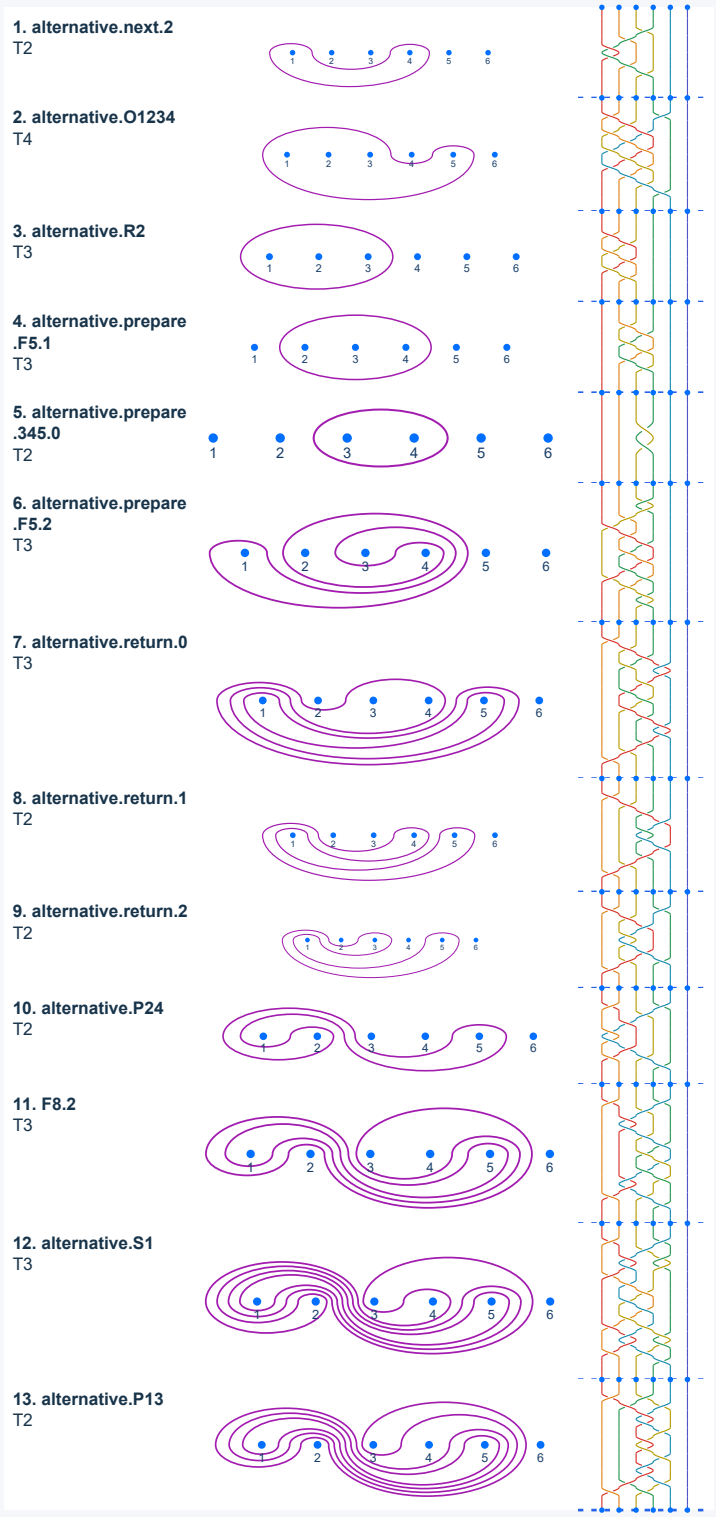
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

0 unchanged displayed factors above; 0 below. Only the changed interval is shown. Explicit common conjugation; its full braid frame is saved.

Before



After



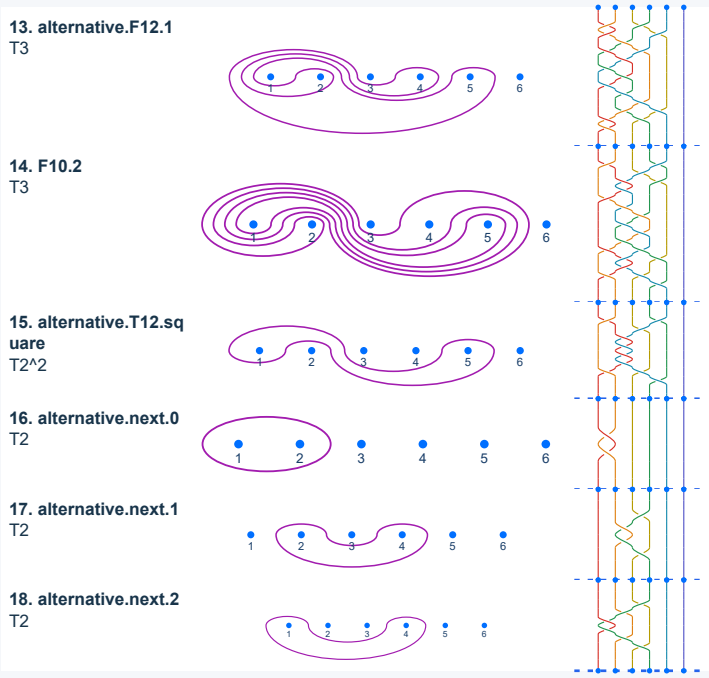
III - step 049 - continued

Choose the cyclic start at the actual Q15

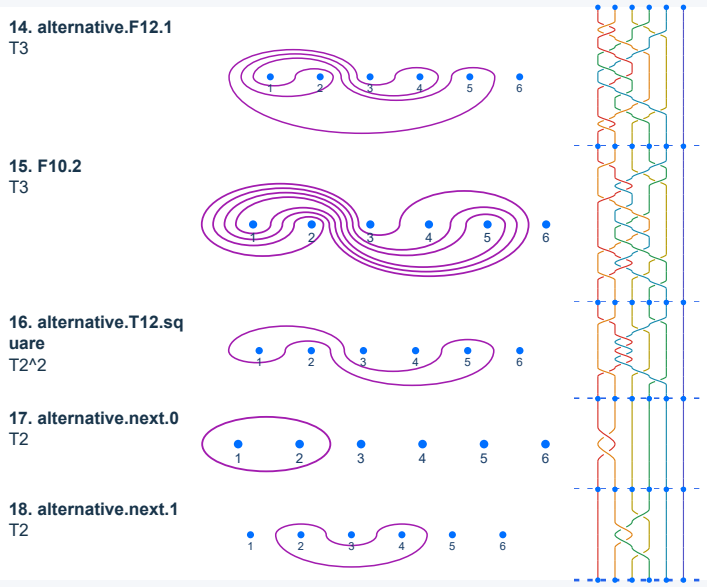
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

0 unchanged displayed factors above; 0 below. Only the changed interval is shown. Explicit common conjugation; its full braid frame is saved.

Before



After



III - step 050

Carry Q15 through the combined four-point prefix and the two roots

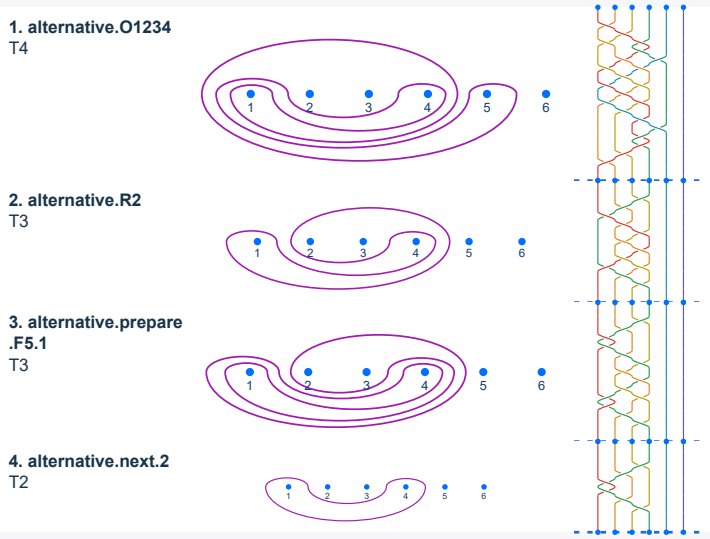
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

0 unchanged displayed factors above; 14 below. Only the changed interval is shown. Retained adjacent checks: 3.

Before



After



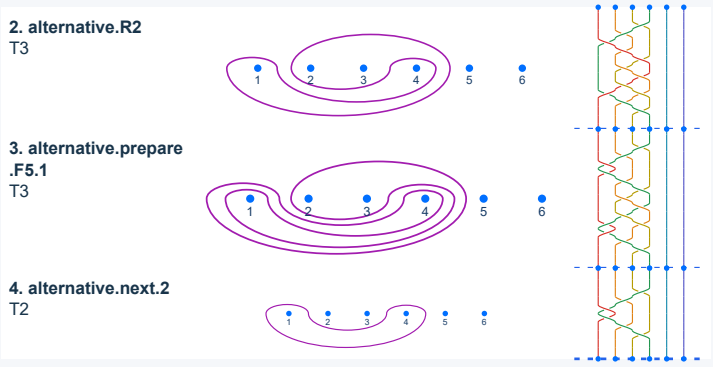
III - step 051

PC cluster lantern gives the second four-point twist

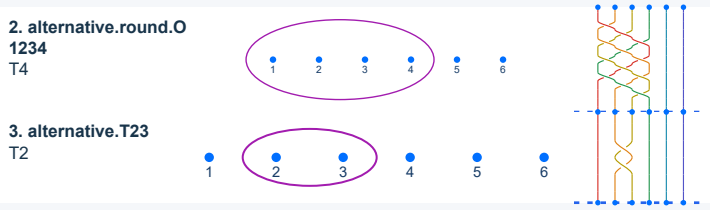
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

1 unchanged displayed factors above; 14 below. Only the changed interval is shown.

Before



After



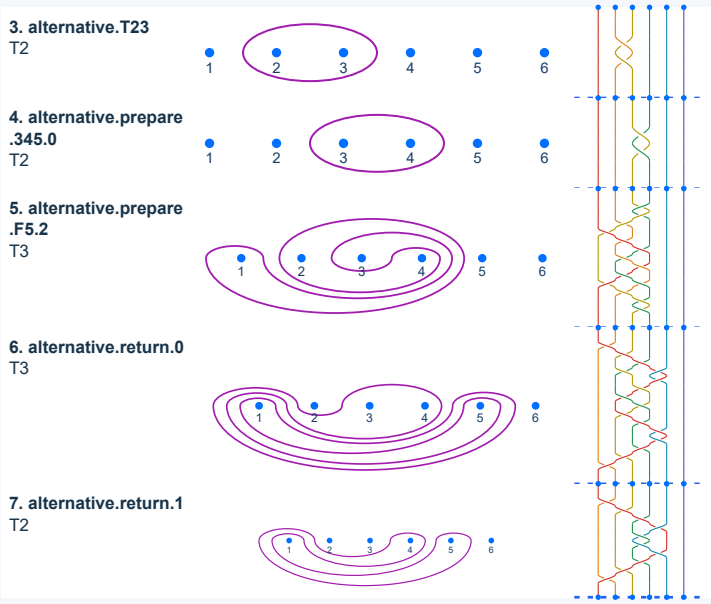
III - step 052

Gather the ACTUAL missing 45 petal by the short side

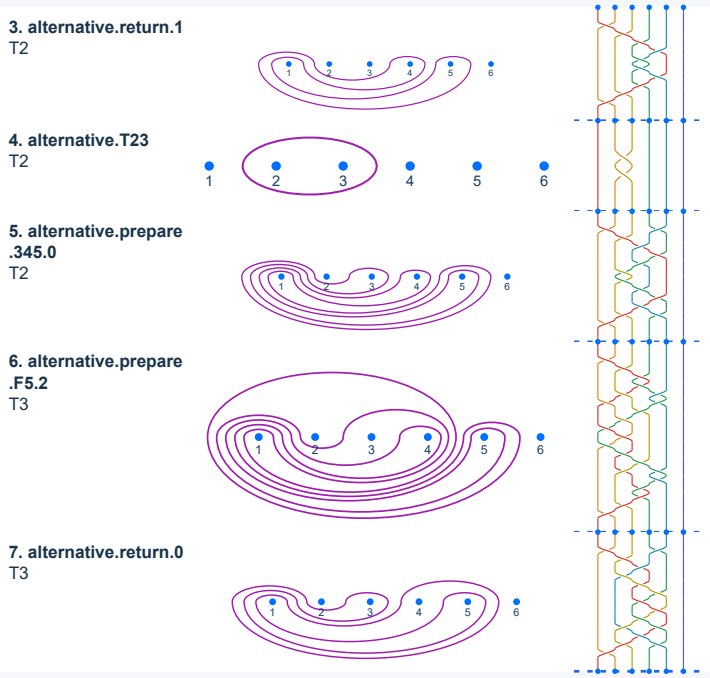
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

2 unchanged displayed factors above; 10 below. Only the changed interval is shown. Retained adjacent checks: 4.

Before



After



III - step 053

Cabled PC lantern on the ACTUAL two four-point curves and supplied 45 petal

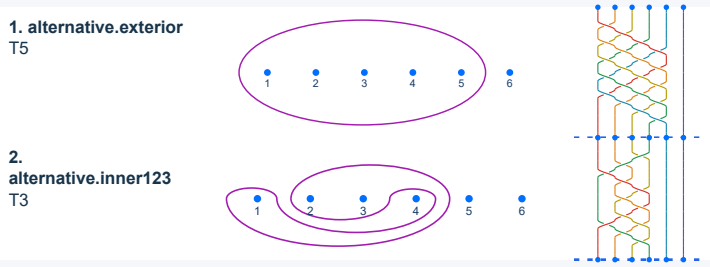
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

0 unchanged displayed factors above; 14 below. Only the changed interval is shown.

Before



After



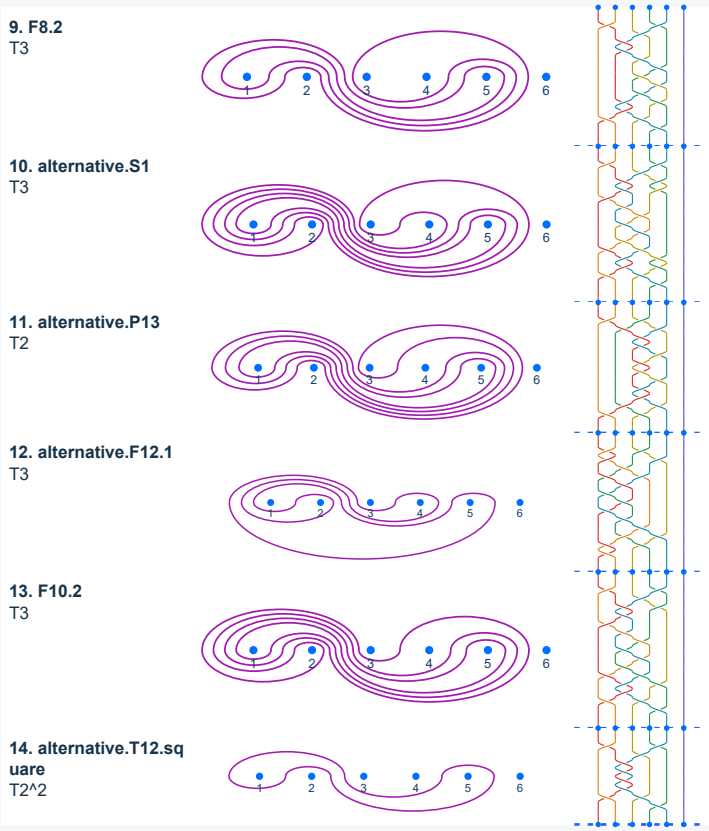
III - step 054

Unwind five actual residual curves with the fixed positive square

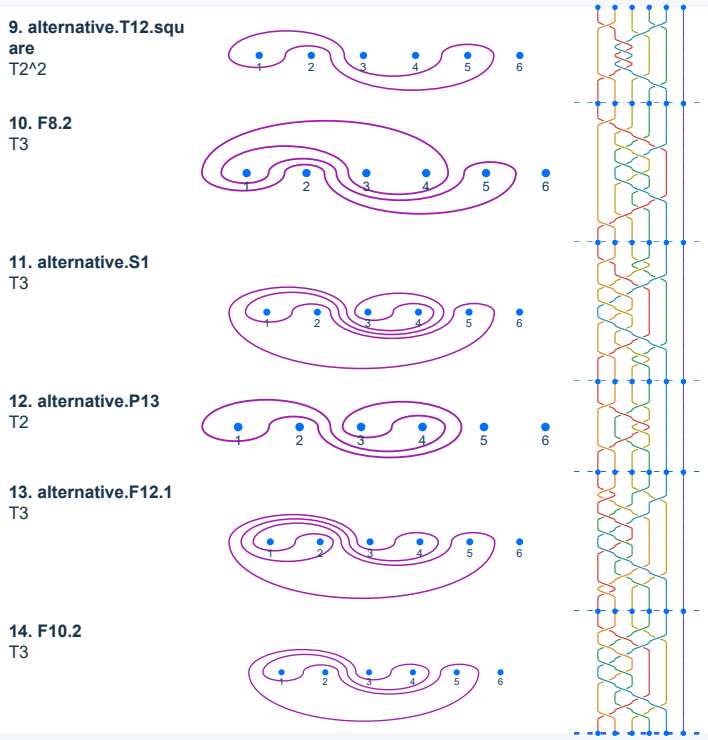
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

8 unchanged displayed factors above; 2 below. Only the changed interval is shown. Retained adjacent checks: 5.

Before



After



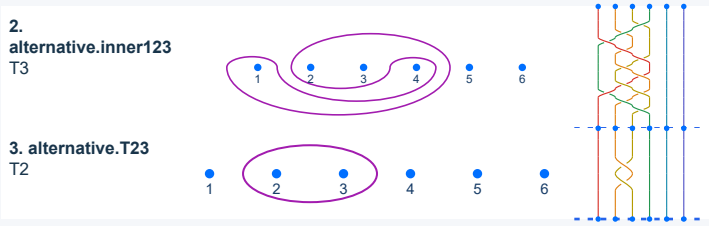
III - step 055

Commute the 123 root past T23 to its actual 34 and 124 companions

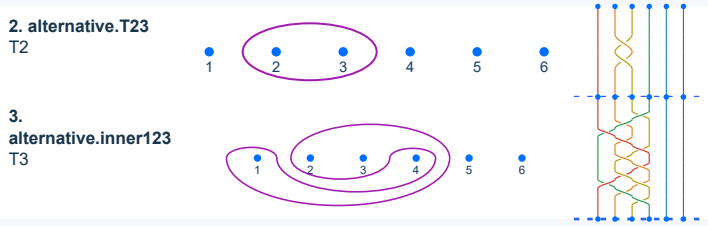
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

1 unchanged displayed factors above; 13 below. Only the changed interval is shown. Retained adjacent checks: 1.

Before



After



III - step 056

Cabled PC lantern contracts the ACTUAL 123, 34 and 124 curves

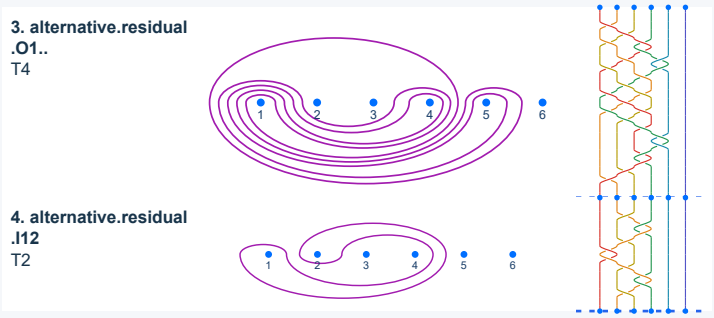
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

2 unchanged displayed factors above; 11 below. Only the changed interval is shown.

Before



After



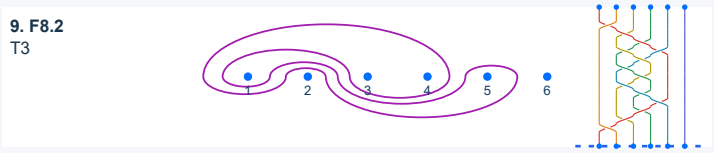
III - step 057

The actual 345 daisy lends its round 34 petal

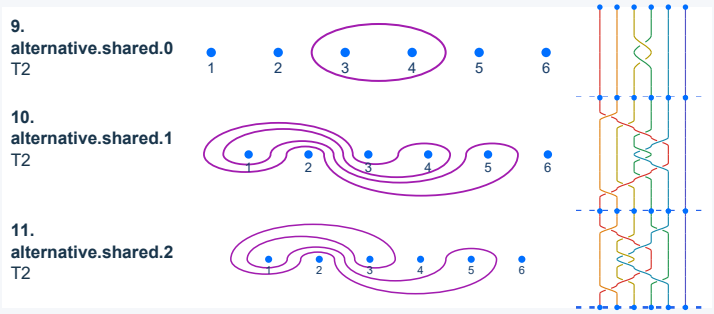
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

8 unchanged displayed factors above; 6 below. Only the changed interval is shown.

Before



After



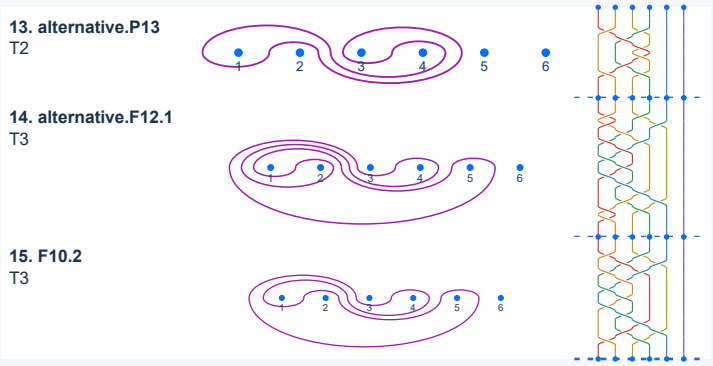
III - step 058

Carry the 245 root up across the 145 root and the 13 pair

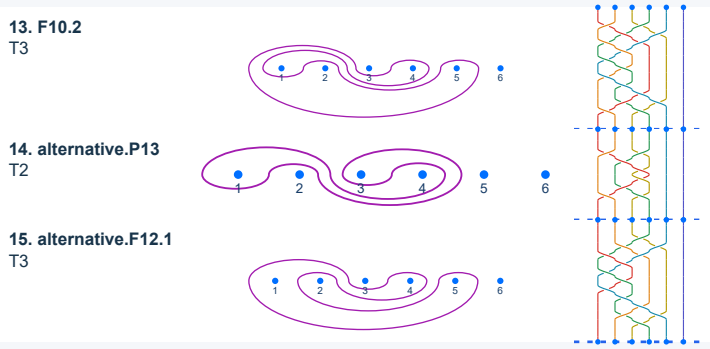
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

12 unchanged displayed factors above; 2 below. Only the changed interval is shown. Retained adjacent checks: 2.

Before



After



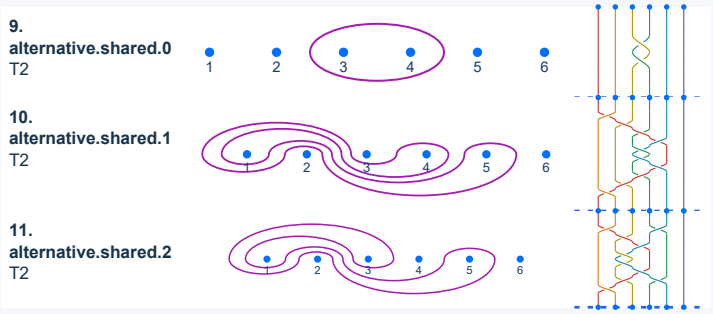
III - step 059

Carry round 34 down through its own two remaining petals

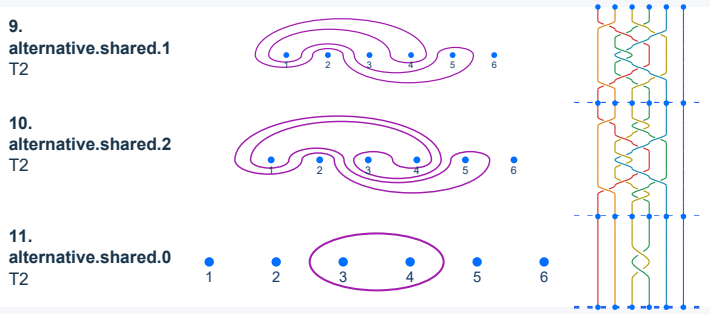
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

8 unchanged displayed factors above; 6 below. Only the changed interval is shown. Retained adjacent checks: 2.

Before



After



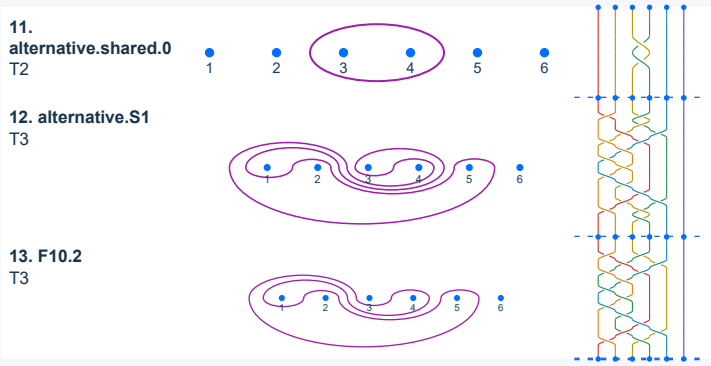
III - step 060

Cabled PC lantern closes the actual 34, 235 and 245 cluster

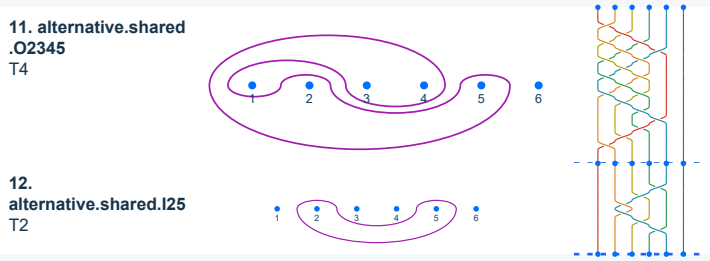
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

10 unchanged displayed factors above; 4 below. Only the changed interval is shown.

Before



After



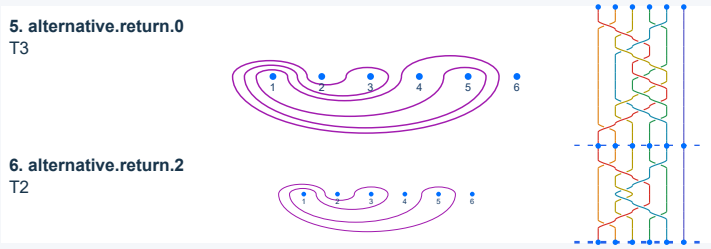
III - step 061

The actual 35 pair unwinds the 134 root to a short chart

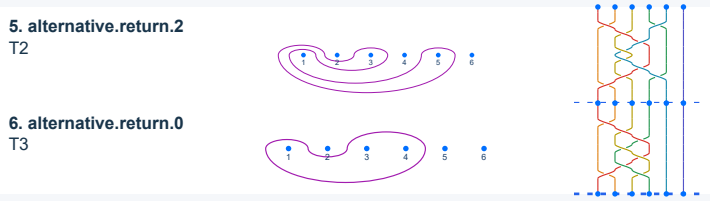
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

4 unchanged displayed factors above; 10 below. Only the changed interval is shown. Retained adjacent checks: 1.

Before



After



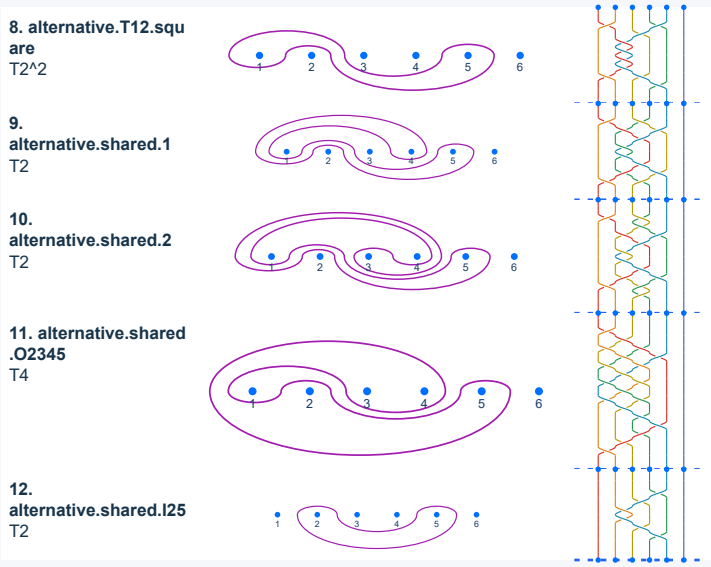
III - step 062

Carry the shared 25 pair up; unwind the square and its two petals

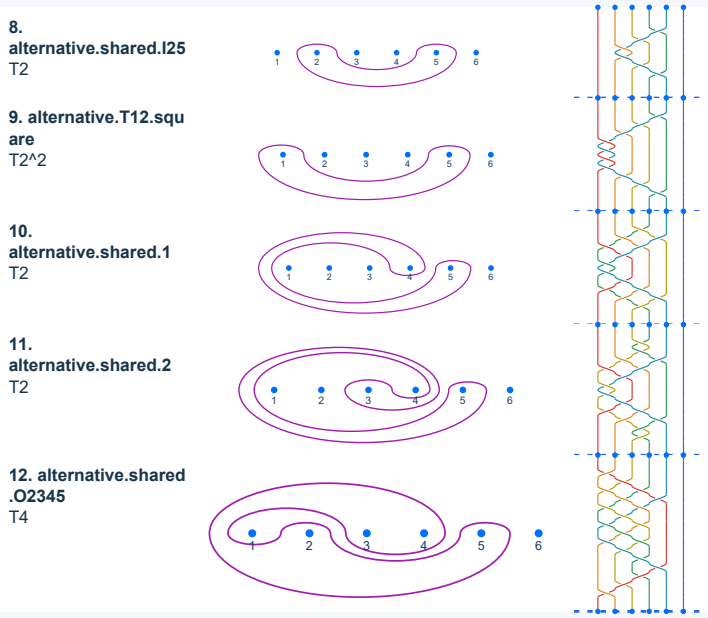
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

7 unchanged displayed factors above; 4 below. Only the changed interval is shown. Retained adjacent checks: 4.

Before



After



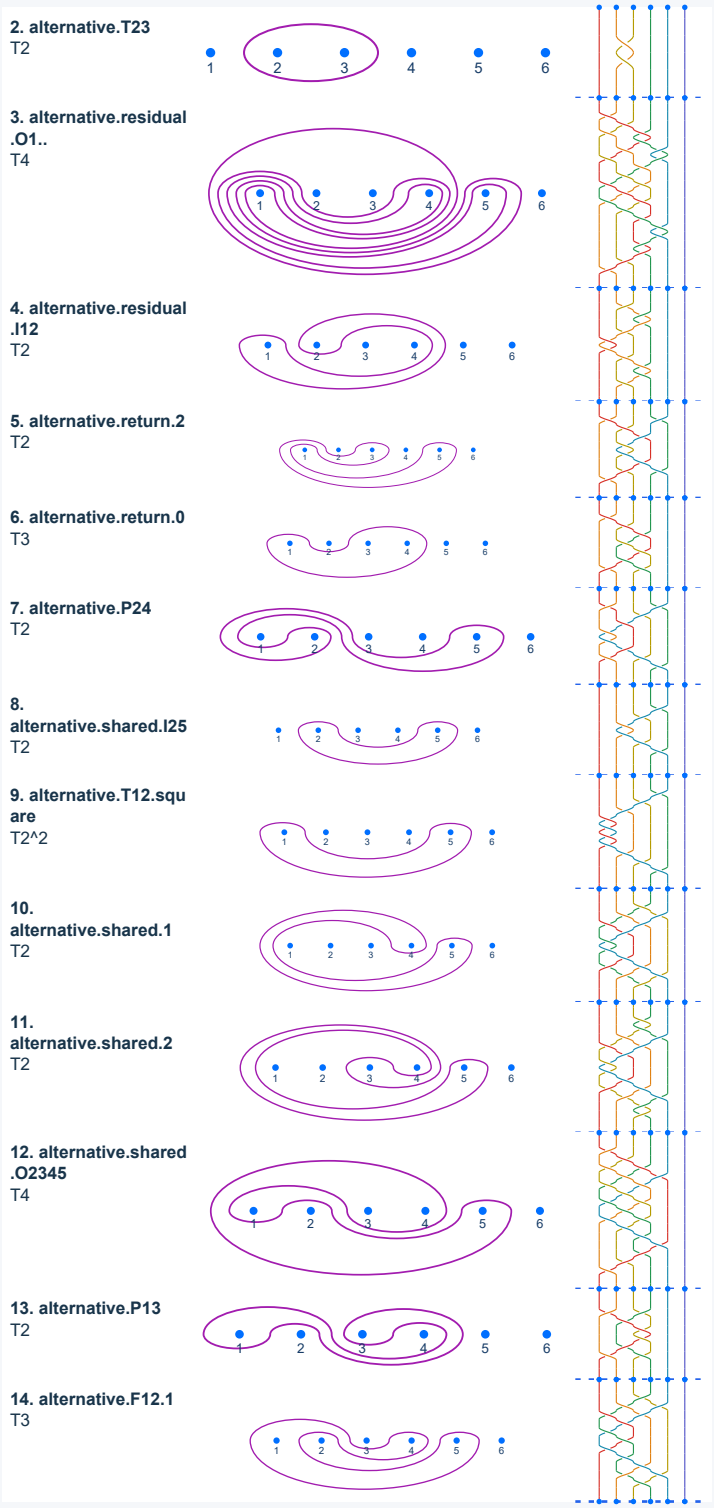
III - step 063

Common disk frame keeps the inside-four daisy readable

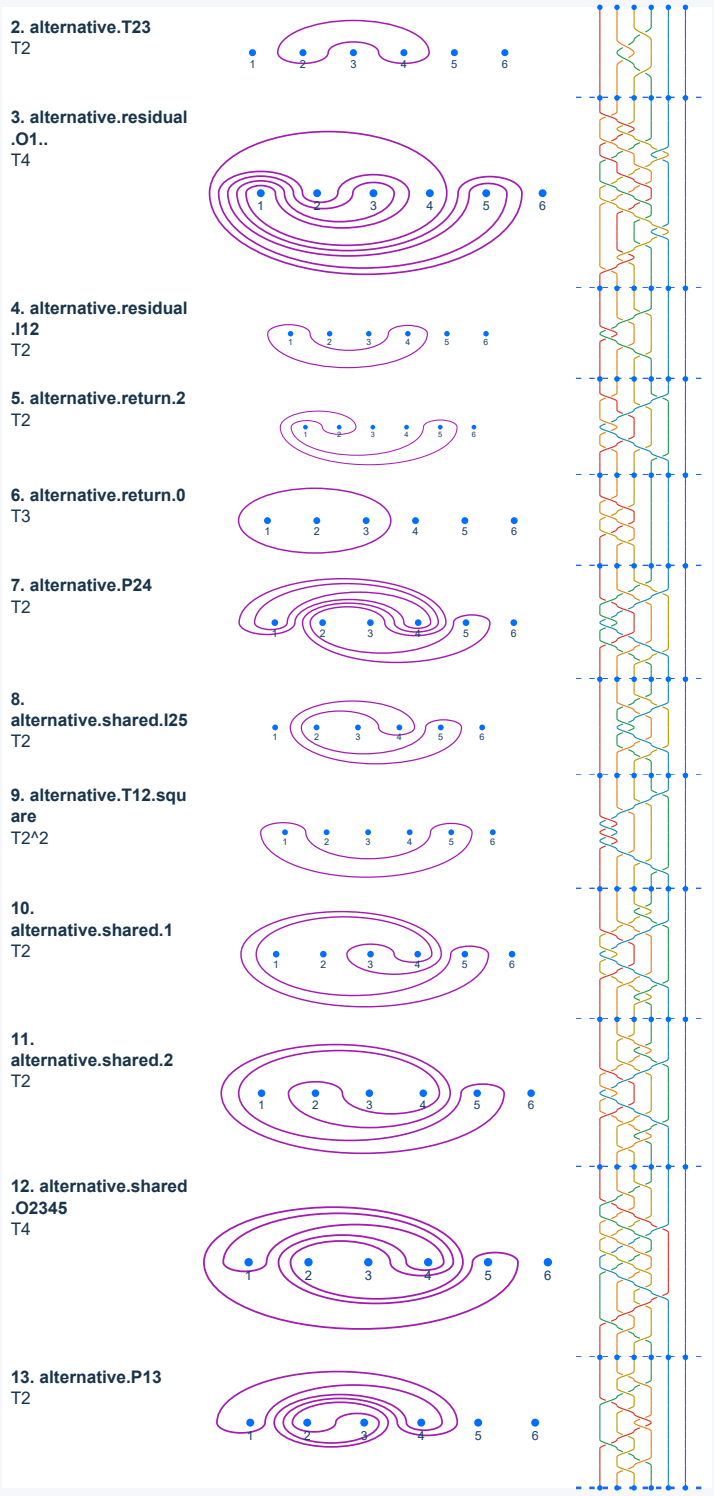
EXPLICIT GLOBAL CONJUGATION: a change of the whole product by the recorded frame.

1 unchanged displayed factors above; 0 below. Only the changed interval is shown.

Before



After



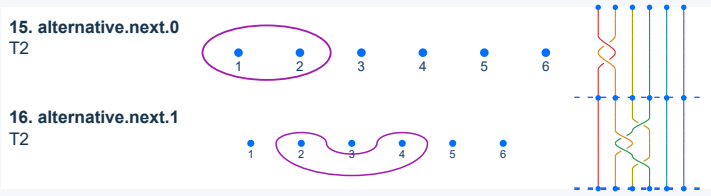
III - step 063 - continued

Common disk frame keeps the inside-four daisy readable

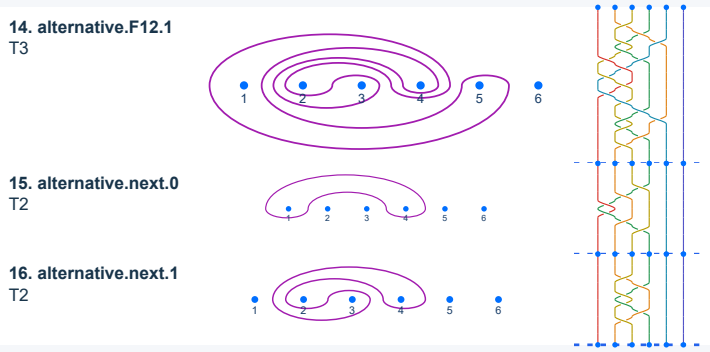
EXPLICIT GLOBAL CONJUGATION: a change of the whole product by the recorded frame.

1 unchanged displayed factors above; 0 below. Only the changed interval is shown.

Before



After



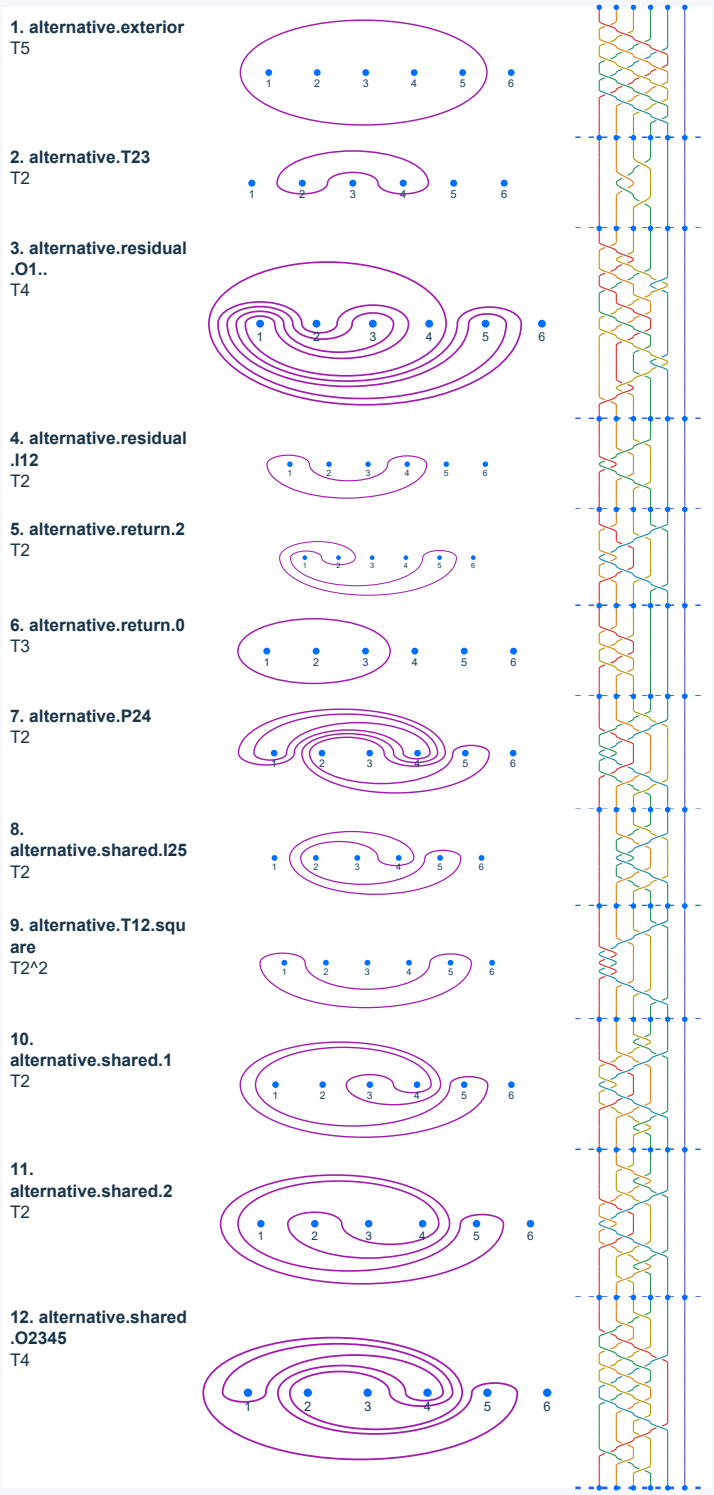
III - step 064

Choose the cyclic start at the actual three-point root

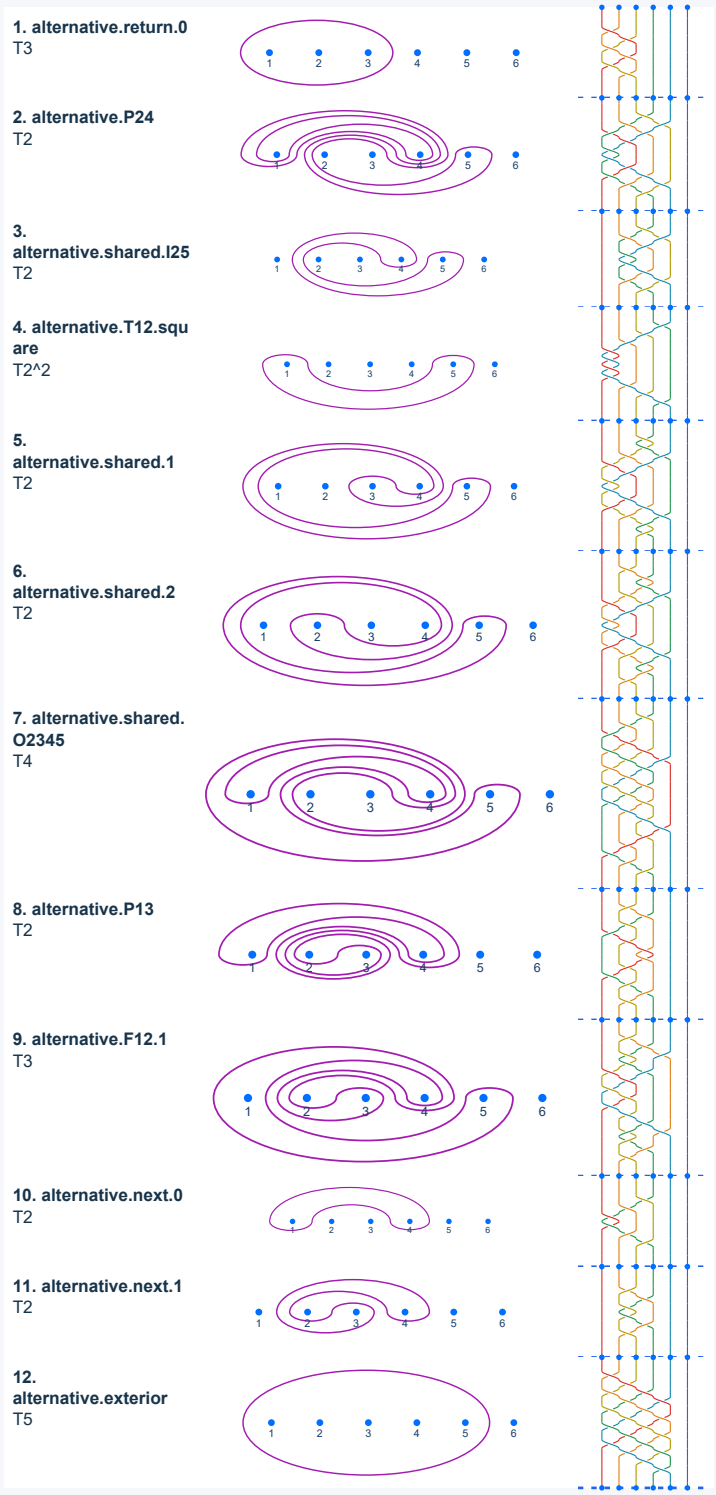
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

0 unchanged displayed factors above; 0 below. Only the changed interval is shown. Explicit common conjugation; its full braid frame is saved.

Before



After



III - step 064 - continued

Choose the cyclic start at the actual three-point root

DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

0 unchanged displayed factors above; 0 below. Only the changed interval is shown. Explicit common conjugation; its full braid frame is saved.

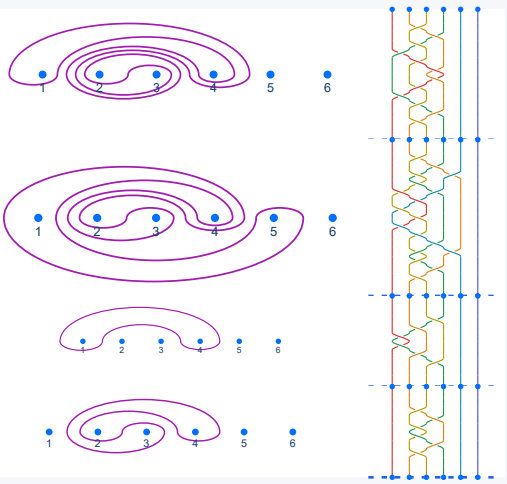
Before

13. alternative.P13
T2

14. alternative.F12.1
T3

15. alternative.next.0
T2

16. alternative.next.1
T2



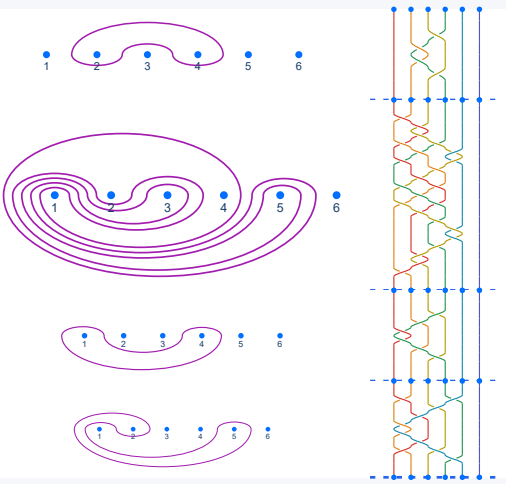
After

13. alternative.T23
T2

14. alternative.residual.O1..
T4

15. alternative.residual.I12
T2

16. alternative.return.2
T2



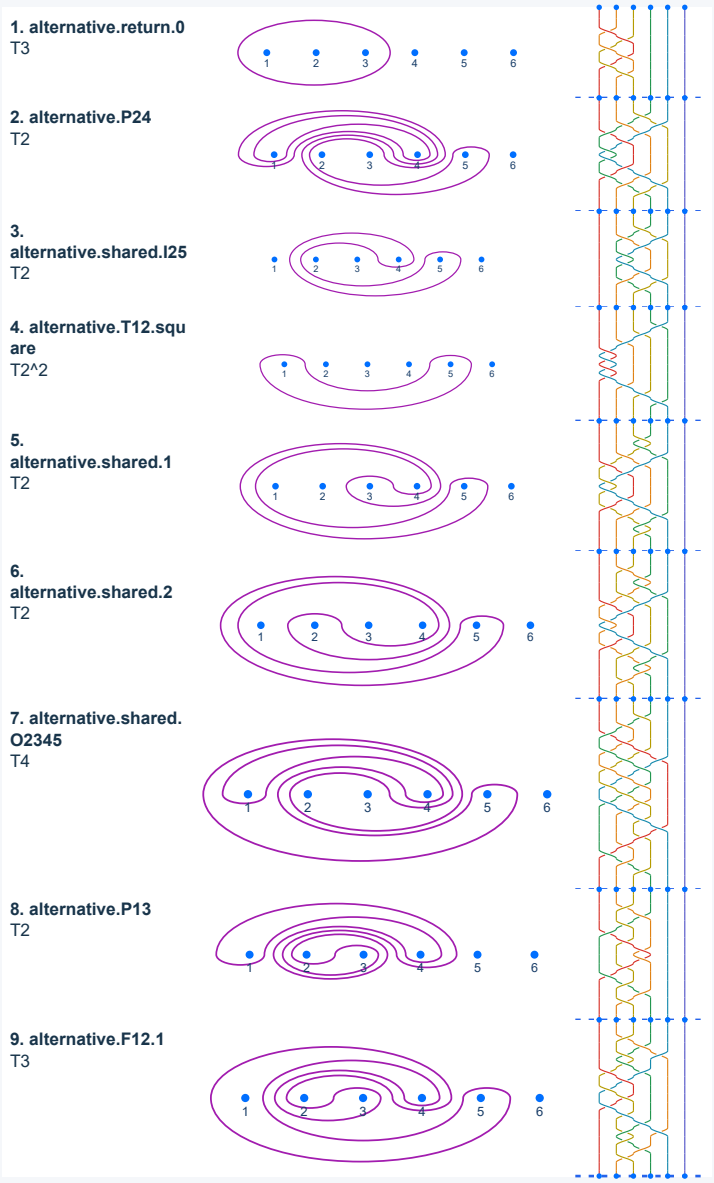
III - step 065

Carry the three-point root down through the eight residual factors

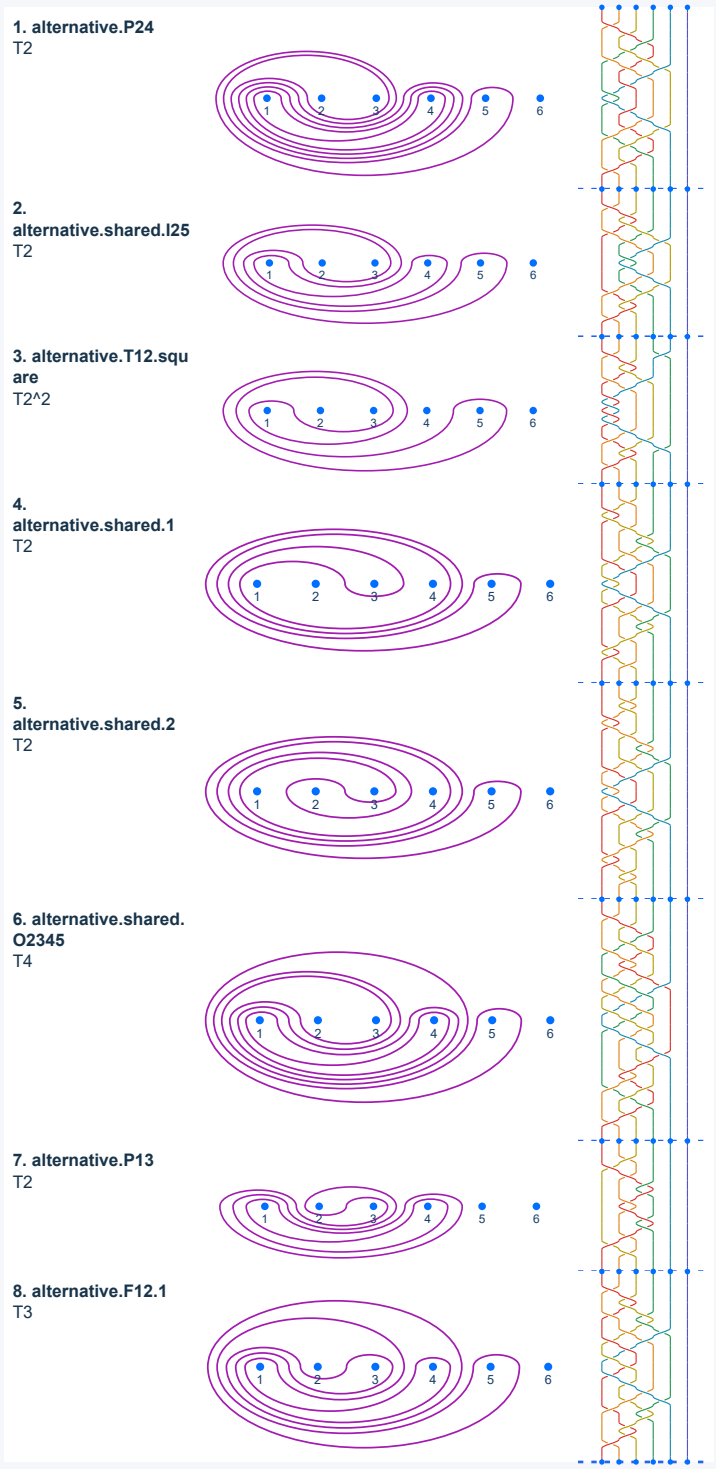
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

0 unchanged displayed factors above; 7 below. Only the changed interval is shown. Retained adjacent checks: 8.

Before



After



III - step 065 - continued

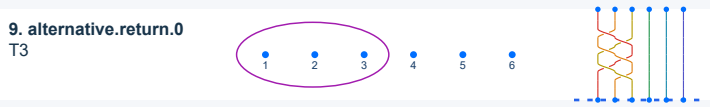
Carry the three-point root down through the eight residual factors

DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

0 unchanged displayed factors above; 7 below. Only the changed interval is shown. Retained adjacent checks: 8.

Before

After



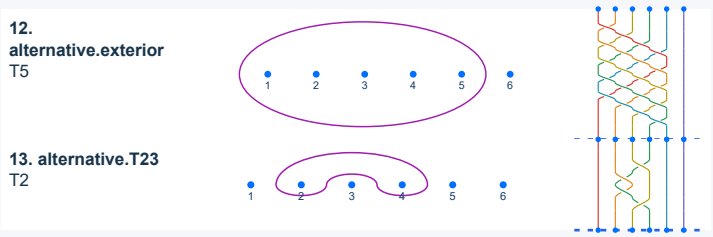
III - step 066

Commute the final pair petal past the visible exterior

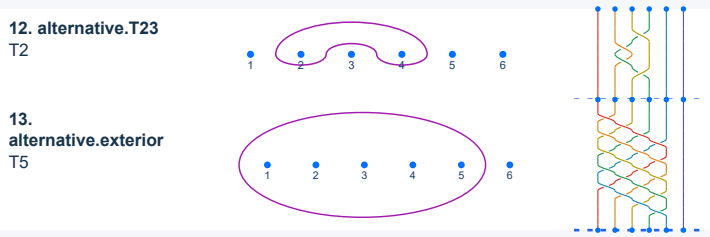
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

11 unchanged displayed factors above; 3 below. Only the changed interval is shown. Retained adjacent checks: 1.

Before



After



III - step 067

Transported PC daisy closes the actual root and its three pair petals

DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

8 unchanged displayed factors above; 4 below. Only the changed interval is shown.

Before



After



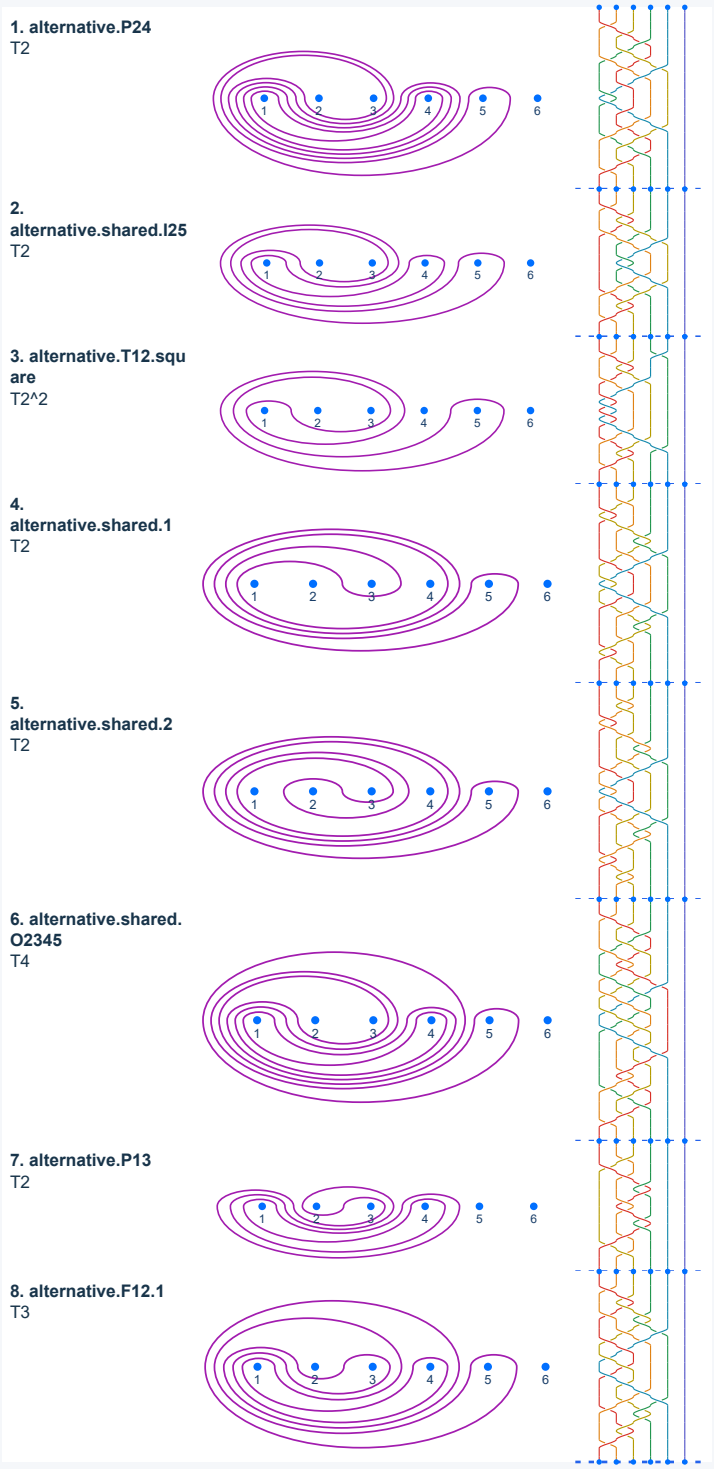
III - step 068

Carry the new round four-point twist up to unwind the residual block

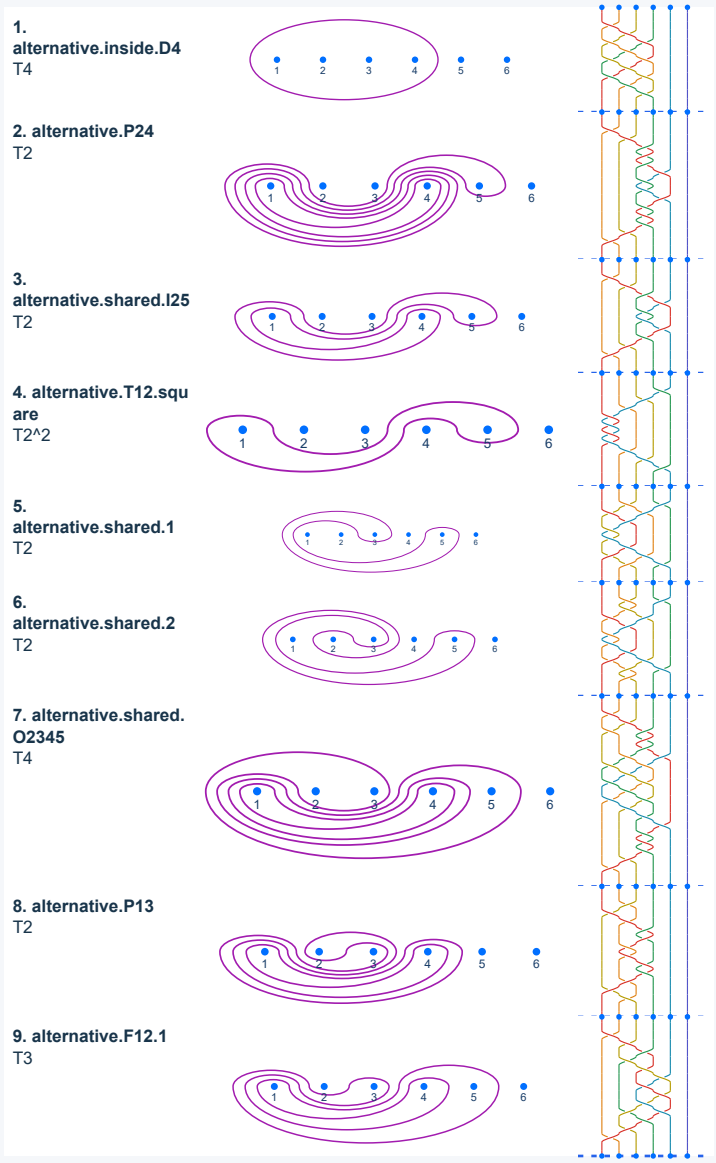
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

0 unchanged displayed factors above; 4 below. Only the changed interval is shown. Retained adjacent checks: 8.

Before



After



III - step 068 - continued

Carry the new round four-point twist up to unwind the residual block

DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

0 unchanged displayed factors above; 4 below. Only the changed interval is shown. Retained adjacent checks: 8.

Before

9.
alternative.inside.D4
T4



After

IV. Reach the second exterior twist

Actual ordered factor supports and their connected braids. All unchanged context is counted, and every retained move is shown.

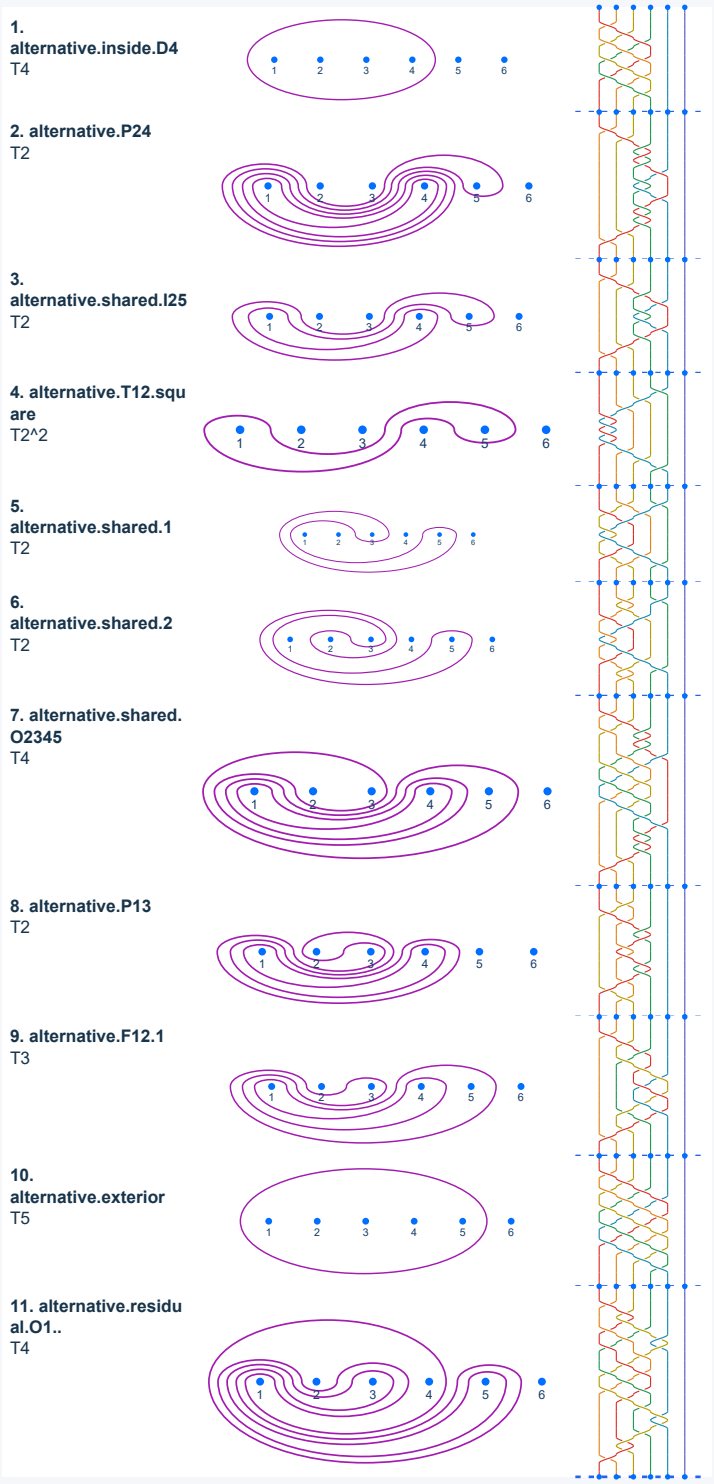
IV - step 069

Common disk frame keeps the second exterior lantern readable

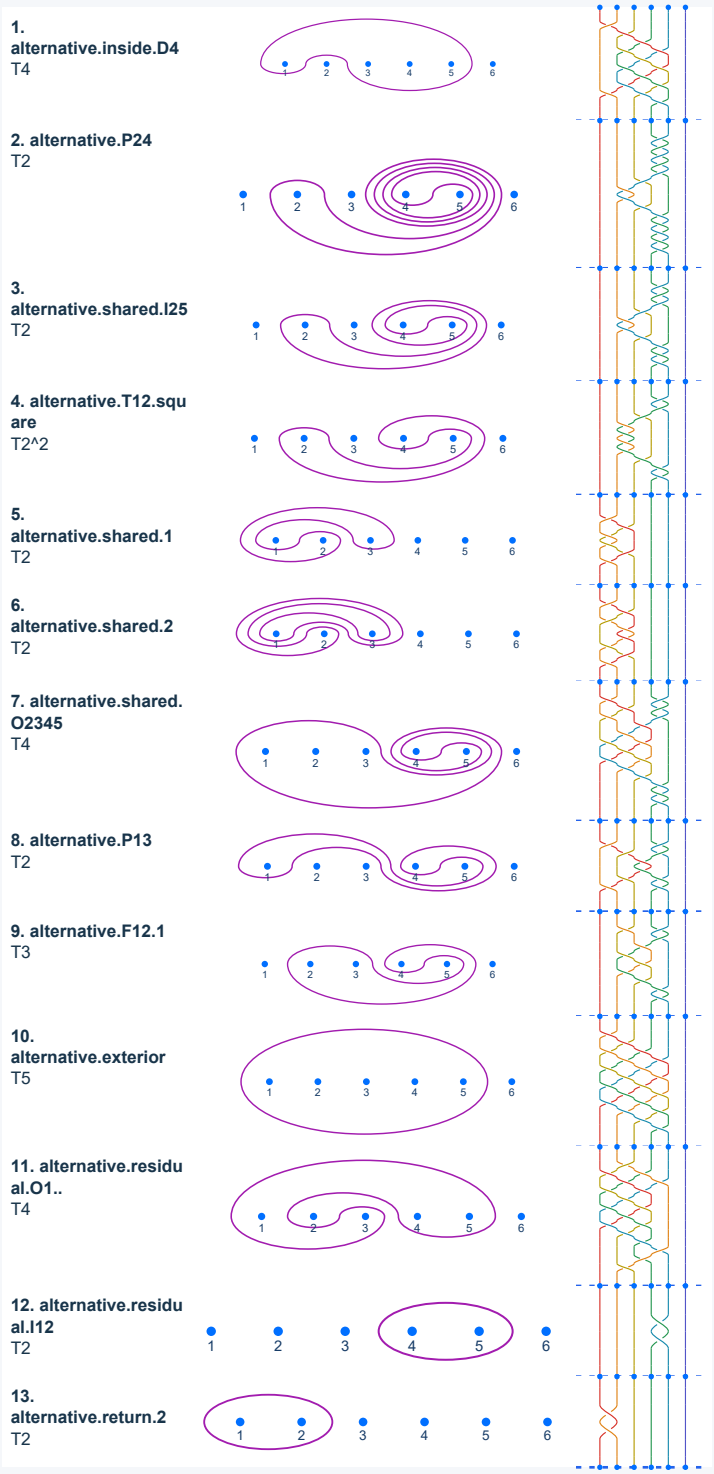
EXPLICIT GLOBAL CONJUGATION: a change of the whole product by the recorded frame.

0 unchanged displayed factors above; 0 below. Only the changed interval is shown.

Before



After



IV - step 069 - continued

Common disk frame keeps the second exterior lantern readable

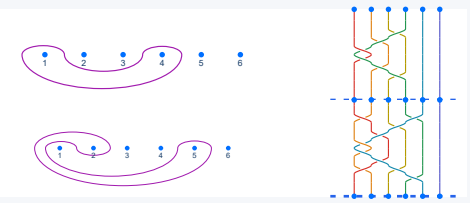
EXPLICIT GLOBAL CONJUGATION: a change of the whole product by the recorded frame.

0 unchanged displayed factors above; 0 below. Only the changed interval is shown.

Before

12. alternative.residual.l12
T2

13. alternative.return.2
T2



After

IV - step 070

The four-point PC daisy lends its actual three-point root

DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

0 unchanged displayed factors above; 12 below. Only the changed interval is shown.

Before

1. alternative.inside.D4
T4



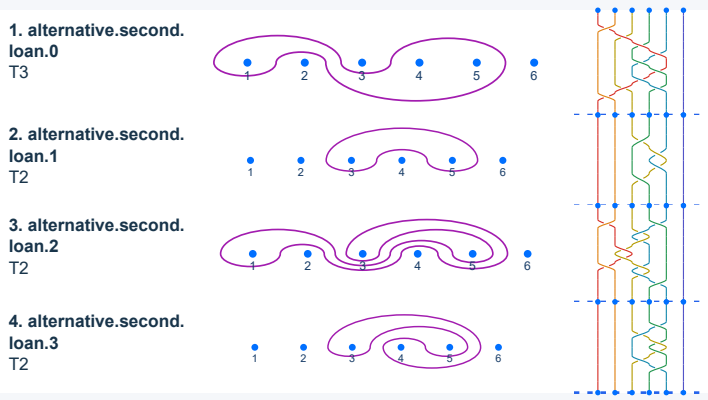
After

1. alternative.second.loan.0
T3

2. alternative.second.loan.1
T2

3. alternative.second.loan.2
T2

4. alternative.second.loan.3
T2



IV - step 071

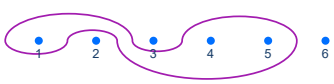
Cyclic start at the actual pair that transports the borrowed root

DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

0 unchanged displayed factors above; 0 below. Only the changed interval is shown. Explicit common conjugation; its full braid frame is saved.

Before

1. alternative.second.loan.0
T3



2. alternative.second.loan.1
T2



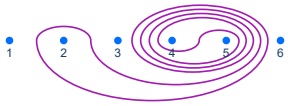
3. alternative.second.loan.2
T2



4. alternative.second.loan.3
T2



5. alternative.P24
T2



6. alternative.shared.I25
T2



7. alternative.T12.square
T2^2



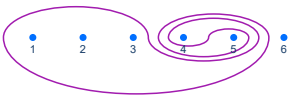
8. alternative.shared.1
T2



9. alternative.shared.2
T2



10. alternative.shared.O2345
T4



11. alternative.P13
T2

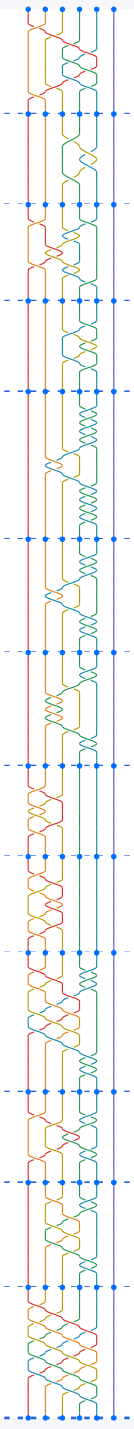


12. alternative.F12.1
T3



13. alternative.exterior
T5





After

1. alternative.return.2
T2



2. alternative.second.loan.0
T3



3. alternative.second.loan.1
T2



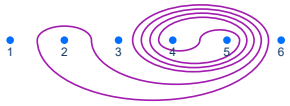
4. alternative.second.loan.2
T2



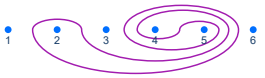
5. alternative.second.loan.3
T2



6. alternative.P24
T2



7. alternative.shared.I25
T2



8. alternative.T12.square
T2^2



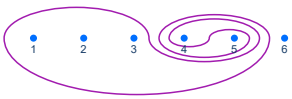
9. alternative.shared.1
T2



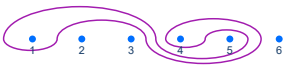
10. alternative.shared.2
T2



11. alternative.shared.O2345
T4



12. alternative.P13
T2

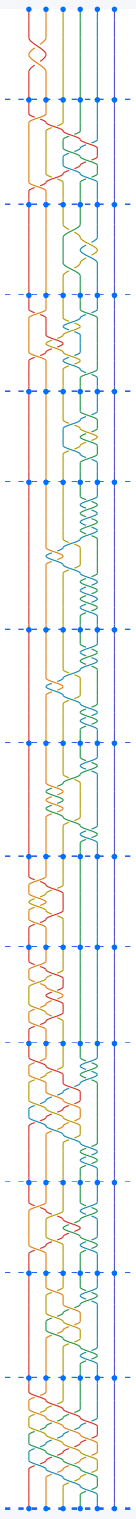


13. alternative.F12.1
T3



14. alternative.exterior
T5





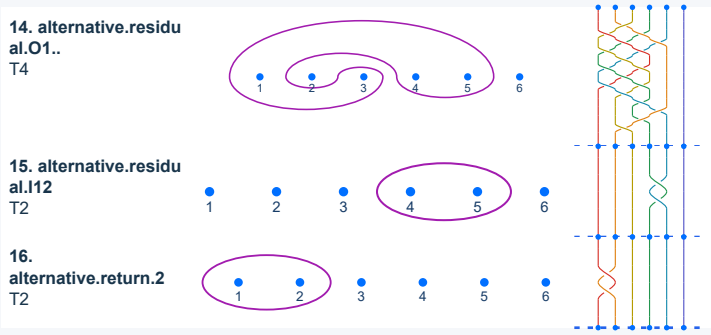
IV - step 071 - continued

Cyclic start at the actual pair that transports the borrowed root

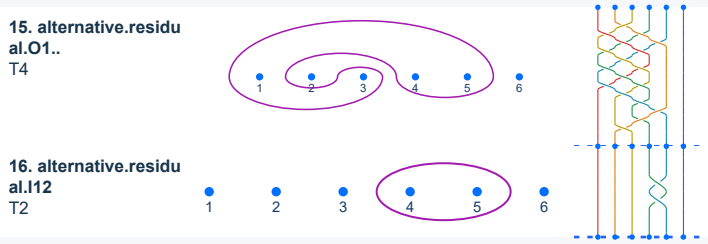
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

0 unchanged displayed factors above; 0 below. Only the changed interval is shown. Explicit common conjugation; its full braid frame is saved.

Before



After



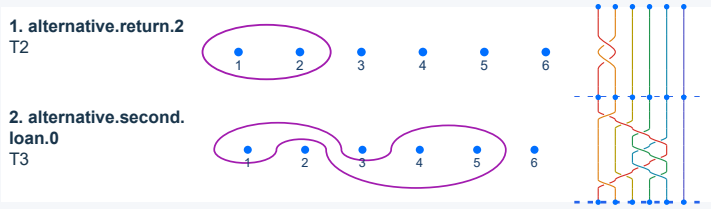
IV - step 072

The actual pair carries the borrowed root to the exact missing class

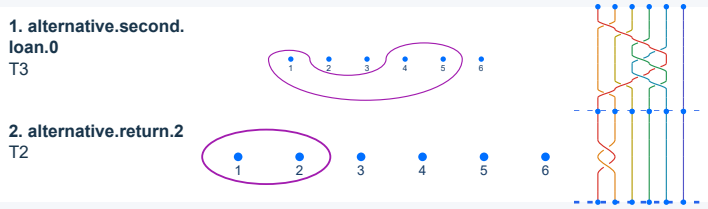
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

0 unchanged displayed factors above; 14 below. Only the changed interval is shown. Retained adjacent checks: 1.

Before



After



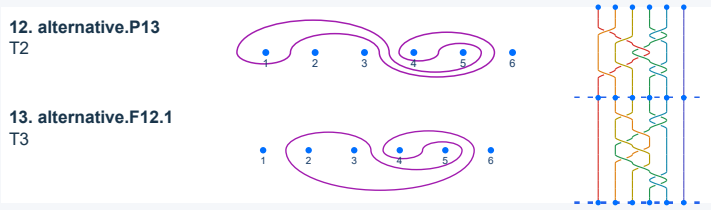
IV - step 073

Bring the three-point curve up across one neighboring pair

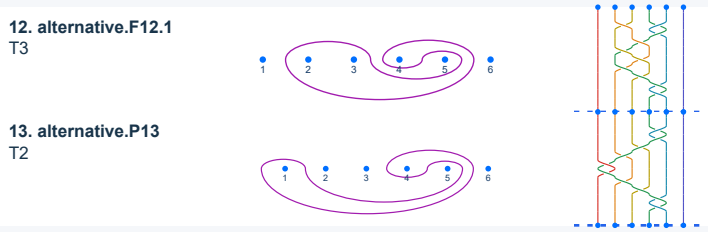
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

11 unchanged displayed factors above; 3 below. Only the changed interval is shown. Retained adjacent checks: 1.

Before



After



IV - step 074

Cyclic start at the matched four-point curve

DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

0 unchanged displayed factors above; 0 below. Only the changed interval is shown. Explicit common conjugation; its full braid frame is saved.

Before

1. alternative.second.loan.0
T3

2. alternative.return.2
T2



3. alternative.second.loan.1
T2

4. alternative.second.loan.2
T2

5. alternative.second.loan.3
T2

6. alternative.P24
T2

7. alternative.shared.I25
T2

8. alternative.T12.square
T2^2

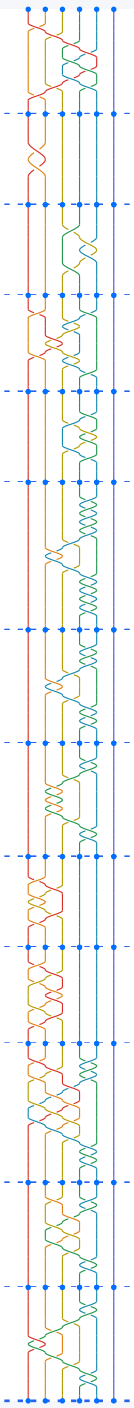
9. alternative.shared.1
T2

10. alternative.shared.2
T2

11. alternative.shared.O2345
T4

12. alternative.F12.1
T3

13. alternative.P13
T2



After

1. alternative.shared.O2345
T4

2. alternative.F12.1
T3

3. alternative.P13
T2

4. alternative.exterior
T5

5. alternative.residual.O1..
T4

6. alternative.residual.I12
T2

7. alternative.second.loan.0
T3

8. alternative.return.2
T2

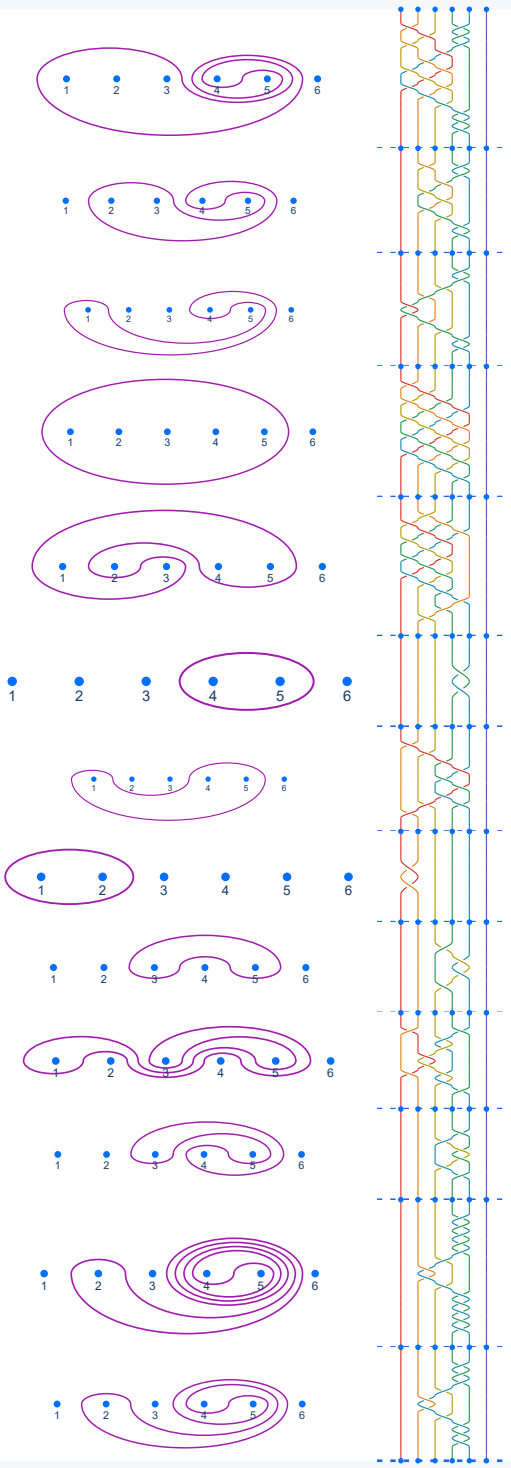
9. alternative.second.loan.1
T2

10. alternative.second.loan.2
T2

11. alternative.second.loan.3
T2

12. alternative.P24
T2

13. alternative.shared.I25
T2



IV - step 074 - continued

Cyclic start at the matched four-point curve

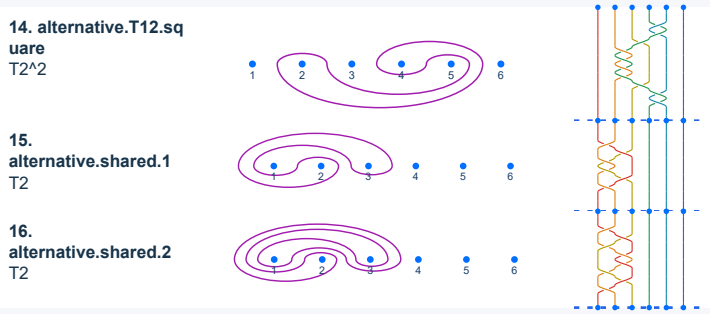
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

0 unchanged displayed factors above; 0 below. Only the changed interval is shown. Explicit common conjugation; its full braid frame is saved.

Before



After



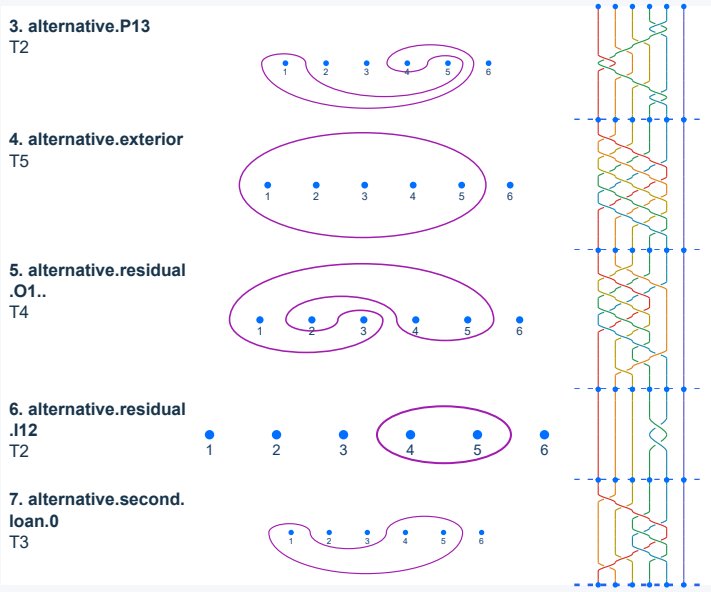
IV - step 075

Gather the borrowed root up the four-neighbor short side

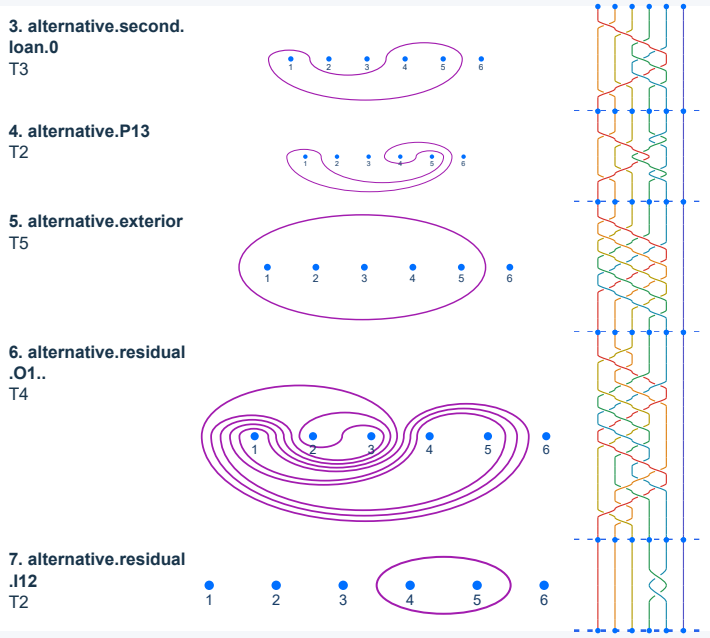
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

2 unchanged displayed factors above; 9 below. Only the changed interval is shown. Retained adjacent checks: 4.

Before



After



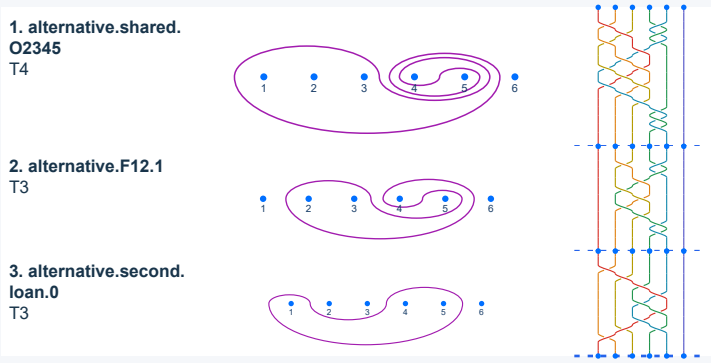
IV - step 076

Cabled PC lantern reaches a SECOND visible exterior twist

DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

0 unchanged displayed factors above; 13 below. Only the changed interval is shown.

Before



After



V. Return the actual borrowed root

Actual ordered factor supports and their connected braids. All unchanged context is counted, and every retained move is shown.

Complete factorization before Return the actual borrowed root

FULL ORDERED FACTORIZATION. Read downward. Braids remain connected, with strand colors retained; disjoint crossings may share a level.

1. alternative.second.exte..
T5



2. alternative.second.l35
T2



3. alternative.second.l24
T2



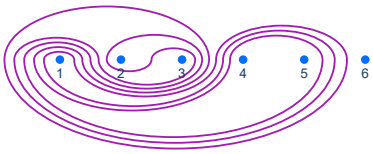
4. alternative.P13
T2



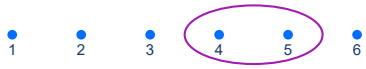
5. alternative.exterior
T5



6. alternative.residual.O1..
T4



7. alternative.residual.l12
T2



8. alternative.return.2
T2



9. alternative.second.loan.1
T2



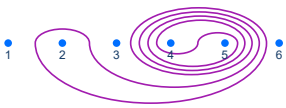
10. alternative.second.loan.2
T2



11. alternative.second.loan.3
T2



12. alternative.P24
T2



13. alternative.shared.l25
T2

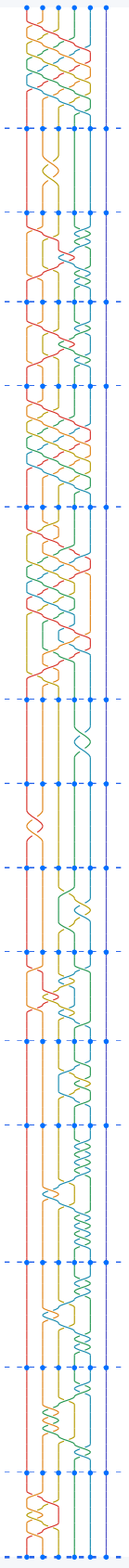


14. alternative.T12.square
T2^2



15. alternative.shared.1
T2



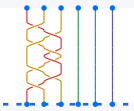


Complete factorization before Return the actual borrowed root - continued

FULL ORDERED FACTORIZATION. Read downward. Braids remain connected, with strand colors retained; disjoint crossings may share a level.

16. alternative.shared.2
T2





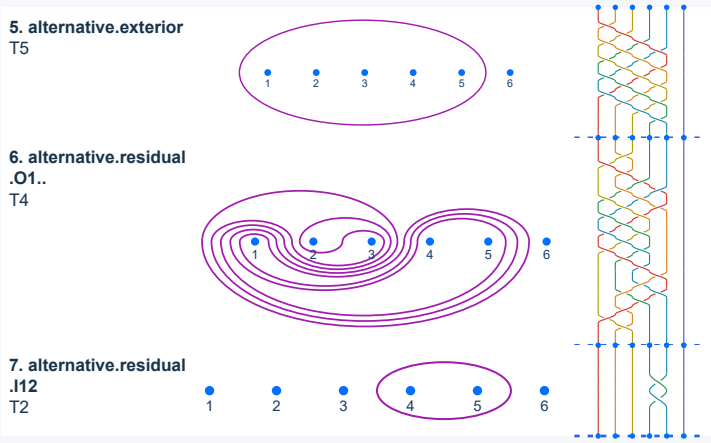
V - step 077

Drag alternative.residual.I12 up

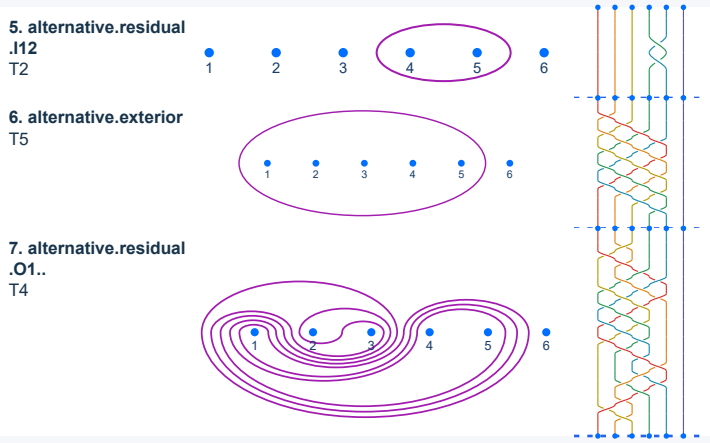
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

4 unchanged displayed factors above; 9 below. Only the changed interval is shown. Retained adjacent checks: 2.

Before



After



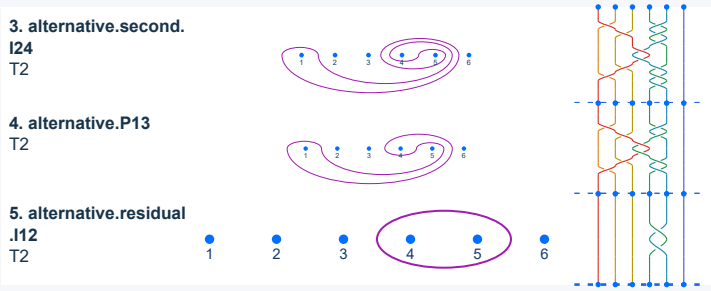
V - step 078

A transported PC rose returns the actual borrowed root

DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

2 unchanged displayed factors above; 11 below. Only the changed interval is shown.

Before



After



V - step 079

Drag alternative.return.2 down

DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

5 unchanged displayed factors above; 5 below. Only the changed interval is shown. Retained adjacent checks: 3.

Before



After



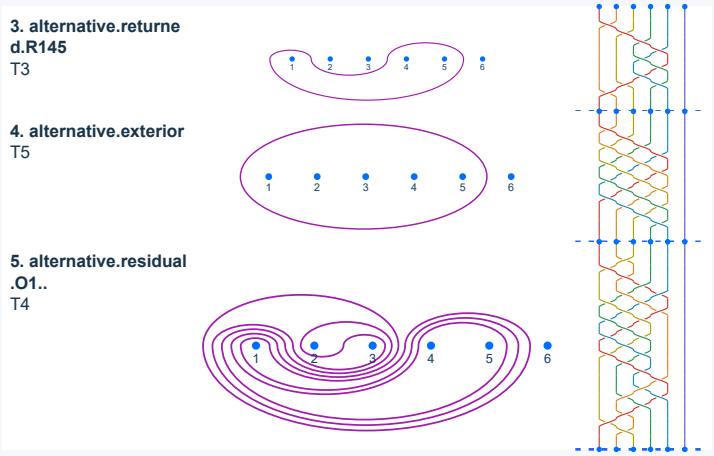
V - step 080

Drag alternative.returned.R145 down

DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

2 unchanged displayed factors above; 9 below. Only the changed interval is shown. Retained adjacent checks: 2.

Before



After



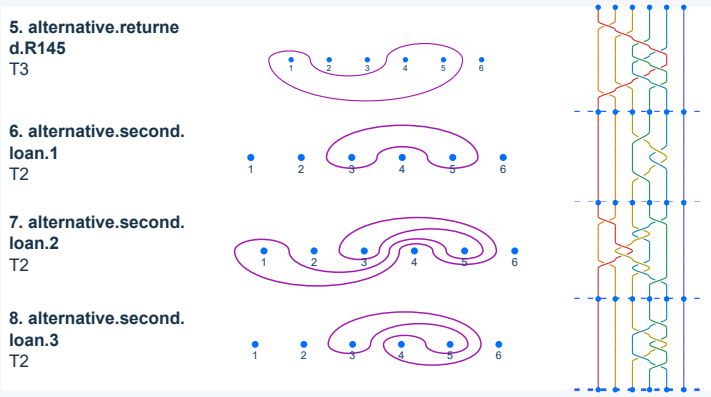
V - step 081

Close the actual returned PC daisy

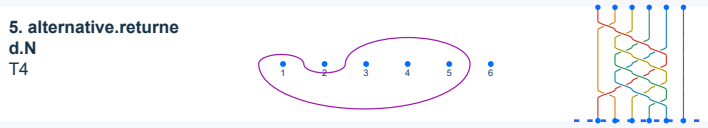
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

4 unchanged displayed factors above; 6 below. Only the changed interval is shown.

Before



After



VI. Align the four-point twists

Actual ordered factor supports and their connected braids. All unchanged context is counted, and every retained move is shown.

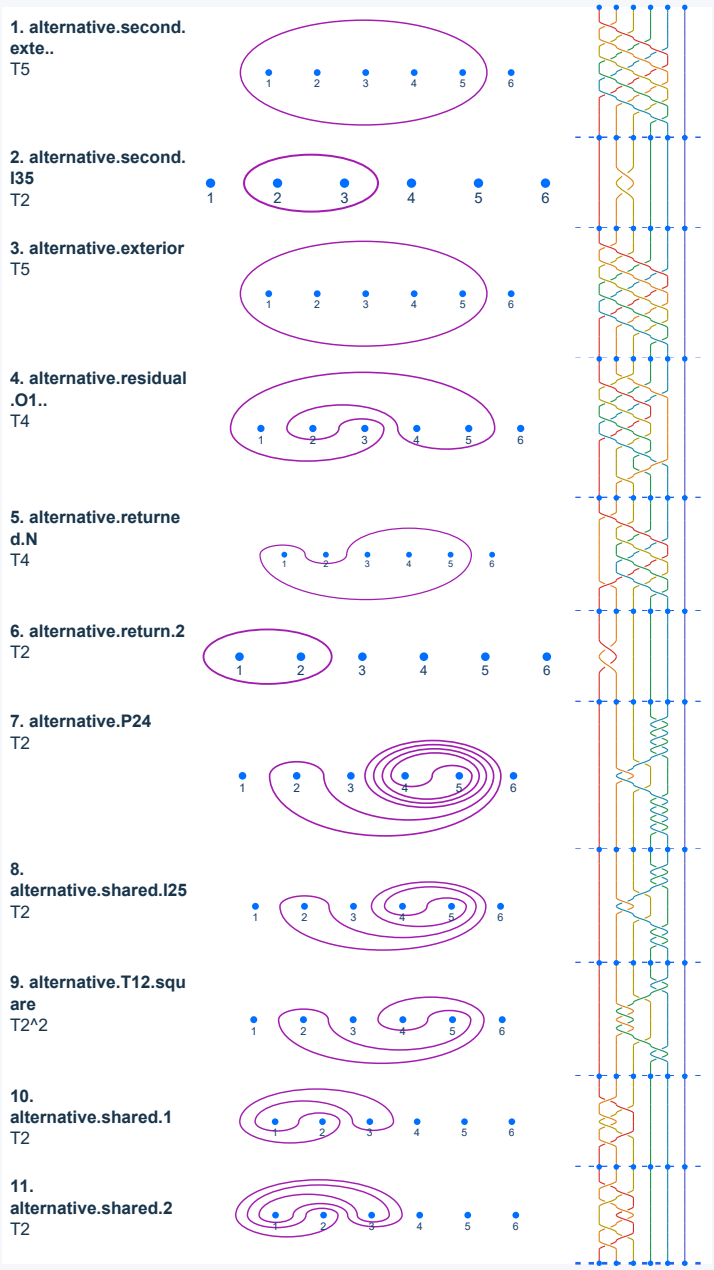
VI - step 082

Cyclic start at the two-petal alignment supplier

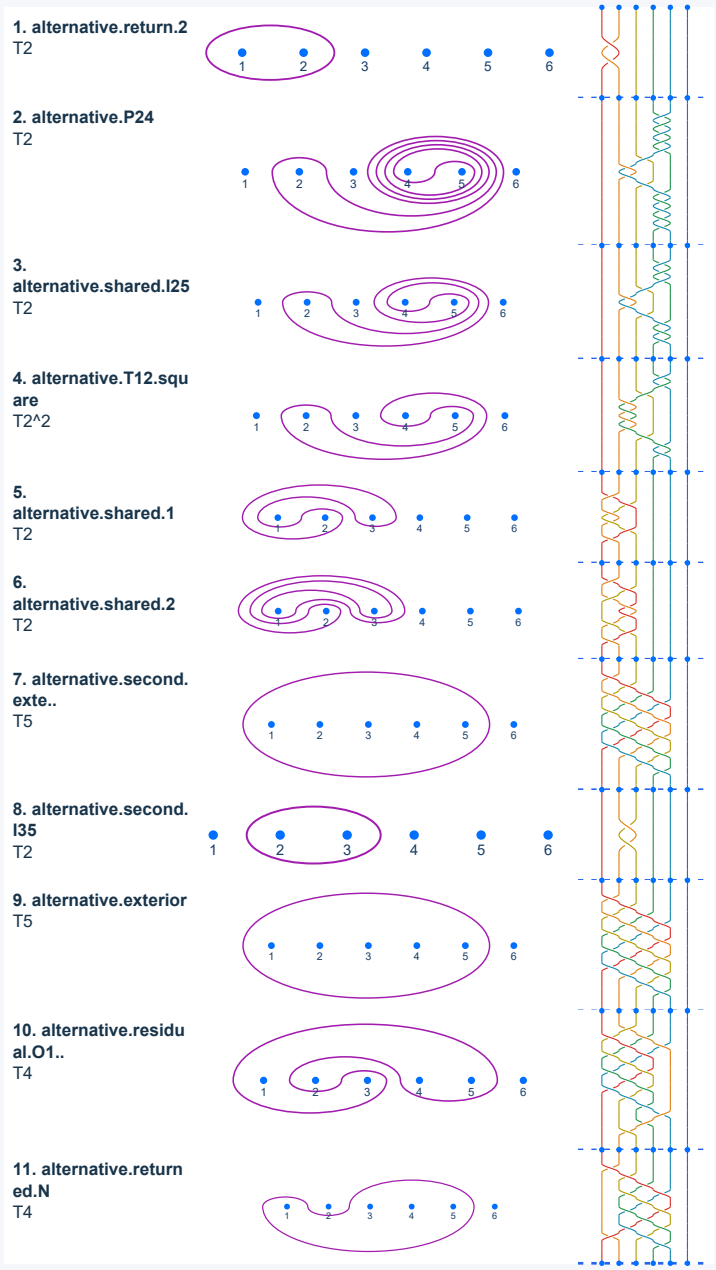
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

0 unchanged displayed factors above; 0 below. Only the changed interval is shown. Explicit common conjugation; its full braid frame is saved.

Before



After



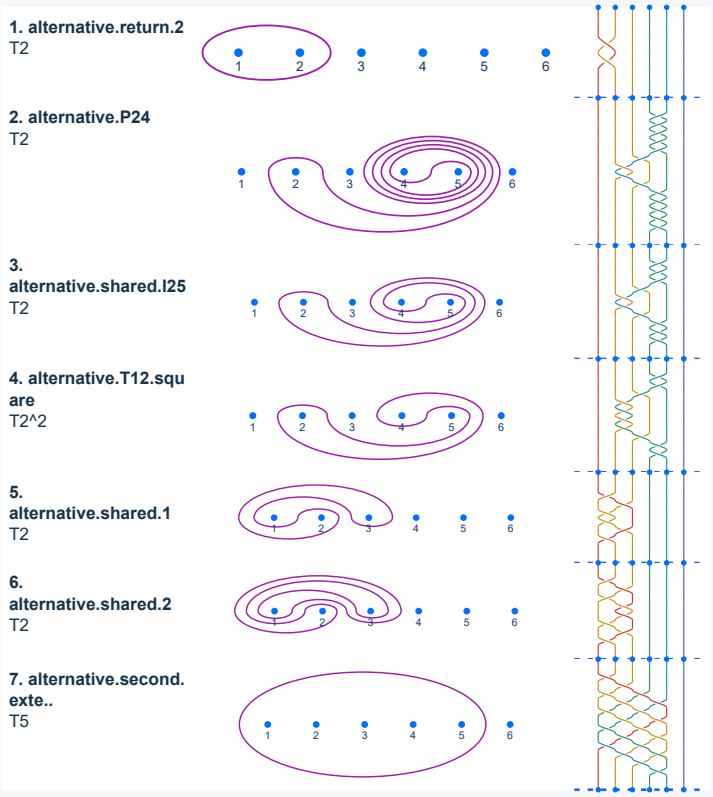
VI - step 083

Drag alternative.return.2 down

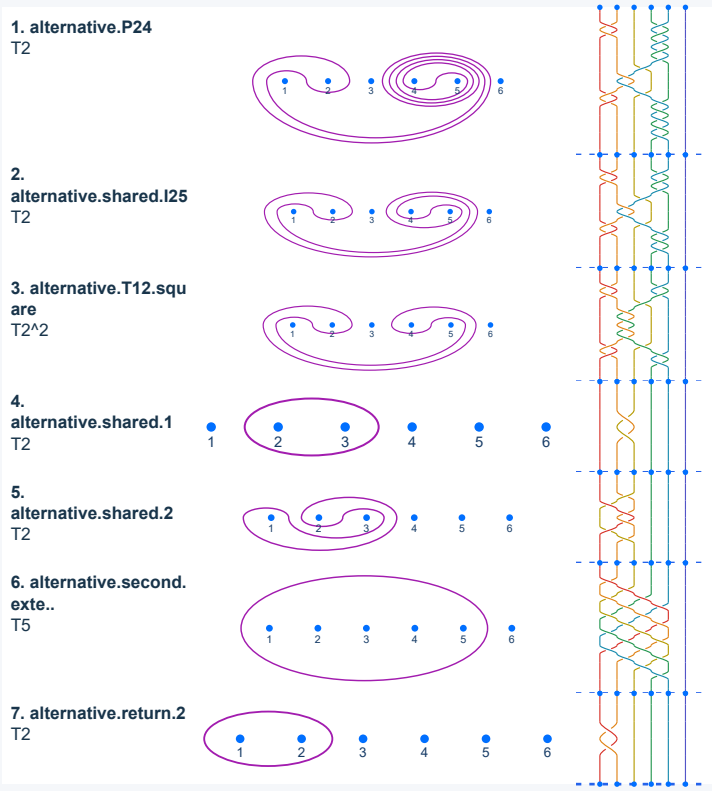
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

0 unchanged displayed factors above; 4 below. Only the changed interval is shown. Retained adjacent checks: 6.

Before



After



VI - step 084

Drag alternative.second.I35 down

DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

7 unchanged displayed factors above; 1 below. Only the changed interval is shown. Retained adjacent checks: 2.

Before



After



VI - step 085

Drag alternative.return.2 down

DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

6 unchanged displayed factors above; 2 below. Only the changed interval is shown. Retained adjacent checks: 2.

Before



After



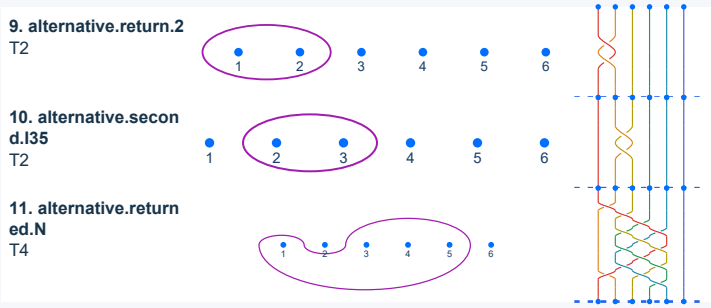
VI - step 086

Drag alternative.returned.N up

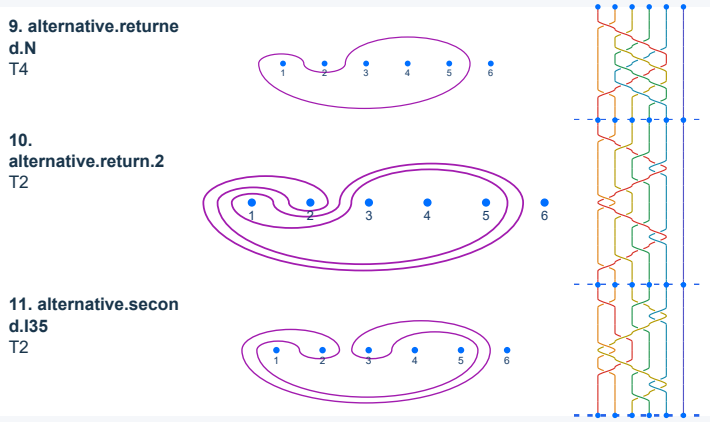
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

8 unchanged displayed factors above; 0 below. Only the changed interval is shown. Retained adjacent checks: 2.

Before



After



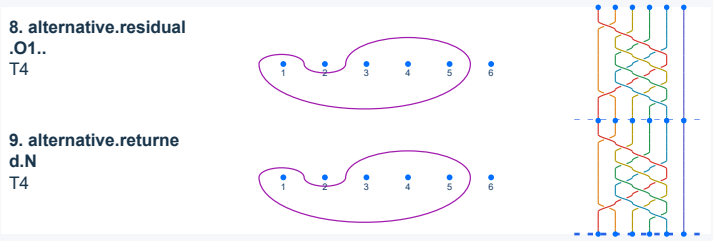
VI - step 087

Combine the matching alternative.residual.O1234 and alternative.returned.N

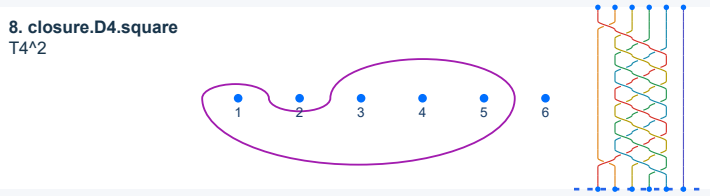
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

7 unchanged displayed factors above; 2 below. Only the changed interval is shown.

Before



After



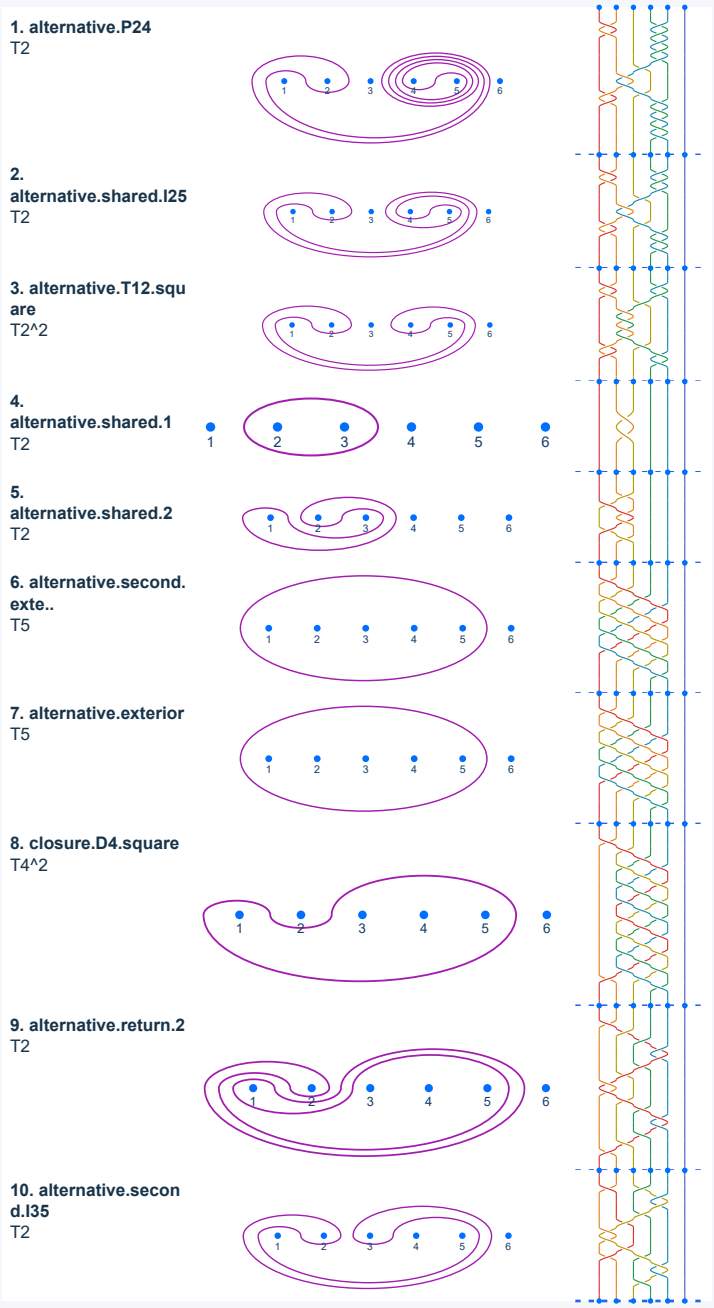
VI - step 088

Common disk frame rounds the actual four-point square

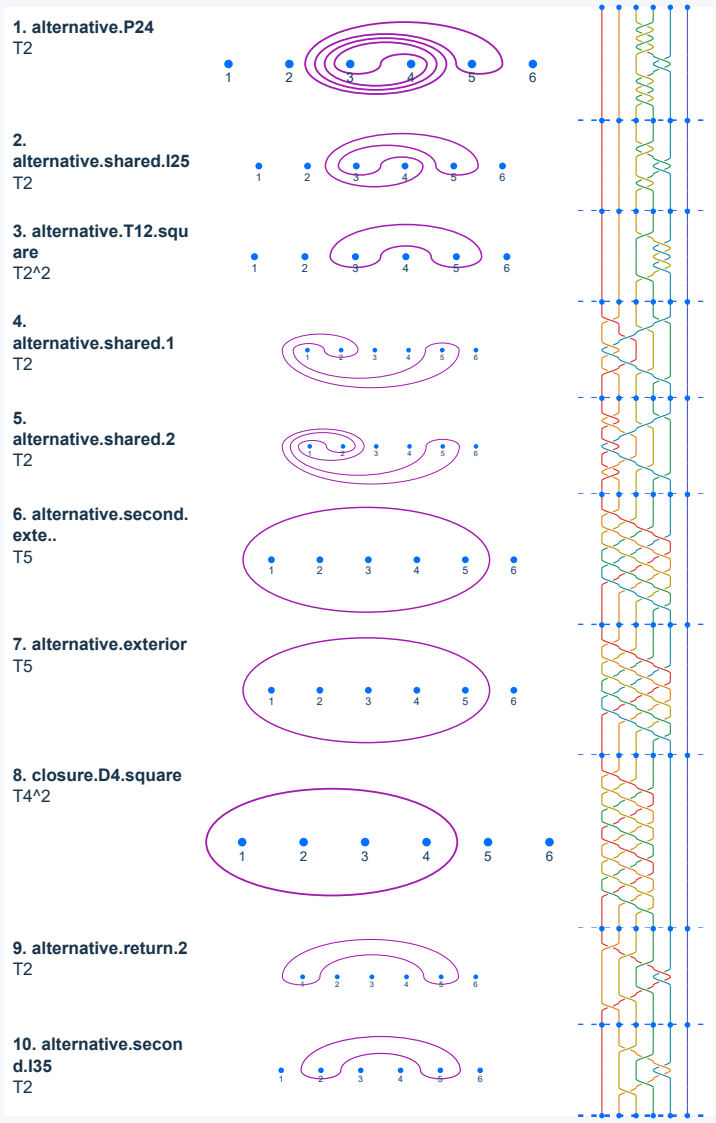
EXPLICIT GLOBAL CONJUGATION: a change of the whole product by the recorded frame.

0 unchanged displayed factors above; 0 below. Only the changed interval is shown.

Before



After



VII. Match the actual daisy petals

Actual ordered factor supports and their connected braids. All unchanged context is counted, and every retained move is shown.

VII - step 089

Split alternative.T12.square into its two positive units

DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

2 unchanged displayed factors above; 7 below. Only the changed interval is shown.

Before

3. alternative.T12.square
T2^2



After

3. closure.E1
T2

4. closure.E2
T2



VII - step 090

Drag alternative.shared.I25 down

DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

1 unchanged displayed factors above; 8 below. Only the changed interval is shown. Retained adjacent checks: 1.

Before

2. alternative.shared.I25
T2

3. closure.E1
T2



After

2. closure.E1
T2

3. alternative.shared.I25
T2



VII - step 091

Drag closure.E1 up

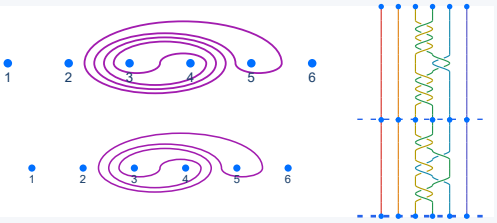
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

0 unchanged displayed factors above; 9 below. Only the changed interval is shown. Retained adjacent checks: 1.

Before

1. alternative.P24
T2

2. closure.E1
T2



After

1. closure.E1
T2

2. alternative.P24
T2



VII - step 092

Drag alternative.P24 up

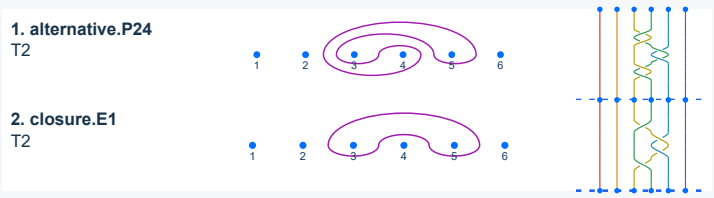
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

0 unchanged displayed factors above; 9 below. Only the changed interval is shown. Retained adjacent checks: 1.

Before



After



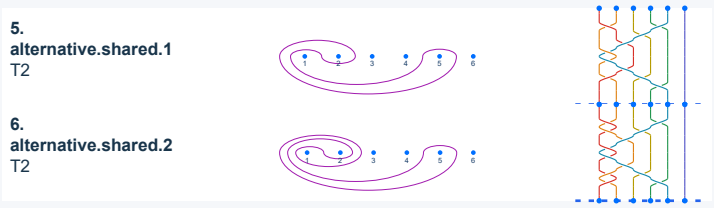
VII - step 093

Drag alternative.shared.1 down

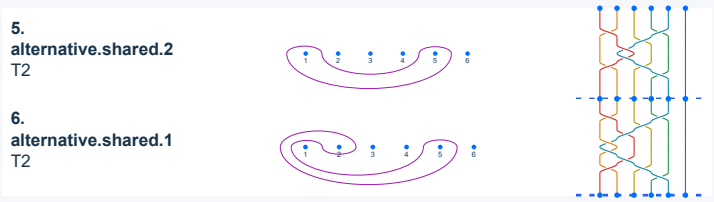
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

4 unchanged displayed factors above; 5 below. Only the changed interval is shown. Retained adjacent checks: 1.

Before



After



VII - step 094

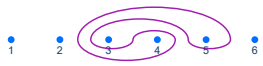
Cyclic start at the round four-point square

DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

0 unchanged displayed factors above; 0 below. Only the changed interval is shown. Explicit common conjugation; its full braid frame is saved.

Before

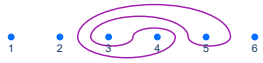
1. alternative.P24
T2



2. closure.E1
T2



3. alternative.shared.I25
T2



4. closure.E2
T2



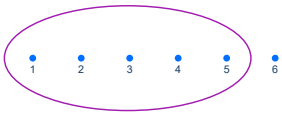
5. alternative.shared.2
T2



6. alternative.shared.1
T2



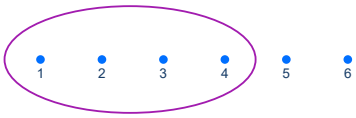
7. alternative.second. exte..
T5



8. alternative.exterior
T5



9. closure.D4.square
T4^2

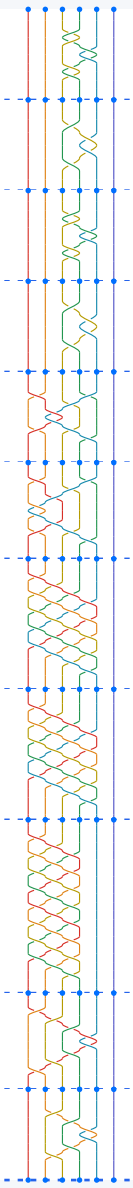


10. alternative.return.2
T2



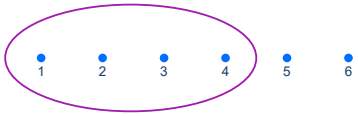
11. alternative.second.I35
T2





After

1. closure.D4.square
T4^2



2. alternative.return.2
T2



3. alternative.second. I35
T2



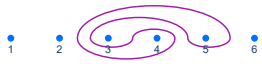
4. alternative.P24
T2



5. closure.E1
T2



6. alternative.shared.I25
T2



7. closure.E2
T2



8. alternative.shared.2
T2



9. alternative.shared.1
T2

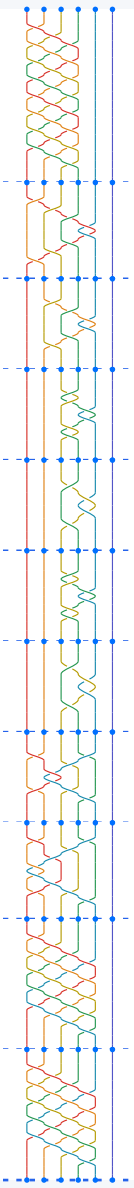


10. alternative.second.exte..
T5



11. alternative.exterior
T5





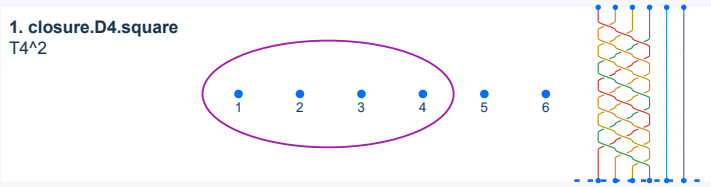
VII - step 095

Split closure.D4.square into its two positive units

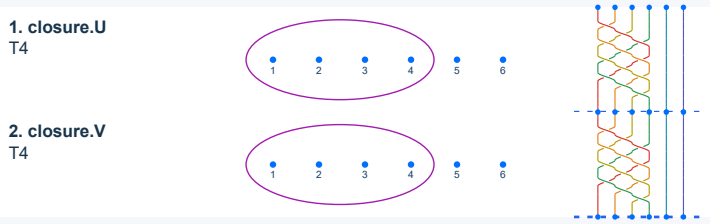
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

0 unchanged displayed factors above; 10 below. Only the changed interval is shown.

Before



After



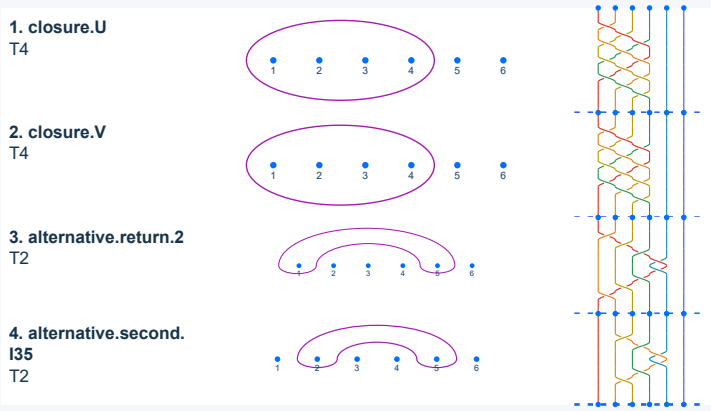
VII - step 096

Drag closure.U down

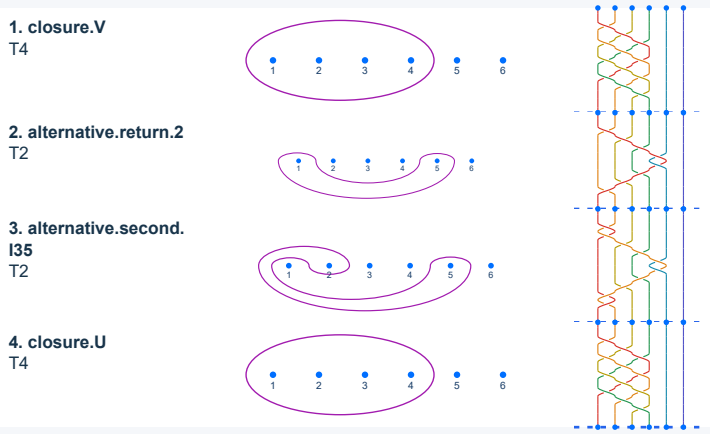
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

0 unchanged displayed factors above; 8 below. Only the changed interval is shown. Retained adjacent checks: 3.

Before



After



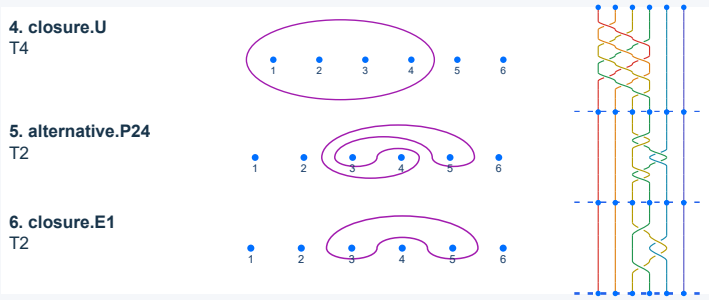
VII - step 097

Drag closure.U down

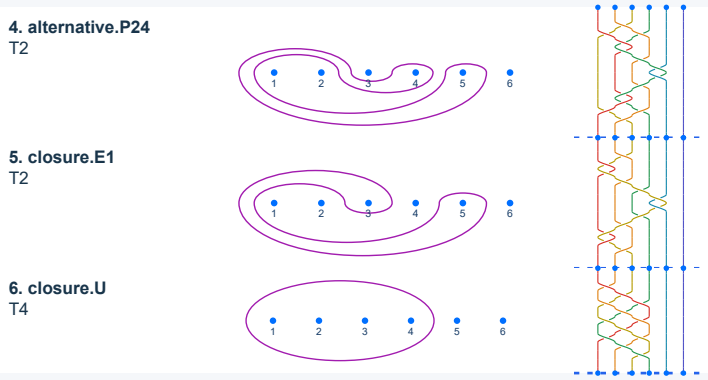
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

3 unchanged displayed factors above; 6 below. Only the changed interval is shown. Retained adjacent checks: 2.

Before



After



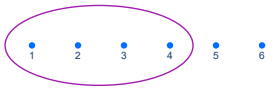
VIII. Two daisies, four exteriors and the sphere cap

Actual ordered factor supports and their connected braids. All unchanged context is counted, and every retained move is shown.

Complete factorization before Two daisies, four exteriors and the sphere cap

FULL ORDERED FACTORIZATION. Read downward. Braids remain connected, with strand colors retained; disjoint crossings may share a level.

1. closure.V
T4



2. alternative.return.2
T2



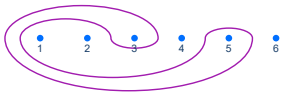
3. alternative.second.l35
T2



4. alternative.P24
T2



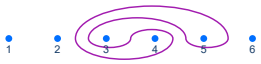
5. closure.E1
T2



6. closure.U
T4



7. alternative.shared.l25
T2



8. closure.E2
T2



9. alternative.shared.2
T2



10. alternative.shared.1
T2

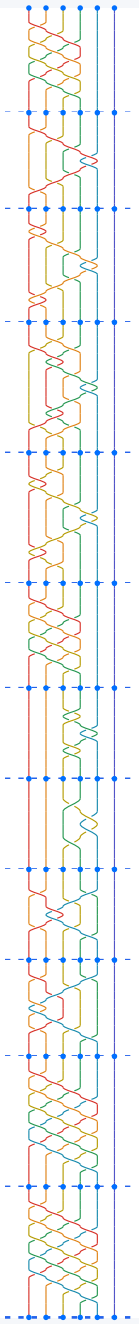


11. alternative.second.exte..
T5



12. alternative.exterior
T5





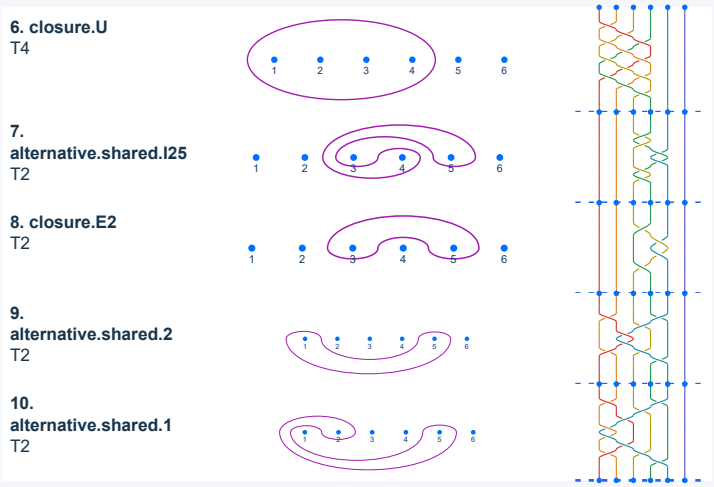
VIII - step 098

First actual transported five-point PC daisy

DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

5 unchanged displayed factors above; 2 below. Only the changed interval is shown.

Before



After



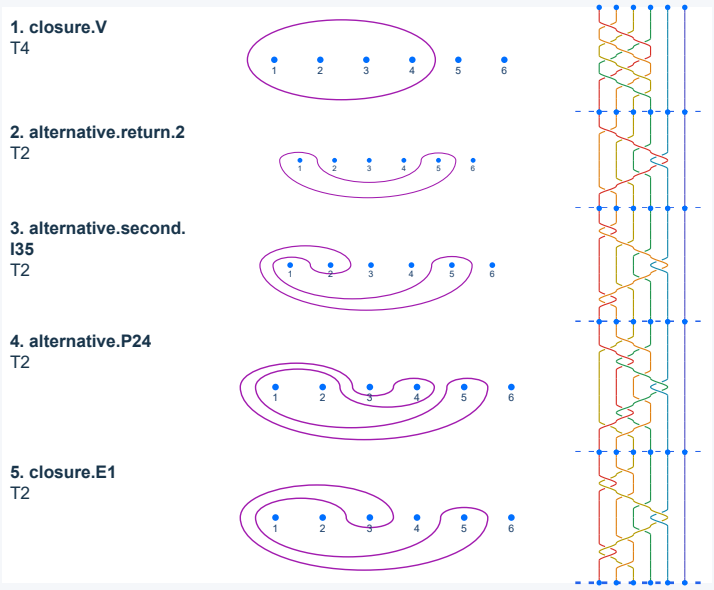
VIII - step 099

Second actual transported five-point PC daisy

DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

0 unchanged displayed factors above; 3 below. Only the changed interval is shown.

Before



After



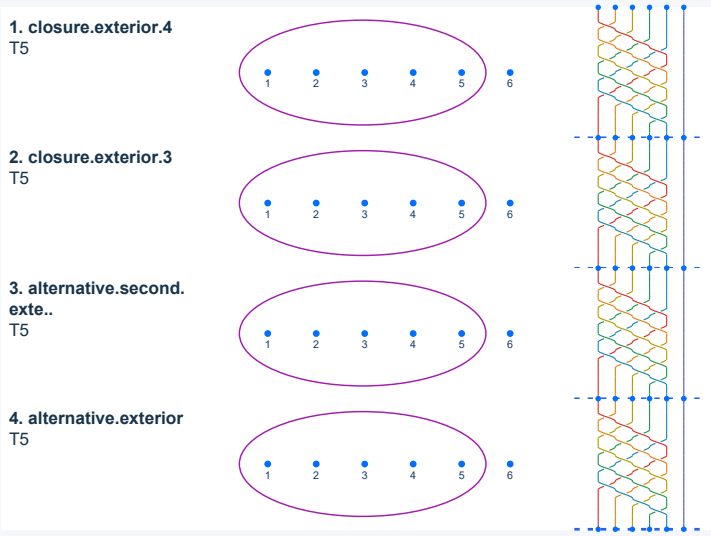
VIII - step 100

Combine the FOUR identical visible exterior twists

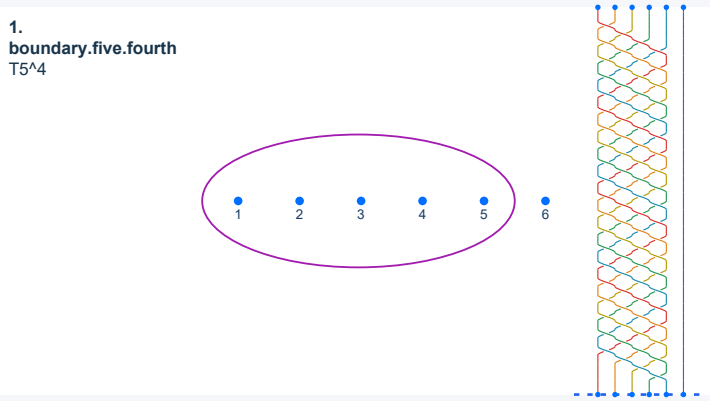
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

0 unchanged displayed factors above; 0 below. Only the changed interval is shown.

Before



After



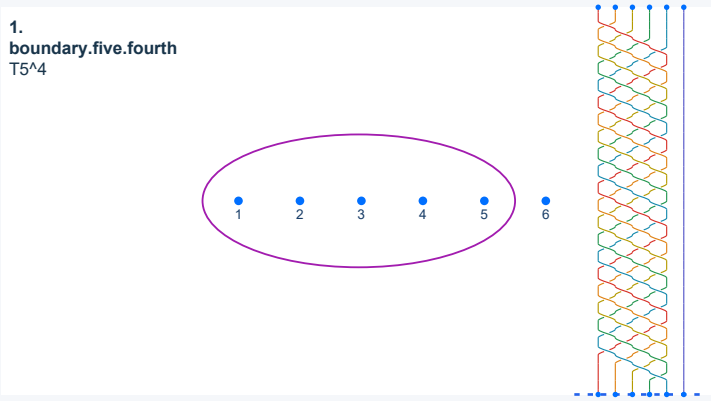
VIII - step 101

SPHERE MOVE: cap the FOURTH boundary power

SPHERE MOVE: exact sphere action and +I lift checked; disk discrepancy retained.

0 unchanged displayed factors above; 0 below. Only the changed interval is shown. SPHERE MOVE: full disk discrepancy appears in Appendix IX; chosen lift is +I.

Before



After

Identity / no factors in this interval.

IX. Sphere discrepancies and certificates

The following are ALL FOUR sphere operations in this named route. They are not disk braid equalities. Each discrepancy is written in signed generator indices; the exact sphere certificate and +I lift are retained online.

SPHERE MOVE at step 14

SPHERE LANTERN: absorb the matched pair and the two prefix roots

Exact sphere action certified. Chosen genus-two lift discrepancy: +I. This is not an equality in the ordinary disk braid group.

FULL DISK DISCREPANCY (66 signed braid letters):

5 -4 2 3 2 3 2 3 4 -5 -4 -2 3 3 2 -5 -2 -3 4 2 3 2 3 2
3 -4 3 2 5 4 2 3 2 5 -3 -3 4 1 2 3 4 5 1 2 3 4 -5 -1
-1 -5 -4 -3 -2 -1 -5 -4 -3 -2 -1 -4 3 3 -5 -2 -3 -2

SPHERE MOVE at step 16

SPHERE RECHART: use the five-point disk; retain SIX boundary corrections

Exact sphere action certified. Chosen genus-two lift discrepancy: +I. This is not an equality in the ordinary disk braid group.

FULL DISK DISCREPANCY (482 signed braid letters):

5 4 4 1 2 3 4 5 1 2 3 4 5 1 1 5 -4 -3 -2 -1 -5 -4 -3 -2
-1 -4 3 3 -5 -2 -3 -2 -2 3 4 5 4 5 4 5 4 5 4 5 -3 2
3 2 4 4 -3 2 1 -2 3 -4 -4 -2 -3 -2 3 -4 2 3 2 3 2 3 2 3
2 3 2 3 4 -3 2 -4 -2 -3 2 4 -5 -3 -3 4 3 3 5 -4 -2 5 4 2
3 2 3 2 3 -4 -5 3 2 4 -2 5 3 4 3 4 3 4 -5 2 -5 -4 2 3
2 3 3 -2 1 2 -3 2 3 4 5 4 5 4 5 4 -1 -2 3 -4 3 4 -3 2
1 5 -1 -2 3 4 3 4 3 4 3 4 3 4 3 4 2 1 -4 2 3 2 3 2
3 2 3 2 3 2 3 4 2 3 2 5 -3 -3 -4 -5 -4 -4 -3 -2 -1 -2 -1 -2
-1 -2 -1 -2 -1 -2 -1 3 -4 -3 -2 -1 -2 -1 -2 -1 -2 -1 -2 -1 -2 -1 3 1
1 1 -4 -3 -2 -1 2 3 4 -1 -1 -1 4 4 4 1 1 2 -4 -3 -4 -3 -4 -3
-2 -1 -1 -4 -3 -4 -3 1 2 2 1 -4 -3 -2 -1 2 3 4 -1 -2 -2 -1 3 4
3 -4 -4 -3 -2 -1 -2 -1 -2 -1 3 4 4 1 -3 -2 -3 -2 -4 -3 -2 -3 -2 -3
-2 4 -3 -2 -1 4 -2 3 -4 -3 -3 2 -4 -3 -2 3 4 -2 3 3 4 -3 2 -4
-3 -2 -3 -2 -3 -2 -3 -2 -3 -2 -3 -2 4 -1 -2 -4 -3 -3 -2 3 3 4 2 1
-2 -1 -2 -1 -2 -1 -2 -1 -2 -1 -2 -1 -1 -1 -4 2 1 1 2 3 -4 -4 -3 -2
-1 -1 -2 -3 -2 -1 -4 -3 -2 -1 -4 -3 -2 -1 -4 -3 -2 -1 -4 -3 -2 -1 -4 -3 -2 -1 -4 -3
-2 -1 -4 -3 -2 -1 -4 -3 -2 -1 -4 -3 -2 -1 -4 -3 -2 -1 -4 -3 -2 -1 -4 -3 -2 -1 -4 -3
-2 -1 -4 -3 -2 -1 -4 -3 -2 -1 -4 -3 -2 -1 -4 -3 -2 -1 -4 -3 -2 -1 -4 -3 -2 -1 -4 -3
-2 -1 -4 -3 -2 -1 -4 -3 -2 -1 -4 -3 -2 -1 -4 -3 -2 -1 -4 -3 -2 -1 -4 -3 -2 -1 -4 -3
-2 -1 -4 -3 -2 -1 -4 -3 -2 -1 -4 -3 -2 -1 -4 -3 -2 -1 -4 -3 -2 -1 -4 -3 -2 -1 -4 -3
-2 -1

SPHERE MOVE at step 17

SPHERE MOVE: cap the SIX framing corrections

Exact sphere action certified. Chosen genus-two lift discrepancy: +I. This is not an equality in the ordinary disk braid group.

FULL DISK DISCREPANCY (120 signed braid letters):

1 2 3 4 1 2 3 4 1 2 3 4 1 2 3 4 1 2 3 4 1 2 3 4
1 2 3 4 1 2 3 4 1 2 3 4 1 2 3 4 1 2 3 4 1 2 3 4
1 2 3 4 1 2 3 4 1 2 3 4 1 2 3 4 1 2 3 4 1 2 3 4
1 2 3 4 1 2 3 4 1 2 3 4 1 2 3 4 1 2 3 4 1 2 3 4
1 2 3 4 1 2 3 4 1 2 3 4 1 2 3 4 1 2 3 4 1 2 3 4

SPHERE MOVE at step 101

SPHERE MOVE: cap the FOURTH boundary power

Exact sphere action certified. Chosen genus-two lift discrepancy: +I. This is not an equality in the ordinary disk braid group.

FULL DISK DISCREPANCY (80 signed braid letters):

1 2 3 4 1 2 3 4 1 2 3 4 1 2 3 4 1 2 3 4 1 2 3 4 1 2 3 4 1 2 3 4
1 2 3 4 1 2 3 4 1 2 3 4 1 2 3 4 1 2 3 4 1 2 3 4 1 2 3 4 1 2 3 4
1 2 3 4 1 2 3 4 1 2 3 4 1 2 3 4 1 2 3 4 1 2 3 4 1 2 3 4 1 2 3 4
1 2 3 4 1 2 3 4

Certificates and exact scope

Complete named graphical route:
https://richardbuckman-math.github.io/surface-diagrams/docs/proofs/six-seven-fifty-blocks/index.html?phase=namedfull

Full raw certificates:
https://richardbuckman-math.github.io/surface-diagrams/docs/proofs/six-seven-fifty-blocks/certificate.json

The independent six-point sphere cut comparison has six marked points and five consecutive-point arcs. It is separately jointly certified; the individual factor supports in this book are not joint cut complexes.

The new 23-operation named closure of the existing-root branch has 28 retained external adjacent checks, stays positive and never exceeds support-word maximum 16. Its rose and two final daisies also retain all internal named-pattern interchanges. Four actual exterior twists remain visible. The goal of an under-twenty complete derivation remains open.

The archived 125-page/177-step PDF is preserved separately. This edition contains the new 101-step named proof, with no action-only boundary-recognition step.

Independent jointly certified six-point sphere cut comparison

